

Three-Body Approaches to Nuclear Reactions

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Weakly bound projectiles turn many nominally two-body nuclear reactions into three-body problems. This review develops a unified account of three major descriptions of such reactions: exact Faddeev-AGS theory, continuum-discretized coupled-channels methods, and the Ichimura-Austern-Vincent inclusive-breakup formalism. Using a common notation for the projectile-target coordinate, the projectile internal coordinate, fragment optical potentials, and Green's functions, I write the core operator equations explicitly: the AGS transition operator equations, the CDCC coupled radial equations, DWBA breakup and transfer matrix elements, and the post-form IAV nonelastic-breakup cross section. The formal discussion is tied to benchmark weakly bound and halo systems, including ${}^6\text{Li}$, ${}^{11}\text{Be}$, ${}^6\text{He}$, ${}^9\text{Be}$, and ${}^8\text{B}$, where elastic breakup, nonelastic breakup, transfer, complete-fusion suppression, and indirect astrophysical observables can be compared. A central conclusion is that convergence of a three-body calculation is not the same as a quantified reaction prediction: fragment optical potentials, continuum truncations, off-shell nonlocality, and omitted channels remain leading sources of uncertainty. I therefore connect the traditional reaction formalism to modern optical-potential uncertainty quantification, Fisher information geometry, Bayesian calibration, and projection-based emulators. The open problems are concrete: exact Coulomb-heavy benchmarks, four-body IAV for Borromean projectiles, nonlocal dynamic polarization potentials from continuum elimination, microscopic optical potentials usable in three-body calculations, and differentiable emulators for posterior inference across elastic, breakup, transfer, and inclusive observables.

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I. INTRODUCTION

Weakly bound projectiles make nuclear reaction theory honest. A tightly bound projectile can often be represented by an optical potential between its center of mass and the target, with all internal excitations hidden in an absorptive term. A deuteron, a halo nucleus, or a cluster projectile does not permit that reduction without conditions. Its constituents can separate on the time scale of the collision, one fragment can be captured while another is observed, the continuum can polarize the elastic channel, and the measured cross section may be inclusive over an enormous set of unobserved target states. In that situation the reaction is not a two-body collision with a small correction. It is a three-body problem.

The phrase “three-body problem” is used in several ways in the reaction literature. In its exact form it means Faddeev or Alt-Grassberger-Sandhas equations for the rearrangement operators of three interacting clusters [1–3]. In its practical continuum form it means the continuum-discretized coupled-channels method, where breakup states of the projectile are retained explicitly in a finite basis [4]. In its inclusive form it means the Ichimura-Austern-Vincent sum rule for nonelastic breakup, where an observed spectator fragment is measured and all final states of the participant-target subsystem are summed [5, 6]. These are often reviewed separately. Here they are treated as three answers to one question: given the Hamiltonian of a composite projectile interacting with a target, which degrees of freedom must remain explicit for the observable being analyzed?

The common starting point is the effective Hamiltonian introduced in Sec. II,

$$H_{3b} = T_{\mathbf{R}} + h_{bx}(\mathbf{r}) + U_{bA} + U_{xA},$$

with $\mathbf{R}$ the projectile-target coordinate and $\mathbf{r}$ the internal b - x coordinate. The same symbols are used throughout this review. This choice is not cosmetic. The history of the subject contains many apparent disagreements that are partly notational: post and prior forms of inclusive breakup, CDCC versus transfer bases, elastic breakup versus nonelastic breakup, complete fusion versus incomplete fusion. Keeping one notation makes it clear which object is a wave function, which is an optical potential, which channel is projected out, and which final states are being summed.

Figure 1 summarizes this organization. The same effective three-body Hamiltonian is projected differently depending on the measured observable, while optical-potential uncertainty and numerical verification enter every projection.

The review is organized around three physics questions. First, when does an optical-model reduction fail badly enough that the continuum must be treated explicitly? Second, once the three-body Hamiltonian is

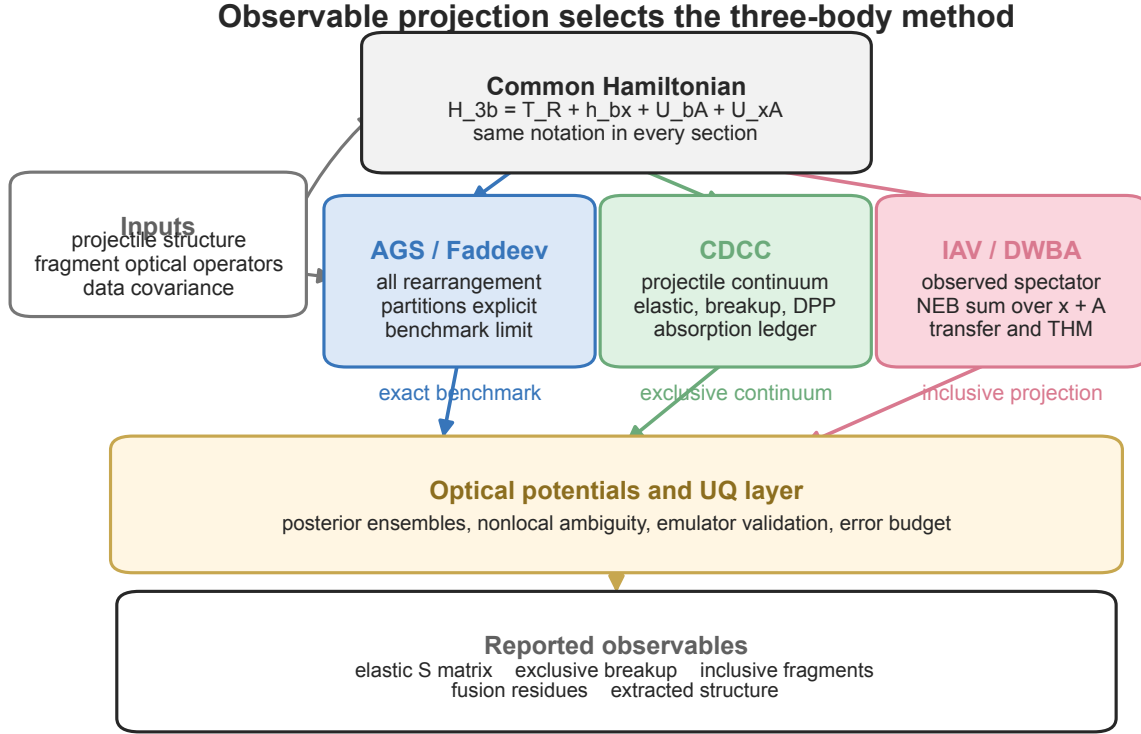


FIG. 1. Schematic organization of the review. A common three-body Hamiltonian underlies exact AGS and Faddeev benchmarks, CDCC calculations of explicit continuum dynamics, and IAV or DWBA projections for inclusive and transfer observables. Optical potentials, uncertainty quantification, and emulator validation form a common layer because they control how strongly each observable is constrained by data.

accepted, which formal description is appropriate for a given observable: exact AGS, CDCC, IAV, DWBA, or an adiabatic reduction? Third, how much of the resulting conclusion is determined by data, rather than by an unconstrained optical potential or a chosen continuum truncation? The first question is answered by scale separation and benchmark phenomenology. The second is answered by operator identities and coupled equations. The third requires uncertainty quantification, information geometry, and emulators.

The organizing thesis can be stated more sharply. Three-body reaction theory is needed when four capabilities are required at the same time: the projectile must be allowed to break, the elastic channel must feel the back-reaction of that breakup, the measured observable must be able to condition on one fragment while summing over the other, and the optical inputs must be treated as uncertain effective operators rather than as fixed numbers. Exact Faddeev or AGS theory supplies the first capability without truncating rearrangement channels. CDCC supplies the second capability in a form usable for heavy targets and radioactive beams. IAV supplies the third capability through the post-form inclusive sum over nonelastic participant states. UQ and emulation supply the fourth capability by turning the fragment op-

tical potentials and their uncertainties into part of the calculation rather than background assumptions.

This division is not a taxonomy of methods for its own sake. It is a way to avoid the most common misreadings of weakly bound reactions. A fitted two-body optical potential can reproduce elastic scattering and still have the wrong mechanism. A CDCC calculation can converge the elastic channel and still not be an exact transfer calculation. An IAV calculation can correctly sum nonelastic participant states and still inherit an unconstrained participant optical potential. A Bayesian posterior can assign a credible interval to Woods-Saxon parameters and still miss a functional ambiguity in a phase-equivalent nonlocal operator. These are not peripheral caveats. They are where spectroscopic factors, complete-fusion suppression, Trojan-Horse rates, and knockout quenching become physics claims rather than model outputs.

The benchmark systems make the thesis concrete. The deuteron is the minimal case, because its breakup continuum is simple enough to compare exact AGS, CDCC, adiabatic, and DWBA descriptions. ^6Li and ^7Li show how stable weakly bound projectiles can generate large incomplete-fusion and inclusive-alpha yields without a large elastic-breakup branch. ^{11}Be shows that halo breakup and core excitation can both be necessary before

a near-barrier reaction becomes a structure probe. ^6He and ^9Be show where the projectile itself is a three-body continuum, so a two-cluster reaction model is at best an effective reduction. ^8B shows that three-body reaction dynamics is not a nuisance correction to astrophysical inference, but the condition under which Coulomb dissociation can be used at all.

Several excellent reviews define parts of this landscape. The CDCC review of Austern and collaborators remains the foundation for deuteron and weakly bound projectile breakup [4]. Faddeev and few-body scattering reviews explain the exact three-cluster formulation and its numerical representations [7–9]. Reviews of weakly bound fusion and radioactive-beam reactions collect the experimental phenomenology of breakup, threshold anomalies, Coulomb dissociation, and complete-fusion suppression [10–12]. Recent UQ and emulator reviews have clarified Bayesian calibration, optical potentials, and model reduction [13–15]. The gap is a synthesis that puts exact three-body theory, CDCC, inclusive breakup, weakly bound benchmarks, and optical-potential UQ into one operator story.

The need for that synthesis is partly historical. The exact few-body community developed a language of transition operators, partitions, and screening limits. The coupled-channel community developed a language of bins, polarization potentials, and continuum convergence. The inclusive-breakup community developed a language of spectator fragments, source functions, and absorptive participant Green functions. The weakly bound fusion community developed a language of complete fusion, incomplete fusion, and breakup thresholds. The UQ community now adds posterior ensembles, effective dimension, and model discrepancy. These languages are all useful, but they can obscure the fact that the underlying reduced Hamiltonian is often the same. The purpose of using one notation in this review is to make the translation explicit enough that a result in one language can be tested in another.

A simple example is the word “breakup”. In an exclusive CDCC calculation it usually means population of discretized continuum states whose fragments are asymptotically observed. In an IAV calculation it is split into elastic breakup and nonelastic breakup, the latter summing over all states of the participant-target subsystem. In a fusion experiment it may be inferred from the absence of complete-fusion flux, even if the measured residue is incomplete fusion or transfer-induced capture. In a THM analysis it may be the entrance reaction used to access a low-energy binary subreaction. These are not interchangeable observables. A central claim of this review is that many apparent disagreements in weakly bound reactions become clearer once the observable projection is named before the mechanism.

Another example is the optical potential. In an elastic two-body analysis it is an efficient parametrization of elastic loss. In CDCC it becomes a fragment-target operator appearing in diagonal and off-diagonal continuum

couplings. In IAV its imaginary part is the inclusive sum over eliminated participant-target channels. In DWBA transfer it controls interior distorted waves and therefore can carry off-shell sensitivity. In Bayesian calibration it is a family of correlated uncertain operators, not a single curve. A review that treats the optical potential as a fixed background input misses why modern three-body reaction theory and uncertainty quantification have become inseparable.

The present article therefore reads the field less as a chronology than as a sequence of projections. The exact projection keeps all three rearrangement channels. The CDCC projection keeps the projectile continuum explicitly but chooses a projectile partition. The IAV projection observes one fragment and sums the other fragment over nonelastic outcomes. The DWBA projection keeps only the transition amplitude to a specified final configuration. The Bayesian projection turns uncertain optical and structure inputs into probability distributions. These projections are not ranked by prestige. They are selected by the observable and by the accuracy required for the physics claim.

That synthesis matters because several current controversies are not resolved by better prose or by another fit. The breakup threshold anomaly can look like a new physical effect when fitted optical potentials are read too literally, and can disappear when their uncertainty is propagated [16, 17]. Complete-fusion suppression of $^6,7\text{Li}$ and ^9Be on heavy targets cannot be understood by equating the missing flux with elastic breakup alone [18–20]. Halo-induced reactions such as $^{11}\text{Be}+^{64}\text{Zn}$ require core excitation and nonelastic breakup before they can be used as structure probes [21–23]. Knockout quenching and THM astrophysical rates raise the same issue in different language: the extracted structure quantity is only as reliable as the three-body reaction model that creates the measured observable [24–26].

The structure of the paper follows the logic of increasing realism. Section II reviews why the optical model works and where it breaks for weakly bound systems. Section III gives the exact Faddeev-AGS formulation, including the Coulomb difficulty, separable representations, and few-body universality. Section IV derives the CDCC coupled equations and their extensions to core excitation, four-body projectiles, microscopic inputs, and emulators. Section V presents the IAV post-form NEB formula, its post-prior relation, its DWBA and CDCC implementations, and its connections to fusion, THM, and surrogate reactions. Section VI compares the benchmark projectiles ^6Li , ^{11}Be , ^6He , ^9Be , and ^8B . Section VII then turns to optical potentials, Fisher information, Bayesian calibration, and differentiable emulators. The final sections collect open problems and the conclusions.

The intended standard is deliberately equation-level. AGS equations, CDCC coupled equations, DWBA amplitudes, and IAV source terms are written explicitly, with formulae tied to their source literature and uncertain normalizations marked in the source file. A review of this

subject that does not write the operators risks reducing reaction theory to terminology. The goal here is the opposite: make the approximations visible enough that their domain of validity, their numerical cost, and their uncertainty can be discussed on the same footing.

II. WHEN A NUCLEAR REACTION BECOMES A THREE-BODY PROBLEM

A. The optical model and its formal foundation

The optical model is the oldest and most successful reduction in reaction theory: the many-body collision of a nucleon with a target of A nucleons is replaced by the motion of a single particle in a complex mean field,

$$\left[-\frac{\hbar^2}{2\mu} \nabla^2 + U(r) - E \right] \chi(\mathbf{R}) = 0, \quad (1)$$

$$U(r) = V(r) + iW(r),$$

where the imaginary part W removes flux from the elastic channel to account for all nonelastic processes. Introduced by Feshbach, Porter, and Weisskopf to describe average neutron cross sections below 20 MeV [27], the model matured into global parametrizations of Woods-Saxon wells whose predictive power for nucleon scattering is remarkable: the potentials of Becchetti and Greenlees [28], the CH89 systematics [29], and the Koning-Delaroché analysis [30] describe elastic angular distributions, analyzing powers, and total cross sections from keV to 200 MeV over most of the nuclear chart with a single smooth energy dependence. The parameters themselves carry physical meaning; folding interpretations relate the real geometry to the matter distribution [31, 32], and causality imposes a dispersion relation that links the energy dependence of the real and imaginary parts [33].

The formal license for this reduction was given by Feshbach [34, 35]. Let P project onto the elastic channel and $Q = 1 - P$ onto everything else. The many-body Schrödinger equation $(E - H)\Psi = 0$ is exactly equivalent to a one-channel equation for $P\Psi$,

$$\left[E - H_{PP} - H_{PQ} \frac{1}{E^+ - H_{QQ}} H_{QP} \right] P\Psi = 0, \quad (2)$$

with $E^+ = E + i\epsilon$ and $H_{PP} = PHP$, $H_{PQ} = PHQ$, and so on. The generalized optical potential

$$\mathcal{U}(E) = H_{PP} + H_{PQ} \frac{1}{E^+ - H_{QQ}} H_{QP} \quad (3)$$

is therefore not a phenomenological invention but an exact object. It is nonlocal, complex above the first nonelastic threshold, and strongly energy dependent, with poles at the compound-nucleus resonances. The smooth empirical potential of Eq. (1) corresponds to an energy average of Eq. (3) over an interval containing many resonances; the averaging generates the absorptive

part, and the real and imaginary parts of the averaged potential satisfy the subtracted dispersion relation

$$V(r, E) = V_0(r) + \frac{\mathcal{P}}{\pi} \int dE' \frac{W(r, E')}{E' - E}, \quad (4)$$

where $\mathcal{P}$ denotes the principal value [33, 34]. Equation (4) will return in Sec. II C as the engine of the threshold anomaly.

Two points in this construction deserve emphasis, because they mark exactly where the practical optical model can fail. First, the projector P is not unique; Feshbach's second paper establishes that only its existence matters [35], and this freedom permits sequential eliminations in which different groups of channels are absorbed in different orders, a freedom exploited by modern effective-Hamiltonian constructions for composite systems [36]. Second, in any application to a reaction more complicated than nucleon elastic scattering, the formal potential of Eq. (3) is replaced by an energy-averaged empirical well fitted to elastic data. For a tightly bound probe this substitution is harmless. For a composite, weakly bound projectile it discards precisely the physics that dominates the dynamics, as the remainder of this section documents.

B. The first cracks: deuteron-induced reactions

The deuteron exposed the limits of the two-body reduction almost immediately. Its binding energy, $S_d = 2.224$ MeV, is small compared with every other energy scale in a typical collision, and continuous spectra of breakup neutrons were observed already in the early 1960s [37], at the same time as deuteron elastic data were being fitted with ad hoc surface-absorption optical potentials [38]. A deuteron incident on a nucleus is more honestly described as a three-body system: a proton and a neutron, bound by the known interaction V_{np} , each interacting with the target through its own nucleon optical potential. For a general two-cluster projectile $a = b + x$ incident on a target A , the effective three-body Hamiltonian reads

$$H_{3b} = T_{\mathbf{R}} + h_{bx}(\mathbf{r}) + U_{bA}(\mathbf{R}_b) + U_{xA}(\mathbf{R}_x), \quad (5)$$

$$h_{bx} = T_{\mathbf{r}} + V_{bx}(\mathbf{r}),$$

where $\mathbf{r}$ is the b - x relative coordinate, $\mathbf{R}$ the projectile-target coordinate, V_{bx} the real interaction that binds the projectile, and U_{bA} , U_{xA} the fragment-target optical potentials, complex and taken at fixed energies (conventionally the fragments' share of the incident energy). The reduction of the antisymmetrized many-body problem to Eq. (5) can itself be derived from Feshbach theory, with Pauli effects folded into the effective interactions [39]; the model is therefore a controlled approximation rather than a cartoon, and it defines the arena for everything in Secs. III-V.

The cheapest reduction of Eq. (5) back to a two-body problem is the single-folding potential of Watanabe [40],

$$U_W(\mathbf{R}) = \langle \phi_0(\mathbf{r}) | U_{bA}(\mathbf{R}_b) + U_{xA}(\mathbf{R}_x) | \phi_0(\mathbf{r}) \rangle, \quad (6)$$

where ϕ_0 is the projectile ground state. Equation (6) keeps the projectile frozen in its ground state and ignores the continuum entirely. It fails quantitatively: the coupled-channels analysis of Austern and collaborators showed that for $d+^{58}\text{Ni}$ at 80 MeV the Watanabe elastic cross section differs substantially from the full solution of the three-body model, and that couplings within the breakup continuum, not merely ground-state-to-continuum couplings, are responsible [4]. The projectile is polarized by the field of the target before and during the collision, and a static folded potential cannot represent that.

A far more successful two-body reduction retains the breakup degrees of freedom in closure. Johnson and Soper observed that if the b - x excitation energies that matter are small compared with the collision energy, the internal Hamiltonian h_{bx} may be replaced by a constant, and the three-body wave function factorizes in the region where the transfer and elastic operators act [41]. The resulting adiabatic potential for the deuteron,

$$U_{AD}(\mathbf{R}) = U_{nA}(\mathbf{R}) + U_{pA}(\mathbf{R}), \quad (7)$$

evaluated with nucleon potentials at half the deuteron energy, is emphatically not a fit to deuteron elastic data; it is less diffuse and less absorptive than fitted deuteron potentials, and it describes the part of the wave function relevant to stripping while making no claim to describe elastic scattering [41]. Johnson and Tandy later showed that the adiabatic approximation is the leading term in an expansion of the three-body wave function on Weinberg states, with an a priori small expansion parameter [42], an expansion whose numerical convergence was demonstrated much later [43]. The same adiabatic logic, exported to halo nuclei, underlies the few-body Glauber and frozen-halo treatments of exotic beams [44, 45]. The lesson of this line of work is double edged. On one side, well-chosen two-body reductions of the three-body Hamiltonian can be quantitatively useful for specific observables. On the other, the reduction is observable dependent: the potential that describes transfer is not the potential that describes elastic scattering, which is another way of saying that no single energy-independent two-body potential represents a weakly bound projectile. Austern, Farrell, and Vincent made the point sharply by iterating the elastic-breakup coupling to self-consistency: the back reaction of breakup on the elastic channel nearly halves the low-partial-wave reflection coefficients [46]. A channel carrying half the flux cannot be integrated out and forgotten.

C. Dynamic polarization potentials and the threshold anomaly

When the eliminated channels matter, their shadow appears in the effective two-body potential as a dynamic polarization potential (DPP). Eliminating the breakup channels from a coupled-channels description, in the spirit of Eq. (2), adds to the bare (folded) potential a nonlocal, energy-dependent, complex correction $\Delta U(E)$. A convenient local representation is the trivially equivalent local potential, obtained by dividing the source term generated by the couplings by the elastic wave function,

$$\Delta U_{\text{pol}}(R) = \frac{\sum_{i \neq 0} V_{0i}(R) \chi_i(R)}{\chi_0(R)}, \quad (8)$$

where χ_i are the channel wave functions of the full coupled-channels solution and V_{0i} the ground-state-to-continuum coupling potentials. For breakup coupling of a weakly bound projectile the DPP is systematically repulsive in its real part and absorptive in its imaginary part, with a long range set by the low binding [4, 47]. This single result resolves an old puzzle of heavy-ion folding models: double-folding potentials that describe tightly bound projectiles must be renormalized by $N_R \approx 0.5$ to fit ^6Li and ^7Li elastic scattering, and the missing physics is exactly the repulsive breakup DPP; a microscopic calculation that couples to the continuum needs no renormalization at all [47]. Polarization potentials of comparable importance arise from transfer channels as well [48], and for halo systems the dipole coupling alone, computable analytically from the measured $B(E1)$ distribution, suppresses near-barrier elastic scattering by tens of percent [49].

The energy dependence of the DPP near the Coulomb barrier is the threshold anomaly. For tightly bound systems, nonelastic channels close as the energy drops toward the barrier, so $W(E)$ falls rapidly; the dispersion relation, Eq. (4), then forces a localized attractive peak in the real potential [50, 51]. Satchler and collaborators showed that this behavior is the one-channel shadow of channel couplings [52], and the systematics are well established for stable projectiles [53, 54]. For weakly bound projectiles the anomaly was argued to invert: breakup remains open below the barrier, the absorption persists or even grows as the energy decreases, and the dispersive real part turns repulsive. This is the breakup threshold anomaly (BTA) proposed by Hussein and collaborators for ^6Li scattering [16], and a decade of near-barrier experiments has been interpreted in these terms [55–59]. The empirical situation is, however, less settled than the number of papers suggests. Fits at individual energies carry strong parameter correlations, and a bootstrap uncertainty analysis of the $^6\text{Li}+^{209}\text{Bi}$ and $^6\text{He}+^{208}\text{Pb}$ data showed that the reported near-threshold rise of W is statistically consistent with a fitting-precision artifact [17]; an earlier systematic study had already conceded that data uncertainties prevent definitive conclusions in several systems [60]. The honest statement is that the

energy dependence of the weakly bound optical potential near threshold sits at the edge of what elastic data can determine, which is itself an argument for treating the breakup channels explicitly rather than reading their physics from the energy dependence of a fitted well.

That elastic data underdetermine the potential is an old theme with a modern quantitative form. Strong absorption means that composite-projectile elastic scattering probes only the surface: Igo showed that alpha-nucleus potentials are fixed only in an exponential surface region, with the interior left completely ambiguous [61], and global potentials for ${}^6,{}^7\text{Li}$ [62] and even for the Borromean halo ${}^6\text{He}$ [63] fit elastic data comfortably with Woods-Saxon forms; the success of such fits is not evidence that the underlying physics is two-body. A Fisher-information analysis of nucleon optical potentials sharpens the statement: elastic angular distributions constrain only an effective dimensionality $D_{\text{eff}} \approx 1.5$ to 1.7 of the nine to thirteen parameter combinations, independent of projectile, target, and energy, so the Igo ambiguity is a scale-invariant information limit rather than an accident of early fitting [64]. Bayesian calibration of a deuteron coupled-channels model against elastic and breakup data leads to the same moral from the opposite direction: the data demand a surface absorption about 36% above the global nucleon potential, a quantified measure of how far the two-body input must deform once it is embedded in an explicit three-body model [65]; these developments are taken up in Sec. VII.

D. Criteria for explicit three-body treatment

The evidence assembled above suggests operational criteria for when the two-body optical model is a safe reduction and when a nuclear reaction must be treated as a genuine three-body problem.

The first criterion is a separation of scales. The optical-model reduction assumes that the projectile's internal excitation energies are large compared with the coupling strengths in the collision, so that virtual excitation is brief and perturbative. The relevant comparison is between the cluster separation energy S_a of the projectile ($S_d = 2.224$ MeV for the deuteron, 1.474 MeV for ${}^6\text{Li} \rightarrow \alpha + d$, $S_n = 0.501$ MeV for ${}^{11}\text{Be}$, $S_{2n} = 0.975$ MeV for ${}^6\text{He}$) and the scales of the collision, the Coulomb-barrier height and the incident energy per nucleon. When S_a is small on those scales, continuum admixtures are neither brief nor perturbative: the coupling halves reflection coefficients [46], doubles reaction cross sections for halo projectiles [21], and renormalizes folded potentials by factors of two [47]. Reviews of the near-barrier phenomenology of weakly bound and exotic beams document the systematics [10, 11, 66].

The second criterion is the observable. Energy-averaged, inclusive observables (total and reaction cross sections, average elastic scattering of tightly bound probes) are exactly the quantities the energy-averaged

optical potential was built for. Observables that resolve the projectile's internal degrees of freedom (exclusive and inclusive breakup, transfer populating specific states, fusion split into complete and incomplete components, knockout spectroscopic strengths) probe the parts of the wave function that the two-body reduction integrates out. The dependence is not marginal. In knockout reactions, treating the effective interactions of a three-body system as an additive sum of two-body optical potentials omits both a nonadditive term and an induced polarization term in the effective Hamiltonian; their neglect masquerades as a dynamical quenching of spectroscopic strength [36]. In the analysis of core destruction in knockout, two groups reached opposite conclusions about the same data traceable to different choices of the valence-core optical potential at low energy [24, 25], a modern case in which optical-potential ambiguity blocks a physics conclusion.

The third criterion is diagnostic rather than a priori: compute the three-body dynamics once, then ask what the effective two-body picture would have missed. The DPP of Eq. (8) and its per-channel refinements quantify which continuum channels matter [67]. Exact flux decompositions make the absorption ledger explicit: in a coupled-channels model with absorptive potentials in all channels, a generalized optical theorem splits the absorption cross section as

$$\sigma_A = \sigma_D + \sigma_B + \sigma_{\text{int}}, \quad (9)$$

where σ_D and σ_B are the direct-channel and breakup-channel absorption components and σ_{int} is an interference term generated by the off-diagonal imaginary couplings; the common simplification of dropping those couplings biases the components by 5% to 57% in representative systems, and the interference term reaches -26% of σ_A for ${}^6\text{Li} + {}^{59}\text{Co}$ [68, 69]. A companion decomposition with an ingoing-wave boundary condition separates exactly the flux that fuses from the flux absorbed peripherally, $\sigma_{\text{abs}} = \sigma_{\text{fus}} + \sigma_W$, and locates the crossover between the two at the Coulomb barrier for ${}^6\text{Li} + {}^{209}\text{Bi}$ [70]. Feshbach theory can also be pushed through exactly for the continuum couplings themselves, yielding the optical potential that a two-body model would need in order to be equivalent to the three-body one [71]. None of these ledgers exists until the three-body channels are explicit; the two-body imaginary potential is one number where the dynamics has several.

A fourth criterion is a time-scale test. A projectile with separation energy S_a has an intrinsic time scale

$$\tau_{\text{int}} \sim \frac{\hbar}{S_a}, \quad (10)$$

while a near-barrier collision samples the target field for a time of order

$$\tau_{\text{coll}} \sim \frac{R_B}{v}, \quad (11)$$

where R_B is a barrier radius and v the relative velocity. If $\tau_{\text{int}} \ll \tau_{\text{coll}}$, the projectile can adjust adiabatically and

its internal excitations may be represented by a polarization potential. If $\tau_{\text{int}} \gtrsim \tau_{\text{coll}}$, the projectile is slow internally and the collision can freeze or break its configuration. This estimate is deliberately crude, because resonances, Coulomb tails, and absorption radii matter, but it captures why deuterons, ${}^6\text{Li}$, ${}^{11}\text{Be}$, and ${}^6\text{He}$ sit outside the ordinary optical-model comfort zone [11, 41, 44]. It also explains why the same projectile can look different at different energies: high-energy breakup can be sudden and eikonal-like, while near-barrier breakup is shaped by Coulomb acceleration, tunneling, and fragment absorption.

The time-scale argument has a companion in impact parameter. Breakup and transfer are often peripheral, but not always peripheral in the same way. Long-range Coulomb breakup acts at large R and can modify the elastic rainbow without bringing either fragment into the strong-absorption region. Nuclear breakup and transfer act closer to the surface. Complete fusion requires flux to reach still smaller radii. A single reaction cross section integrates over all these radii, but a fragment spectrum or fusion residue does not. This is why the same imaginary optical potential can be a reasonable description of total reaction loss and a poor description of which fragment was absorbed. The radial support of the source or absorption kernel is therefore part of the criterion for explicit three-body treatment, not an implementation detail.

There are two common false positives. The first is a successful elastic fit. Because strong absorption hides the interior, a two-body optical potential can fit elastic scattering even when the physical mechanism is breakup, transfer, or incomplete fusion [61–63]. The second is a smooth energy dependence of fitted parameters. A smooth fit may simply mean that the parameterization is flexible enough to absorb missing channels. It does not prove that the missing channels are dynamically small. Both false positives occur in weakly bound systems: ${}^6\text{He}$ elastic scattering can be fit with a global halo OMP, and ${}^6,{}^7\text{Li}$ elastic data can be represented by renormalized folding potentials, yet CDCC and IAV show that continuum couplings and nonelastic breakup are the mechanisms behind those effective potentials [10, 19, 47].

There are also false negatives. A small exclusive breakup cross section does not imply that breakup degrees of freedom are irrelevant. Virtual breakup can generate a large DPP while the measured EBU yield remains modest. Nonelastic breakup can dominate an inclusive spectator spectrum even when the asymptotic two-fragment breakup channel is small. Complete-fusion suppression can be driven by direct participant capture and transfer-like processes, not by free breakup followed by capture [20, 72, 73]. Therefore the decision to use a three-body method should not be based only on whether an exclusive breakup peak is large. It should be based on whether the observable is formed by channels that the two-body reduction has summed over.

A useful operational test is to ask whether the two-

body optical potential would have to be different for two observables measured in the same system. If the elastic data require one surface absorption, the inclusive fragment data require a different absorption, and the fusion residues require a third, then the two-body reduction is hiding resolved physics. This is not a failure of the optical model in its original domain. It is a sign that a single projectile-target potential is being asked to represent several different projections of the many-body final space. The remedy is not another local renormalization unless the goal is only interpolation. The remedy is to keep the fragment degrees of freedom that distinguish the observables.

The same test can be phrased in terms of conditional probabilities. Elastic scattering asks for the probability amplitude that the projectile returns to its ground state. Exclusive breakup asks for the probability amplitude that the fragments emerge with specified relative energy and angle. Inclusive breakup asks for the probability density of one fragment after summing the other over all nonelastic states. Complete fusion asks whether all projectile charge is absorbed into the compound system. Incomplete fusion asks whether only a subset is absorbed. These are different conditional statements about the same collision. A two-body optical potential is adequate only when the conditioning is not observed or when its effect can be represented by one energy-dependent elastic operator.

The criteria are also tied to the desired inference. If the aim is to interpolate an elastic angular distribution for use in a detector simulation, a two-body optical potential may be sufficient even for a weakly bound projectile. If the aim is to extract an ANC, a spectroscopic factor, a sub-Coulomb astrophysical rate, or a complete-fusion suppression mechanism, then the reaction model becomes part of the physics result. The tolerance for hidden channels is much smaller. This is why indirect methods demand more than visually acceptable fits. The extracted quantity is not measured directly; it is inferred through a reaction filter whose selectivity must be known.

One can summarize the decision by comparing three scales. The first is the projectile internal scale, set by separation energies, resonance positions, and continuum phase shifts. The second is the collision scale, set by beam energy, Coulomb barrier, target charge, and absorption radius. The third is the resolution scale of the observable, set by which fragments, residues, and energies are measured. Three-body theory becomes unavoidable when the observable resolution is fine enough to see a degree of freedom whose internal scale is comparable to the collision scale. This statement is intentionally broader than a threshold rule. It includes deuteron transfer, halo Coulomb dissociation, weakly bound fusion, THM, and knockout.

The scale comparison also explains why the same projectile can move between descriptions. A deuteron at high energy in an inclusive reaction may be treated by an eikonal or adiabatic approximation. The same deuteron

in a precision transfer reaction may need breakup and rearrangement benchmarks. ${}^6\text{Li}$ elastic scattering may be captured by a renormalized folding potential, while inclusive alpha production from the same system requires IAV. ${}^8\text{B}$ Coulomb dissociation at very forward angles may be close to a first-order electromagnetic process, while less restrictive cuts require all-order CDCC. The projectile label does not select the theory; the observable and accuracy goal do.

These criteria define the program of this review. When they are violated, one needs the machinery of the following sections: the exact Faddeev formulation (Sec. III) when rigor and benchmarking are the issue, the continuum-discretized coupled-channels method (Sec. IV) as the workhorse for elastic scattering and exclusive breakup, and the inclusive-breakup theory of Ichimura, Austern, and Vincent (Sec. V) when the experiment detects only one fragment.

III. EXACT SOLUTIONS: FADDEEV AND ALT-GRASSBERGER-SANDHAS EQUATIONS

A. Formal theory

The three-body Hamiltonian introduced in Eq. (5) is not merely a more detailed optical model. It has a different scattering theory. If one writes the total interaction as a sum of pair interactions,

$$H = H_0 + v_1 + v_2 + v_3, \quad (12)$$

where H_0 is the free Hamiltonian in any one set of Jacobi coordinates and v_α acts in the pair opposite to spectator α , the naive Lippmann-Schwinger equation

$$\begin{aligned} \Psi^{(+)} &= \Phi_\alpha + G_0(E) (v_1 + v_2 + v_3) \Psi^{(+)}, \\ G_0(E) &= \frac{1}{E^+ - H_0}, \end{aligned} \quad (13)$$

has disconnected pieces in its kernel. The iteration of Eq. (13) repeatedly scatters the same pair while the third particle is a spectator, so compactness is lost and convergence cannot be read as a controlled three-body calculation. Faddeev's central step was to decompose the wave function into components,

$$\begin{aligned} \Psi_\alpha^{(+)} &= \Phi_\alpha + \sum_{\beta=1}^3 \psi_\beta, \\ \psi_\beta &= G_0(E) v_\beta \Psi_\alpha^{(+)}, \end{aligned} \quad (14)$$

with Φ_α the incident channel state in partition α . The components for an incident channel α obey the coupled equations

$$\begin{aligned} \psi_\beta^{(\alpha)} &= \delta_{\beta\alpha} \Phi_\alpha + G_0(E) t_\beta(E) \sum_{\gamma \neq \beta} \psi_\gamma^{(\alpha)}, \\ t_\beta(E) &= v_\beta + v_\beta G_0(E) t_\beta(E), \end{aligned} \quad (15)$$

where t_β is the two-body transition operator embedded in the three-body space. Equation (15) is the mathematical point at which the exact formulation separates itself from coupled-channel approximations: all pair rescatterings inside a subsystem are summed into the pair operators t_β , and the remaining kernel only transfers the collision between different pair partitions. This removes the disconnected part and gives the three-body problem the same analytic discipline that the two-body Lippmann-Schwinger equation has for short-range forces [1, 3].

For reactions, the most useful version is the Alt-Grassberger-Sandhas (AGS) form, which promotes the component equations to transition-operator equations between asymptotic channels [2, 74]. Let

$$\bar{\delta}_{\beta\alpha} = 1 - \delta_{\beta\alpha} \quad (16)$$

and let $U_{\beta\alpha}$ denote the operator that takes an incoming channel α into an outgoing channel β . The AGS equations, in the notation of Alt, Grassberger, and Sandhas [2], are

$$U_{\beta\alpha}(E) = \bar{\delta}_{\beta\alpha} G_0^{-1}(E) + \sum_{\gamma=1}^3 \bar{\delta}_{\beta\gamma} t_\gamma(E) G_0(E) U_{\gamma\alpha}(E). \quad (17)$$

The breakup operator is obtained from the same set [2],

$$U_{0\alpha}(E) = G_0^{-1}(E) + \sum_{\gamma=1}^3 t_\gamma(E) G_0(E) U_{\gamma\alpha}(E), \quad (18)$$

where the label 0 denotes the three-free-particle channel. Once $U_{\beta\alpha}$ is known, the on-shell transition amplitude is

$$T_{\beta\alpha} = \langle \Phi_\beta^{(-)} | U_{\beta\alpha}(E) | \Phi_\alpha^{(+)} \rangle, \quad (19)$$

with channel states normalized to unit incident flux. Elastic scattering, transfer, rearrangement, and breakup are therefore not separate models. They are different matrix elements of the same unitary three-body operator, up to the non-Hermitian optical interactions deliberately introduced when the target is not treated explicitly.

B. Observables, unitarity, and what the exact solution buys

The AGS equations are often introduced as formal integral equations, but their real value for reaction theory is that they define all observable channels from one operator. For a two-cluster channel α with relative momentum k_α and reduced mass μ_α , the partial-wave S matrix may be written schematically as

$$S_{\beta\alpha}^J = \delta_{\beta\alpha} - 2\pi i \delta(E_\beta - E_\alpha) \langle \Phi_\beta^J | U_{\beta\alpha}(E) | \Phi_\alpha^J \rangle. \quad (20)$$

The precise density-of-states factor depends on the normalization of the channel states, but the physics does not: the same $U_{\beta\alpha}$ gives the elastic amplitude for $\beta = \alpha$, the

transfer or rearrangement amplitude for $\beta \neq \alpha$, and the breakup amplitude through $U_{0\alpha}$. In a spinless notation the angle-integrated two-cluster transition cross section is then of the form

$$\sigma_{\beta\alpha} = \frac{\pi}{k_\alpha^2} \sum_J (2J+1) |S_{\beta\alpha}^J - \delta_{\beta\alpha}|^2, \quad (21)$$

while the fully exclusive breakup cross section is obtained from

$$T_{0\alpha} = \left\langle \Phi_0^{(-)} | U_{0\alpha}(E) | \Phi_\alpha^{(+)} \right\rangle, \quad (22)$$

where $\Phi_0^{(-)}$ denotes a three-free-particle scattering state with the proper asymptotic boundary condition. These equations make a point that is easy to lose in coupled-channel language. Breakup is not an add-on channel after elastic scattering has been fitted. It is one branch of the same three-body scattering operator that also determines the elastic channel.

For real pair interactions the exact three-body S matrix is unitary,

$$\sum_\beta S_{\beta\alpha}^{J\dagger} S_{\beta\alpha'}^J = \delta_{\alpha\alpha'}, \quad (23)$$

where the sum includes elastic, rearrangement, and breakup sectors. This identity is more than a formal check. It is the reason exact three-body calculations can decide whether an apparent loss of elastic flux is breakup, transfer, or a missing channel in the Hamiltonian. When phenomenological fragment-target optical potentials are inserted, unitarity is intentionally broken because target excitations and compound processes have been removed. The absorption cross section for the model Hamiltonian is then controlled by the imaginary parts of the pair interactions,

$$\sigma_{\text{abs}} = -\frac{2}{\hbar v_\alpha} \left\langle \Psi_\alpha^{(+)} \left| \sum_{\gamma=1}^3 W_\gamma \right| \Psi_\alpha^{(+)} \right\rangle, \quad W_\gamma = \text{Im} v_\gamma. \quad (24)$$

This is the exact-theory version of the flux ledgers that reappear in CDCC and IAV. It also states the limitation of an “exact” calculation with optical potentials. The wave function is exact for the chosen reduced Hamiltonian, not for the original many-body problem. If v_γ is a local optical potential fitted to elastic data, the AGS solution cannot repair the missing nonlocality, energy dependence, or target-structure information hidden in that fit. Exactness and input realism are distinct.

In many few-nucleon applications this distinction is mild, because the pair interactions are real nucleon-nucleon potentials or chiral interactions, and the reaction calculation genuinely tests the Hamiltonian [7, 75]. In reactions involving a composite projectile on a medium or heavy target, the same equations usually contain complex nucleon-target, core-target, or fragment-target optical potentials. The resulting calculation is still essential,

because it treats breakup and rearrangement without a CDCC truncation. But the comparison with data must be read as a test of the reduced three-body model. This is why the exact benchmarks against CDCC in deuteron reactions are so valuable: they isolate the error due to the continuum discretization while holding the same fragment optical input fixed [76, 77].

This unity is the main reason AGS calculations have become the standard against which approximate reaction theories are judged. The Bochum momentum-space program established this status for the three-nucleon continuum, with n - d and p - d elastic and breakup observables computed from realistic two-nucleon forces and, later, consistent three-nucleon forces [7, 75]. The same calculations also sharpened the field’s anomalies. The triton underbinding of modern two-nucleon forces, the roughly 30% underprediction of the vector analyzing power A_y , and the space-star discrepancy are not artifacts of an uncontrolled scattering solver. They are force or current problems exposed by an exact reaction calculation. This diagnostic role is exactly what is needed in nuclear reaction theory: an exact three-body solution may be too expensive for daily analysis, but it tells the approximate methods what they are approximating.

C. The Coulomb problem

The short-range formalism above is not enough for the reactions that motivate this review, because charged fragments and charged targets are almost always present. Long-range Coulomb forces spoil the compactness properties on which the Faddeev proof relies. If the Coulomb interaction between particles β and γ is denoted by w_α^C , the corresponding two-body transition operator is singular on shell and its partial-wave series converges slowly. The problem is not cosmetic. For a proton, deuteron, alpha particle, or halo core on a medium-mass target, the Coulomb field acts at distances much larger than the nuclear interaction range, precisely where breakup channels are already being formed.

One production strategy is screening and renormalization [78, 79]. The Coulomb potential is replaced by a screened interaction

$$w_{\alpha R}^C(r) = w_\alpha^C(r) \exp \left[- \left(\frac{r}{R} \right)^n \right], \quad (25)$$

with screening radius R and exponent n . For finite R all interactions are short range, so Eq. (17) can be solved in the usual momentum basis. The physical amplitude is recovered by subtracting the screened pure Coulomb amplitude and renormalizing the external charged channels,

$$f_{\beta\alpha} = f_{\beta\alpha}^C + \lim_{R \rightarrow \infty} Z_{\beta R}^{-1/2} \left[f_{\beta\alpha}^R - f_{\beta\alpha}^{C,R} \right] Z_{\alpha R}^{-1/2}. \quad (26)$$

Here $f_{\beta\alpha}^R$ is the full screened amplitude, $f_{\beta\alpha}^{C,R}$ is the screened two-body Coulomb amplitude between the

charged clusters, $f_{\beta\alpha}^C$ is the exact Rutherford amplitude, and $Z_{\alpha R}$ is the divergent phase factor that converts the screened Coulomb wave into the unscreened Coulomb wave in channel α . In partial waves this phase factor may be written as

$$Z_{\alpha R}(k_\alpha) = \exp[-2i(\sigma_\ell^\alpha(k_\alpha) - \eta_{\ell R}^\alpha(k_\alpha))], \quad (27)$$

where σ_ℓ^α is the exact Coulomb phase shift and $\eta_{\ell R}^\alpha$ is the phase shift generated by the screened potential. This route has produced high-precision proton-deuteron and deuteron-induced benchmarks [76, 77, 80]. Its limitation is also clear: increasing charge and decreasing momentum transfer push the required screening radii and partial waves upward, and calculations that are routine for light targets become stretched for ^{12}C and unfavorable for heavier nuclei [76].

A second route keeps the Coulomb field in the basis from the beginning. In the Coulomb-distorted momentum-space formulation, the short-range nuclear operators are sandwiched between Coulomb scattering states, and the Coulomb-distorted form factors replace the ordinary plane-wave form factors [81, 82]. This avoids the limiting procedure in Eq. (26) and is conceptually attractive for heavy targets, but it moves the difficulty into the off-shell Coulomb transition operator and into the regularization of singular form factors. The generalized AGS program with target excitation, developed for transfer reactions, follows this line [83, 84]. Its value for this review is not that it replaced screening in everyday calculations. Rather, it shows that the difficulty of Coulomb in three-body reactions is a representation problem, not a failure of the exact formalism.

The third route is the configuration-space Faddeev-Merkuriev formulation [85, 86]. In Jacobi coordinates (x_α, y_α) for partition α , each Coulomb interaction is split by a smooth cutoff function ζ_α ,

$$\begin{aligned} w_\alpha^C(x_\alpha) &= w_\alpha^s(x_\alpha, y_\alpha) + w_\alpha^l(x_\alpha, y_\alpha), \\ w_\alpha^s(x_\alpha, y_\alpha) &= w_\alpha^C(x_\alpha)\zeta_\alpha(x_\alpha, y_\alpha), \\ w_\alpha^l(x_\alpha, y_\alpha) &= w_\alpha^C(x_\alpha)[1 - \zeta_\alpha(x_\alpha, y_\alpha)]. \end{aligned} \quad (28)$$

The function ζ_α tends to one in the two-body cluster asymptotic region of partition α and to zero in the genuine three-body breakup region. The long-range part is then retained in the channel Hamiltonian,

$$\begin{aligned} (E - H_0 - W_\alpha - v_\alpha^s)\psi_\alpha &= v_\alpha^s \sum_{\beta \neq \alpha} \psi_\beta, \\ W_\alpha &= w_1^l + w_2^l + w_3^l, \end{aligned} \quad (29)$$

so that the coupling terms remain short range. Combined with exterior complex scaling, the method converts scattering boundary conditions into square integrability in a rotated coordinate sector. The important practical point is subtle: after complex scaling the total wave function may decay, while an individual transfer amplitude still has to be extracted by a Green's-theorem surface or volume integral rather than by reading off an asymptotic

coefficient. This is one of the reasons exact Coulomb three-body solvers are powerful benchmarks but demanding production tools.

The three Coulomb strategies should be viewed as complementary rather than as a sequence in which one has defeated the others. Screening and renormalization is algebraically close to the short-range AGS machinery and has produced the most systematic momentum-space benchmarks for deuteron and proton-induced reactions [76, 80]. Coulomb-distorted AGS keeps the physical asymptotic field from the start and is better matched to charged-cluster transfer, but the singular off-shell Coulomb structure makes the numerical representation more specialized [81, 83]. Faddeev-Merkuriev configuration-space theory gives the cleanest coordinate-space separation of two-body Coulomb asymptotics from three-body breakup, at the price of large multidimensional boundary-value problems [85, 86]. Which method is best depends on the benchmark being asked for.

For the weakly bound reactions in this review, the Coulomb problem is not only the Rutherford tail. It is the coexistence of a long-range field with a low breakup threshold. A halo projectile can be polarized at radii where the nuclear interaction is negligible. A charged fragment can tunnel or be accelerated while the neutral fragment is absorbed. A near-barrier reaction can depend on Coulomb phases in both the entrance and exit partitions. These features are precisely what make CDCC attractive for heavy targets, because it builds the correct two-body Coulomb waves into coupled radial equations. They are also what make exact AGS benchmarks rare in the same systems. The absence of an exact benchmark for a high- Z halo reaction should therefore not be read as a formal gap in Faddeev theory. It is a numerical gap at the intersection of long range, low binding, many partial waves, and complex fragment optical potentials.

The practical consequence is that exact calculations should be chosen to stress one approximation at a time. A deuteron on ^{12}C can test CDCC continuum discretization with moderate Coulomb. A halo projectile on a proton target can test the entrance-partition bias of CDCC against exact transfer and breakup. A charged light cluster can test screening radii and Coulomb renormalization. A heavy target may be better used to test CDCC convergence and DPP extraction, with exact theory supplying nearby lighter benchmarks. This is less glamorous than a single universal exact calculation, but it is how exact theory can most effectively control the approximations used in data analysis.

D. Numerical methods and bottlenecks

The numerical burden of the exact theory comes from dimensionality before it comes from sophistication. A three-body scattering wave function depends on two Jacobi vectors, and after angular-momentum coupling the integral equations still contain two continuous momenta.

The standard momentum-space partial-wave representation writes a channel state as

$$|p_\alpha q_\alpha \nu_\alpha\rangle = \left[[Y_{\ell_\alpha}(\hat{p}_\alpha) Y_{\lambda_\alpha}(\hat{q}_\alpha)]_{L_\alpha} [s_\beta s_\gamma]_{S_\alpha} s_\alpha \right]_{JM} |p_\alpha q_\alpha\rangle, \quad (30)$$

where p_α is the relative momentum in the interacting pair, q_α the spectator momentum, ℓ_α and λ_α the two orbital angular momenta, and ν_α abbreviates the spin, isospin, and coupling labels. Matrix elements of Eq. (17) then form coupled two-dimensional integral equations. Their kernels contain moving logarithmic singularities above breakup, and practical calculations rely on contour deformation, complex-energy smoothing, special quadratures, or reformulations that remove the moving singularity from the kernel [7, 9, 87].

Finite-rank separable expansions are the classic reduction that makes the AGS equations tractable for nuclear

reactions. The two-body transition operator in pair α is represented as

$$t_\alpha(E) \simeq \sum_{i,j=1}^{N_\alpha} |g_{\alpha i}\rangle \tau_{ij}^\alpha(E) \langle g_{\alpha j}|, \quad (31)$$

where the form factors $g_{\alpha i}$ and the propagator matrix τ_{ij}^α are chosen so that the exact two-body scattering amplitudes are reproduced at a set of support energies. In the Ernst-Shakin-Thaler construction, this matching is exact on shell and half-off shell at the support points [88, 89]. Substitution of Eq. (31) into Eq. (17) collapses the AGS kernel to coupled one-dimensional equations for the reduced amplitudes, following the Lovelace and AGS separable reduction [2, 74].

$$X_{\beta j, \alpha i}(q_\beta, q_\alpha; E) = Z_{\beta j, \alpha i}(q_\beta, q_\alpha; E) + \sum_{\gamma, n, m} \int_0^\infty dq_\gamma q_\gamma^2 Z_{\beta j, \gamma n}(q_\beta, q_\gamma; E) \tau_{nm}^\gamma \left(E - \frac{q_\gamma^2}{2M_\gamma} \right) X_{\gamma m, \alpha i}(q_\gamma, q_\alpha; E), \quad (32)$$

where M_γ is the spectator reduced mass and Z contains the exchange geometry between partitions. This is the computational logic behind the Faddeev-Lovelace calculations of light cluster systems and behind modern separable optical-potential programs [90–92].

Separable representations are approximations to the pair dynamics, and their validity has to be tested at the three-body level, not only in the two-body subsystems. The $n + p + \alpha$ system is a clean laboratory for this test. Early separable Faddeev calculations already described $d + \alpha$ elastic scattering and the low-lying spectrum of ${}^6\text{Li}$ [93–95]; subsequent work treated resonances as complex eigenvalues of the Faddeev kernel, clarified the role of the Pauli-forbidden α - N state, and showed that local interactions with explicit breakup lead to the same physics when convergence is controlled [96–100]. In the modern benchmark, separable and nonseparable momentum-space Faddeev calculations agree to four significant figures for the ${}^6\text{Li}$ bound state, and the same technology gives roughly one percent agreement for $d + \alpha$ elastic and breakup observables [101, 102]. That result matters beyond ${}^6\text{Li}$. It says that a finite-rank representation of the two-body input can be accurate enough for exact three-body reaction benchmarks, provided the rank, support energies, and off-shell behavior are audited against a nonseparable calculation.

The same reduced equations also show where off-shell information enters. A two-body interaction is usually constrained by phase shifts and bound-state properties, but Eq. (32) samples the half-off-shell and fully off-shell transition operator through the exchange kernel Z and

the spectator momentum integration. This is harmless when the off-shell behavior is fixed by a microscopic Hamiltonian and the omitted many-body force is included consistently. It is less harmless when a fragment-target optical potential is replaced by a phase-equivalent local form. The three-body observable can then move while all two-body elastic observables remain unchanged. The nonlocal optical-potential work based on separable representations makes this visible: an EST representation that is exactly on-shell equivalent to a local OMP is nevertheless a nonlocal operator in coordinate space, and its near-diagonal Perey-Buck-like pocket is only part of the full kernel [91, 92, 103]. This is the exact-theory origin of the off-shell uncertainty discussed again in Sec. VII.

The comparison with continuum-discretized coupled channels is where the exact solver becomes most useful for this review. For deuteron scattering on ${}^{12}\text{C}$ at 56 MeV and ${}^{58}\text{Ni}$ at 80 MeV, CDCC agrees well with AGS for elastic scattering and breakup observables, supporting its use as the workhorse method for weakly bound stable projectiles [76]. The same benchmark is less forgiving for ${}^{11}\text{Be} + p$ at 38.4 MeV per nucleon, where transfer is underestimated and the breakup result depends on how the CDCC basis is arranged. A broader deuteron-induced study found no universal correlation between elastic, transfer, and breakup accuracy: transfer is best near 10 MeV per nucleon and degrades with increasing energy, while breakup shows the opposite trend and can fail at low energy because different Faddeev components interfere destructively [77]. This is the precise sense in which AGS is a benchmark rather than a replacement.

It does not say that CDCC is unreliable. It says which observable and energy domain the CDCC truncation actually controls.

Configuration-space exact methods face the same wall in a different language. Faddeev-Yakubovsky calculations for four particles, hyperspherical-harmonic benchmarks, and complex-scaled configuration-space solutions have reached three- and four-nucleon scattering with impressive precision [104–106]. They also show why the frontier moves slowly. The number of coupled channels can reach tens or hundreds even in ${}^6\text{Li}$ structure calculations, four-body continuum calculations carry millions of unknowns, and realistic Coulomb plus breakup above threshold remains a numerical project rather than a black-box input to data analysis. Recent developments recast the cost problem as a reduction problem: variational emulators built from exact snapshots, complex-energy reduced bases, low-rank Woodbury reductions for affine short-range updates, and neural scattering-state ansatzes all aim to move exact few-body physics into parameter studies [107–110]. Their domains of validity differ. An exact Woodbury reduction is natural when the parameter dependence enters through low-rank separable operators, as in some chiral three-nucleon-force studies, but that same algebra does not automatically carry over to a CDCC optical-potential emulator whose breakup couplings depend on the calibrated interaction. The honest conclusion is therefore conservative: exact solvers define the reference surface, and emulators are beginning to sample that surface, but the route from few-nucleon exactness to routine weakly bound heavy-ion inference is still open.

There is a second bottleneck that is more physical than numerical: consistency of the Hamiltonian. In a few-nucleon calculation, two-nucleon forces, three-nucleon forces, and electromagnetic currents can be varied coherently, so a discrepancy such as the A_y puzzle can be assigned to missing force or current physics rather than to the scattering solver [7, 75]. In a nuclear-reaction calculation with an optical potential, the pair interactions are effective, complex, and often taken from different global fits. The exact three-body solution then tests the reaction approximation, but not a single underlying microscopic Hamiltonian. This is why agreement between AGS and CDCC for a given optical input is only one level of validation. The optical input itself remains part of the theory uncertainty.

The same issue appears in cluster models. The α - N interaction in $n+p+\alpha$ calculations must represent Pauli exclusion between the valence nucleon and the core. One can use repulsive cores, deep potentials with forbidden-state projection, or phase-equivalent interactions that remove the forbidden state. These choices may agree in two-body phase shifts but differ in three-body binding and scattering unless the projection is handled consistently [97, 111, 112]. Exact methods make the dependence visible. They do not eliminate the need to decide which effective interaction represents the many-body

physics.

This is the reason exact benchmarks should be reported with the same care as experimental comparisons. The pair potentials, support points of separable expansions, partial-wave truncations, treatment of Coulomb, Pauli projection, and optical-potential energy prescription all define the benchmark. If two methods disagree, the difference is useful only after these inputs are matched. If they agree, the agreement establishes the accuracy of the reaction approximation for that Hamiltonian, not the correctness of the Hamiltonian itself. This distinction becomes central when exact results are used to justify CDCC or adiabatic reductions in systems where no exact calculation with the physical target is feasible.

E. Adiabatic and Weinberg reductions

Exact three-body theory also explains why much simpler deuteron-reaction models sometimes work. In a (d, p) reaction the transition operator samples the three-body wave function at small neutron-proton separations because the binding interaction V_{np} appears in the transfer matrix element. Johnson and Soper exploited this scale separation by freezing the internal deuteron motion during the collision [41]. In the notation of this review, the adiabatic entrance equation is

$$[E_d - T_{\mathbf{R}} - U_{pA}(\mathbf{R}_p) - U_{nA}(\mathbf{R}_n)] \Psi_d^{\text{AD}}(\mathbf{R}, \mathbf{r}) = 0, \quad (33)$$

with the deuteron binding energy neglected in the energy denominator of the continuum propagation. The simplest Johnson-Soper potential follows by folding the nucleon optical potentials over the deuteron ground-state density,

$$U_{\text{JS}}(\mathbf{R}) = \langle \phi_d(\mathbf{r}) | U_{pA}(\mathbf{R}_p) + U_{nA}(\mathbf{R}_n) | \phi_d(\mathbf{r}) \rangle_{\mathbf{r}}. \quad (34)$$

Equation (34) is not a phenomenological deuteron optical potential. It is the first term of a three-body reduction, and it retains the fact that the proton and neutron see different points in the target field.

The Weinberg-state formulation gives the sharper statement [42, 43]. One expands the three-body wave function as

$$\Psi^{(+)}(\mathbf{R}, \mathbf{r}) = \sum_{i=1}^{\infty} \Phi_i^W(\mathbf{r}) X_i(\mathbf{R}), \quad (35)$$

where the Weinberg states satisfy

$$(T_{\mathbf{r}} + \alpha_i V_{np}) \Phi_i^W(\mathbf{r}) = -\epsilon_d \Phi_i^W(\mathbf{r}), \quad (36)$$

and are normalized by the V_{np} metric,

$$\langle \Phi_i^W | V_{np} | \Phi_j^W \rangle = -\delta_{ij}. \quad (37)$$

The transfer amplitude is then dominated by the first Weinberg component because the same V_{np} metric appears in the transition operator. The adiabatic distorted

wave is the one-component limit of this expansion, not a guess. Higher Weinberg components measure the correction to adiabaticity in the short-range part of the wave function that transfer actually sees.

This reduction is an important bridge between exact theory and CDCC. CDCC tries to describe the projectile continuum globally, including large neutron-proton separations and asymptotic breakup phases. The Weinberg expansion describes the part of the same continuum most strongly weighted by transfer. This is why adiabatic distorted-wave calculations can be reliable for some (d, p) spectroscopic analyses even when they are not reliable exclusive breakup calculations. It is also why AGS benchmarks find that elastic, breakup, and transfer errors do not track each other [77]. They are different projections of the same exact wave function.

The reduction has limits. It relies on the weak binding of the deuteron, on the short range of V_{np} , and on the reaction being dominated by small n - p separations. It also inherits the nucleon optical potentials at effective energies that are not uniquely fixed by the exact three-body Hamiltonian. For halo nuclei the analogous adiabatic models can describe elastic ratios and radius effects when the core-target interaction dominates [44, 45], but they do not provide a complete replacement for CDCC or AGS when resonant continuum structure, core excitation, or transfer rearrangement is the observable. The adiabatic family is best read as an exact-theory-motivated filter: it keeps the continuum component most relevant to one class of matrix elements.

F. Few-body universality

Exact three-body methods do more than provide numerical standards. They also show when apparently different nuclear systems share the same low-energy physics. The simplest expression is the effective-range expansion for an s -wave two-body subsystem,

$$k \cot \delta_0(k) = -\frac{1}{a} + \frac{1}{2}r_e k^2 + \mathcal{O}(k^4), \quad (38)$$

where a is the scattering length and r_e the effective range. If $|a|$ is much larger than the interaction range, the shallow two-body binding energy is fixed at leading order by

$$B_2 = \frac{\hbar^2}{2\mu a^2}, \quad (39)$$

up to effective-range corrections. In the three-body sector, a new scale is needed. For three identical bosons at unitarity the Efimov spectrum obeys

$$\frac{E_{n+1}}{E_n} = \exp\left(-\frac{2\pi}{s_0}\right), \quad s_0 \simeq 1.00624, \quad (40)$$

so successive binding energies differ by the universal factor $\exp(2\pi/s_0) \simeq 515$. Nuclear systems do not sit at this ideal point: spin, isospin, Coulomb, Pauli blocking, and

resonant p waves all weaken the symmetry. Yet the organizing idea survives in correlation lines. The Phillips line correlates the doublet neutron-deuteron scattering length with the triton binding energy, and the Tjon line correlates three- and four-nucleon binding energies [7, 113–116].

For reaction theory, universality matters because it tells which details of the interaction can be varied without changing the low-energy observable. The $n + p + \alpha$ system gives an unusually transparent example. The deuteron is shallow, the α core is compact, and ${}^6\text{Li}$ is bound by only 1.474 MeV relative to $d + \alpha$. Exact Faddeev calculations with different phase-equivalent N - α and N - N interactions show that the d - α scattering length and the ${}^6\text{Li}$ binding energy collapse onto a one-parameter curve, a Phillips-line analog for $n + p + \alpha$ [117]. In this controlled sense, ${}^6\text{Li}$ behaves as a deuteron halo. The phrase does not mean that ${}^6\text{Li}$ is spatially as dilute as ${}^{11}\text{Li}$ or ${}^6\text{He}$. It means that the low-energy $d + \alpha$ dynamics is governed by a small number of near-threshold scales, and that off-shell changes in the α - N P waves move the system along a universal correlation rather than producing arbitrary behavior.

The caveats are as important as the result. The calculation just described was performed in the $J^\pi = 1^+$, $T = 0$ channel without the α - p Coulomb interaction, and the Pauli-forbidden α - N $S_{1/2}$ state has to be removed consistently. These are not small decorations. Work on $\alpha + N + N$ and halo systems shows that repulsive-core approximations, orthogonalizing pseudo-potentials, and attractive interactions with excluded bound states can lead to quantitatively different three-body spectra and continuum signatures [97, 111, 112]. The value of exact three-body theory is that those choices can be isolated. When two phase-equivalent two-body inputs give different three-body observables, the difference is an off-shell or three-body-parameter question, not a failure of the scattering method.

This universality thread also connects the exact few-body section to the rest of the review. CDCC often treats ${}^6\text{Li}$, ${}^{11}\text{Be}$, ${}^6\text{He}$, and ${}^9\text{Be}$ as cluster projectiles because their breakup thresholds and low-lying continuum strength dominate the reaction. IAV theory then asks what part of that continuum is seen when only one fragment is detected. Optical-potential UQ asks which fragment-target interactions are actually constrained by data. Faddeev and AGS calculations sit at the base of that hierarchy. They are the place where the cluster Hamiltonian, the treatment of Coulomb, the Pauli constraints, and the off-shell content of the input interactions can be tested without the additional truncation of a reaction model. For weakly bound projectiles on heavy targets this exact level is rarely the final calculation. It is the reference calculation that keeps the approximations honest.

G. What an exact benchmark should measure

An exact three-body calculation is most useful when it is used to benchmark an observable, not a method name. Saying that CDCC agrees with AGS is meaningful only after specifying the elastic channel, breakup energy sharing, transfer partition, partial waves, and optical inputs being compared. The natural diagnostic is an observable-level residual,

$$\Delta_O(M) = \frac{|O_M - O_{\text{AGS}}|}{\max(|O_{\text{AGS}}|, \delta O_{\text{scale}})}, \quad (41)$$

where O_M is the prediction of an approximate method and δO_{scale} is a chosen scale, for example an experimental uncertainty or a small fraction of the peak cross section. This is a definition, not a universal convention. Its purpose is to prevent a common ambiguity: a small absolute error in a cross-section minimum can be a large relative error, while a visually small relative error in an integrated observable may still change an extracted spectroscopic factor.

Elastic scattering is the easiest benchmark because all rearrangement channels enter only through their back-reaction on the entrance channel. For deuteron scattering, this is where CDCC performs best over a broad energy range [76, 77]. Transfer is harder because the final partition is not a projectile continuum state. A CDCC entrance wave can improve the transfer amplitude by including breakup before transfer, but the final rearrangement channel is still treated by an additional transition operator. Exact AGS includes all partitions on the same footing, so the transfer benchmark measures not only entrance-channel breakup but also the consistency of the transition operator and final-state interaction.

Breakup is still more differential. A calculation may reproduce the integrated breakup cross section while failing in the energy sharing of the fragments, or may reproduce a resonant peak while missing a nonresonant background. AGS benchmarks are therefore most valuable when they compare fully exclusive or semi-exclusive distributions. The $^{11}\text{Be}p$ benchmark literature illustrates the point: basis choices that look acceptable for one projection can matter for transfer or breakup partitions [76]. This is not a weakness of the benchmark. It is the information the benchmark was designed to expose.

Coulomb makes the benchmark hierarchy even more important. Screening and renormalization can be pushed to convergence in light and moderate cases, but large charges and low energies make the numerical limit expensive [8, 80]. A comparison that has not converged the screening radius is not an exact benchmark, even if it solves AGS equations. Conversely, a CDCC calculation with a large Coulomb basis may be more reliable for a specific heavy-target elastic observable than an un-converged nominally exact calculation. Exactness is a property of the converged mathematical limit, not of the acronym attached to the code.

The benchmark should also report which interactions are real and which are optical. Exact three-body theory with real pair interactions tests unitary rearrangement dynamics. Practical reaction benchmarks often use complex fragment-target optical potentials, because target excitation and compound formation are not explicit degrees of freedom. In that case the AGS result is exact only for the chosen effective Hamiltonian. This is still extremely valuable: it isolates the error of CDCC, ADWA, or DWBA relative to the same optical inputs. It does not remove the optical-potential uncertainty discussed in Sec. VII.

This distinction matters for absorption observables. If a complex optical potential removes flux, exact AGS can determine how elastic, transfer, and breakup amplitudes are modified by that absorption, but it cannot by itself tell which target final state received the flux unless those channels are included explicitly. IAV answers a different question by using the imaginary part of the participant optical potential as an inclusive sum over the eliminated subsystem. Thus AGS and IAV are complementary. AGS benchmarks the three-body dynamics generated by a chosen Hamiltonian. IAV supplies the inclusive projection needed when the measured observable is a spectator spectrum.

H. Exact theory as a diagnostic of approximations

The most useful role of exact theory is diagnostic rather than decorative. It tells which approximation failed. If ADWA fails while CDCC agrees with AGS, the adiabatic treatment of the continuum is the likely issue. If CDCC fails for transfer but succeeds for elastic scattering, the missing element is probably the rearrangement partition rather than the entrance breakup coupling. If both CDCC and AGS disagree with data using the same optical potentials, the problem lies in the effective interactions or in omitted target degrees of freedom, not in the three-body solver. If two phase-equivalent inputs give the same elastic observable and different transfer amplitudes, the benchmark has exposed off-shell or nonlocal sensitivity.

This diagnostic use is especially important for halo nuclei. In an inert-core model, the projectile continuum may be described accurately in the dominant single-particle channel. But if the observable populates core-excited states, or if transfer proceeds through a core-excited component, the inert-core AGS or CDCC benchmark is solving the wrong Hamiltonian. The comparison with XCDCC or with an expanded three-body model then tests structure completeness, not numerical exactness [118–120]. Exact few-body theory therefore does not remove the need for structure physics. It identifies which structure degrees of freedom must be promoted from hidden inputs to explicit channels.

The same logic applies to Pauli effects. Cluster models often remove forbidden states by projecting them out or

by using repulsive cores. Exact three-body calculations can reveal whether different removal prescriptions lead to the same low-energy observables or whether they introduce a genuine three-body ambiguity [97, 111, 112]. For weakly-bound projectiles, where the continuum is close to threshold, this is not only a bound-state issue. The Pauli prescription can change phase shifts, continuum-bin wave functions, and transfer form factors. An exact benchmark that ignores this sensitivity may certify a reaction approximation while leaving the projectile Hamiltonian ambiguous.

Finally, exact theory defines the cleanest language for truncation errors. CDCC truncates the projectile continuum and selects a partition. ADWA collapses the continuum into an adiabatic channel. DWBA linearizes the transition operator. IAV replaces an explicit sum over participant-target states by an optical Green function. None of these approximations is intrinsically wrong. Each has a domain where it preserves the observable to the desired accuracy. Exact AGS and Faddeev calculations are the way to draw those domains in systems where the full problem is still solvable. The next sections use CDCC and IAV not as alternatives to exact theory, but as controlled reductions whose errors should be measured whenever exact benchmarks are available.

IV. CONTINUUM-DISCRETIZED COUPLED CHANNELS

A. The CDCC ansatz and coupled equations

The exact Faddeev equations answer what the three-body problem is. The continuum-discretized coupled-channels method answers how much of that problem can be retained in a calculation that still looks like a nuclear reaction analysis. Its starting point is the same Hamiltonian as Eq. (5),

$$H_{3b} = T_{\mathbf{R}} + h_{bx}(\mathbf{r}) + U_{bA}(\mathbf{R}_b) + U_{xA}(\mathbf{R}_x), \quad (42)$$

with $a = b + x$ and with $\mathbf{R}_b$ and $\mathbf{R}_x$ understood as the fragment-target coordinates generated by $\mathbf{R}$ and $\mathbf{r}$. CDCC does not eliminate the projectile continuum into a polarization potential. It projects the three-body wave function onto a finite set of square-integrable projectile states,

$$\Psi_{JM}^{(+)}(\mathbf{R}, \mathbf{r}) = \sum_{\alpha \in \mathcal{P}} \frac{u_{\alpha J}(R)}{R} \left[\Phi_{\alpha}(\mathbf{r}) Y_{L_{\alpha}}(\hat{R}) \right]_{JM}, \quad (43)$$

where JM are the total angular momentum and projection, L_{α} is the projectile-target orbital angular momentum, and the channel label $\alpha \equiv (n_{\alpha}, \ell_{\alpha}, I_{\alpha}, L_{\alpha})$ contains a projectile state index n_{α} , internal orbital angular momentum ℓ_{α} , and projectile spin I_{α} . For a spinless two-cluster projectile $I_{\alpha} = \ell_{\alpha}$; spin couplings can be restored without changing the structure of the equations. The model space $\mathcal{P}$ contains the projectile ground state and

a finite set of continuum representatives. The projector is therefore

$$\mathcal{P} = \sum_{\alpha \in \mathcal{P}} |\Phi_{\alpha}\rangle \langle \Phi_{\alpha}|, \quad \mathcal{P}\Psi^{(+)} = \Psi_{\text{CDCC}}^{(+)}, \quad (44)$$

which makes CDCC a projected three-body calculation rather than a fitted two-body model [4].

Substitution of Eq. (43) into $(E - H_{3b})\Psi^{(+)} = 0$ gives the coupled radial equations [4, 47],

$$\begin{aligned} & \left[\frac{d^2}{dR^2} - \frac{L_{\alpha}(L_{\alpha} + 1)}{R^2} + K_{\alpha}^2 \right] u_{\alpha J}(R) \\ &= \frac{2\mu_{aA}}{\hbar^2} \sum_{\alpha' \in \mathcal{P}} V_{\alpha\alpha'}^J(R) u_{\alpha' J}(R), \end{aligned} \quad (45)$$

where μ_{aA} is the projectile-target reduced mass and

$$K_{\alpha}^2 = \frac{2\mu_{aA}}{\hbar^2} (E - \varepsilon_{\alpha}). \quad (46)$$

Here ε_{α} is the eigenenergy represented by projectile state Φ_{α} , negative for a bound state and positive for a continuum bin or pseudostate. It is useful to define the channel angular function

$$\mathcal{Y}_{\alpha}^{JM}(\mathbf{r}, \hat{R}) = \left[\Phi_{\alpha}(\mathbf{r}) Y_{L_{\alpha}}(\hat{R}) \right]_{JM}. \quad (47)$$

The coupling potential in Eq. (45) is then

$$V_{\alpha\alpha'}^J(R) = \langle \mathcal{Y}_{\alpha}^J | U_{bA}(\mathbf{R}_b) + U_{xA}(\mathbf{R}_x) | \mathcal{Y}_{\alpha'}^J \rangle_{\mathbf{r}, \hat{R}}, \quad (48)$$

where the bracket indicates integration over the projectile internal coordinate and the orientation of $\mathbf{R}$. The diagonal pieces contain the folding potential of Eq. (6); the off-diagonal pieces are the continuum couplings that generate breakup, continuum-continuum rescattering, and the dynamic polarization potential discussed in Sec. II C. The scattering boundary condition is imposed on the radial functions,

$$\begin{aligned} u_{\alpha J}(R \rightarrow \infty) \rightarrow & \frac{i}{2} \left[H_{L_{\alpha}}^{-}(\eta_{\alpha}, K_{\alpha}R) \delta_{\alpha\alpha_0} \right. \\ & \left. - H_{L_{\alpha}}^{+}(\eta_{\alpha}, K_{\alpha}R) S_{\alpha\alpha_0}^J \right], \end{aligned} \quad (49)$$

where α_0 is the entrance ground-state channel, $H_L^{\mp}$ are incoming and outgoing Coulomb functions, η_{α} is the Sommerfeld parameter, and $S_{\alpha\alpha_0}^J$ is the coupled-channels S matrix. This is the object from which elastic and exclusive breakup observables are computed.

For a spinless projectile the elastic scattering amplitude can be read from the entrance-channel element of this matrix,

$$f_{\text{el}}(\theta) = f_C(\theta) + \frac{1}{2iK_0} \sum_L (2L+1) (S_{00}^L - 1) P_L(\cos \theta), \quad (50)$$

where f_C is the point-Coulomb amplitude and the channel label 0 denotes the projectile ground state. The total

reaction cross section in the same one-entrance-channel notation is

$$\sigma_R = \frac{\pi}{K_0^2} \sum_J (2J+1) \left[1 - \sum_{\alpha \in \text{open}} |S_{\alpha 0}^J|^2 \right] + \sigma_{\text{abs}}^{\text{closed}}, \quad (51)$$

where the explicit sum runs over open discretized channels and $\sigma_{\text{abs}}^{\text{closed}}$ denotes loss into the imaginary parts of the fragment-target optical potentials. If the fragment optical potentials are real and the discretized model space is complete, the bracketed quantity vanishes after summing over all open elastic, transfer, and breakup channels. In realistic CDCC applications it does not vanish, because the model space includes complex fragment-target interactions and therefore removes target excitation, transfer to unrepresented states, and compound processes into absorption.

The exclusive breakup observable represented by a bin α is similarly obtained from $S_{\alpha 0}^J$. In a bin calculation the cross section for populating a finite continuum interval is schematically

$$\sigma_{\alpha}^{\text{bin}} = \frac{\pi}{K_0^2} \sum_J (2J+1) |S_{\alpha 0}^J|^2, \quad (52)$$

with spin recoupling and channel degeneracy factors restored in practical calculations. To compare with a differential continuum spectrum one divides by the bin width or smooths the bin amplitudes with the exact continuum states. This step is often more delicate than the elastic calculation. The elastic channel depends on a coherent back-coupling from the full model space, while the exclusive breakup spectrum asks for the distribution of flux inside the discretized continuum. Thus elastic convergence and breakup convergence are related but not identical.

The physical content of the method is already visible in Eq. (45). If only the ground-state channel is kept, CDCC collapses to a folded optical model. If the continuum channels are retained but their mutual couplings are removed, the calculation becomes a one-step breakup model. The foundational deuteron studies showed that this latter shortcut fails: continuum-continuum couplings can change the elastic reflection coefficients by factors of order one, and their omission spoils the elastic channel even when ground-state-to-continuum couplings are included [4, 46]. CDCC became useful because it keeps the continuum as a finite coupled space, not because it treats breakup perturbatively.

B. Discretization schemes and convergence

The continuum in Eq. (43) must be represented by normalizable states. The original and still most transparent representation is the momentum-bin construction [4, 121]. Let $\phi_{\ell}(k, \mathbf{r})$ be an exact scattering state of the projectile Hamiltonian h_{bx} with internal orbital angular momentum ℓ and relative momentum k . A bin state on

the interval $[k_i, k_{i+1}]$ is

$$\Phi_{i\ell m}(\mathbf{r}) = \frac{1}{\sqrt{N_{i\ell}}} \int_{k_i}^{k_{i+1}} dk w_{i\ell}(k) \phi_{\ell}(k, \mathbf{r}) Y_{\ell m}(\hat{r}), \quad (53)$$

with normalization

$$N_{i\ell} = \int_{k_i}^{k_{i+1}} dk |w_{i\ell}(k)|^2. \quad (54)$$

The weight $w_{i\ell}(k)$ is often chosen as $\exp[-i\delta_{\ell}(k)]$ for a nonresonant bin, where δ_{ℓ} is the two-body phase shift, or as a resonance-enhancing weight proportional to the two-body transition amplitude when a narrow resonance lies inside the bin [121]. The bin energy entering Eq. (46) is the expectation value

$$\varepsilon_{i\ell} = \langle \Phi_{i\ell} | h_{bx} | \Phi_{i\ell} \rangle. \quad (55)$$

This construction keeps the connection to two-body scattering explicit: increasing k_{max} , adding internal partial waves, and shrinking $\Delta k_i = k_{i+1} - k_i$ are controlled routes toward convergence.

The alternative is to diagonalize h_{bx} in an L^2 basis and use the positive-energy eigenstates as pseudostates,

$$h_{bx} \Phi_n(\mathbf{r}) = \varepsilon_n \Phi_n(\mathbf{r}), \quad \varepsilon_n > 0, \quad (56)$$

with bases such as Gaussian, transformed harmonic oscillator, or Lagrange-mesh functions [122, 123]. Pseudostates give a compact representation of broad continuum strength and generalize naturally to three-cluster projectiles, while bins keep a direct asymptotic connection to continuum momenta. They are not different physics; they are different finite representations of the same projector $\mathcal{P}$. The relevant question is how quickly observables stabilize as the representation is enlarged.

The answer is system dependent. For polarized ${}^6\text{Li}$ scattering, microscopic CDCC calculations needed non-resonant S , P , and D bins up to $k_{\text{max}} \simeq 1.2 \text{ fm}^{-1}$ with $\Delta k \simeq 0.2 \text{ fm}^{-1}$ to converge analyzing powers [124]. For ${}^{11}\text{Be} + {}^{64}\text{Zn}$ near the barrier, a systematic R-matrix CDCC study found that $l_{\text{max}} \geq 2$ and $k_{\text{max}} \simeq 2.2 \text{ fm}^{-1}$, corresponding to pseudostate energies around 110 MeV, were needed for elastic convergence at a collision energy of only 24.5 MeV [123]. Long-range Coulomb breakup is even more severe: calculations of ${}^{11}\text{Be}$ breakup on ${}^{12}\text{C}$ at 67 MeV per nucleon used bins up to 20 MeV, radial bin wave functions to 60 fm, thousands of projectile-target partial waves, and asymptotic matching around 1000 fm [125]. These numbers are not technical trivia. They show that the low binding and long Coulomb range make convergence an observable-level statement, not a property of the projectile basis alone.

A useful convergence audit therefore cannot be only a statement about the number of bins. For an observable O computed in a sequence of model spaces $\mathcal{P}_N$, the relevant quantity is the model-space residual

$$\Delta_N O = \left| \frac{O[\mathcal{P}_N] - O[\mathcal{P}_{N-1}]}{O[\mathcal{P}_N]} \right|, \quad (57)$$

evaluated separately for elastic scattering, breakup spectra, transfer amplitudes, and absorption components. A model space that gives $\Delta_N O_{el} < 1\%$ need not give the same residual for an exclusive breakup spectrum at a fixed fragment energy. The reason is visible in Eq. (45). Elastic scattering sees the continuum through a sum over back-couplings into the entrance channel, while the breakup spectrum resolves individual continuum regions and their phases. A coarse bin may carry the correct net polarization but the wrong energy distribution.

The convergence audit should also separate three cut-offs that are often merged in prose. The maximum continuum energy controls how much short-range projectile distortion is present. The maximum internal angular momentum controls the multipolarity of the breakup and polarization field. The projectile-target matching radius controls the long-range Coulomb coupling. Their physical meanings differ. In a nuclear-breakup calculation on a light target, increasing k_{max} and adding resonant partial waves may be the dominant tests. In Coulomb breakup of a neutron halo on a heavy target, matching radii of hundreds of fm can matter more than adding very high internal energies. In ${}^6\text{Li}$ near the barrier, the cancellation of the cluster $E1$ operator shifts the burden from long-range dipole convergence to nuclear and resonant-continuum convergence. This is why the benchmark systems in Sec. VI cannot be interchanged.

The same lesson appears in transfer. The adiabatic Johnson-Soper model can describe the part of the three-body wave function sampled by a (d, p) transition, even though it does not describe exclusive breakup phases. The Weinberg-state analysis explains why: the transfer amplitude is dominated by the first Weinberg component of the full CDCC wave function, with higher components contributing only a few percent in representative ${}^{132}\text{Sn}(d, p)$ calculations [42, 43, 126]. Thus adiabatic success is not a general proof that breakup dynamics is simple; it is a statement about the short n - p separations and channel combinations sampled by one class of transfer matrix elements.

C. Benchmarks against exact theory

CDCC is a truncation, so its credibility rests on comparisons with exact three-body calculations. The canonical benchmark is the AGS comparison for $d+{}^{12}\text{C}$ at 56 MeV, $d+{}^{58}\text{Ni}$ at 80 MeV, and ${}^{11}\text{Be}+p$ at 38.4 MeV per nucleon [76]. The result is not a single yes or no. For deuteron elastic scattering and breakup at tens of MeV, CDCC agrees well with AGS. For the halo case, transfer is underestimated and breakup is basis dependent: a basis built on ${}^{11}\text{Be}$ continuum states and a basis built on p - n continuum states each represent different asymptotic partitions well. A later systematic study of $d+{}^{10}\text{Be}$, $d+{}^{12}\text{C}$, and $d+{}^{48}\text{Ca}$ sharpened the conclusion: elastic, transfer, and breakup accuracy are not correlated; transfer is most reliable near 10 MeV per nu-

cleon, while breakup improves at higher energy and can fail at low energy because different Faddeev components interfere destructively [77].

This is the practical boundary of CDCC. It is a controlled expansion in projectile states for observables whose dynamics remains dominated by the entrance partition. It is not an exact rearrangement theory. When transfer to a different mass partition is the observable, the relevant asymptotic channel does not live naturally in the entrance-channel CDCC basis. When breakup observables are extracted from the asymptotic tail of the CDCC wave function, they stress the least converged part of the calculation. This is precisely why post-form transition amplitudes built from the interior CDCC wave function are attractive: they use the part of CDCC that is most stable and delegate the outgoing fragment dynamics to the transition operator. That idea becomes central in Sec. V.

The benchmarks also clarify how to read a successful CDCC calculation. A good elastic fit does not prove that the continuum basis is sufficient for breakup, transfer, or incomplete fusion. Conversely, the failure of a particular breakup observable does not invalidate the CDCC elastic wave function as an entrance-channel input to an inclusive theory. This separation of uses is important for the later IAV-CDCC program, where the question is not whether CDCC gives the exact three-body asymptotic breakup amplitude, but whether it gives a reliable entrance wave function in the region where the post-form source is localized [20, 127].

D. Dynamic polarization potentials inside CDCC

One of the most persistent uses of CDCC is to explain why a fitted optical potential changes when breakup channels are present. The operator statement is a Feshbach elimination inside the CDCC model space. Let P_0 project onto the projectile ground-state channel and let $Q = 1 - P_0$ project onto the discretized continuum. The CDCC Hamiltonian in the projected space can be written in block form,

$$H_{\text{CDCC}} = \begin{pmatrix} H_{00} & H_{0Q} \\ H_{Q0} & H_{QQ} \end{pmatrix}. \quad (58)$$

Eliminating Q gives the exact elastic-channel equation inside the chosen CDCC space,

$$[E - H_{00} - \Delta U_{\text{DPP}}(E)] P_0 \Psi^{(+)} = 0, \quad (59)$$

with the nonlocal, energy-dependent dynamic polarization operator

$$\Delta U_{\text{DPP}}(E) = H_{0Q} \frac{1}{E^+ - H_{QQ}} H_{Q0}. \quad (60)$$

This equation is the CDCC realization of Feshbach's generalized optical potential [4, 35]. It also explains why the

local DPP extracted from an elastic fit is not unique. A local potential that reproduces the elastic S matrix is a representation of Eq. (60), not the operator itself.

The old microscopic ${}^6\text{Li}$ calculations are best read through this equation. The double-folding potential without continuum coupling required an empirical real renormalization of roughly one half. When the $d + \alpha$ continuum was included explicitly, the missing effect appeared as a repulsive real DPP generated by virtual breakup [47]. The same mechanism is visible in deuteron CDCC: breakup channels reduce elastic reflection in low partial waves, generate absorptive polarization in the surface, and make the Watanabe folding potential an incomplete elastic description [4, 46]. The physical lesson is that a good two-body fit may already contain a continuum correction, but it contains it in a compressed, energy-dependent, and nonunique way.

Local DPP diagnostics remain useful because they tell where the eliminated continuum acts. A schematic trivially equivalent local polarization can be written as

$$\Delta U_{\text{loc}}(R) \simeq \frac{\sum_{\alpha \neq 0} V_{0\alpha}(R) u_{\alpha J}(R)}{u_{0J}(R)}, \quad (61)$$

with the understanding that a practical TELP average must be weighted over partial waves and protected against nodes of u_{0J} . The ratio form already shows the danger: a local DPP is tied to the elastic wave function used to construct it. It can reveal a long-range absorptive tail or a repulsive real surface term, but it should not be treated as a unique potential to be moved unchanged into another reaction.

The threshold-anomaly discussion in Sec. II C is one place where this distinction matters. For tightly bound heavy ions, collective couplings generate an attractive real DPP and an imaginary term that follows a dispersion relation. For weakly bound projectiles, breakup can generate a repulsive real DPP and long-range absorption, motivating the breakup threshold anomaly picture [10, 11, 16]. But if the local potential is only a representation of Eq. (60), a claim about a new anomaly must be checked against model-space convergence and optical-potential uncertainty. The bootstrap analysis of ${}^6\text{Li}+{}^{209}\text{Bi}$ and ${}^6\text{He}+{}^{208}\text{Pb}$ showed that an apparent anomaly can be statistically consistent with a precision artifact [17]. The DPP is real physics; the fitted local energy trend is not automatically a well-determined observable.

This operator view also prepares the later absorption discussion. If the continuum channels carry imaginary fragment-target interactions, the eliminated operator in Eq. (60) is complex for two reasons: open breakup channels carry flux away, and absorptive interactions inside those channels remove flux into target excitation, transfer, or compound processes. Separating these two effects is not possible with an elastic fit alone. It requires either a channel-resolved CDCC calculation, an IAV inclusive sum rule, or a generalized optical-theorem decomposition of the coupled-channel flux [67, 68, 70].

E. Core excitation: XCDCC

Weak binding is not the only internal degree of freedom that matters. Halo and weakly bound projectiles often contain a deformed or easily excited core. The extended CDCC construction keeps those core states inside the projectile Hamiltonian. For a projectile made of a valence particle x and a core c , one writes

$$h_{cx} = T_{\mathbf{r}} + V_{cx}(\mathbf{r}, \xi_c) + h_c(\xi_c), \quad (62)$$

where ξ_c denotes the intrinsic coordinates of the core and $h_c \varphi_I(\xi_c) = \epsilon_I \varphi_I(\xi_c)$. Projectile eigenstates are expanded as

$$\Phi_{nJ_p M_p}(\mathbf{r}, \xi_c) = \sum_{\ell s I} \frac{u_{\ell s I}^{nJ_p}(r)}{r} \left[[Y_{\ell}(\hat{r}) \chi_s]_j \varphi_I(\xi_c) \right]_{J_p M_p}, \quad (63)$$

where s is the valence spin, I the core spin, and J_p the projectile spin. The reaction wave function has the same CDCC form as Eq. (43), but with Φ_{α} replaced by the multi-configuration states of Eq. (63). With

$$\mathcal{Y}_{\alpha}^{X,JM}(\mathbf{r}, \xi_c, \hat{R}) = \left[\Phi_{\alpha}(\mathbf{r}, \xi_c) Y_{L_{\alpha}}(\hat{R}) \right]_{JM}, \quad (64)$$

the coupling potential becomes

$$V_{\alpha\alpha'}^J(R) = \left\langle \mathcal{Y}_{\alpha}^{X,J} | U_{xA}(\mathbf{R}_x) + U_{cA}(\mathbf{R}_c, \xi_c) | \mathcal{Y}_{\alpha'}^{X,J} \right\rangle_{\mathbf{r}, \xi_c, \hat{R}}. \quad (65)$$

This expression contains ordinary breakup, projectile core excitation, and their interference in one channel space [118, 119].

The core-target interaction is the term that makes XCDCC more than a spectator correction. For a deformed core one may expand it as

$$U_{cA}(\mathbf{R}_c, \xi_c) = \sum_{\lambda\mu} U_{\lambda}(R_c) Y_{\lambda\mu}(\hat{R}_c) \mathcal{T}_{\lambda\mu}^{\dagger}(\xi_c), \quad (66)$$

where $\mathcal{T}_{\lambda\mu}$ acts on the intrinsic core states. The same multipole can excite the core directly during the collision or indirectly by mixing core-excited configurations into the projectile continuum. These two paths interfere. A calculation that includes the core excitation in the projectile structure but not in the reaction coupling, or the reverse, has not solved the physical problem.

This distinction is visible in ${}^{11}\text{Be}$. The ${}^{10}\text{Be}(2^+)$ core component is not a small decoration to the spectrum: it generates continuum strength in partial waves that a pure single-particle model cannot reproduce [118, 119]. But the same core excitation can appear with different weight in different reactions. Breakup on a light target is sensitive to the continuum resonance content and to nuclear core excitation [125]. Near-barrier elastic scattering on a medium target is sensitive to the long-range halo polarization and to the suppression of the Coulomb rainbow [21, 123]. Transfer induced by core excitation samples still another combination of the projectile wave

function and reaction operator [128, 129]. XCDCC is valuable because it puts all these contributions in one channel basis, not because it guarantees they will have the same size.

There is also a useful negative lesson. If a calculation reproduces elastic scattering without core excitation, this does not prove that core excitation is irrelevant. Elastic scattering can hide missing structure in a fitted optical potential or in a DPP. Breakup spectra and transfer angular distributions are sharper tests because they resolve which continuum components were populated. This is another instance of the review's central theme: the correct model space is observable dependent. The same ^{11}Be projectile may require a minimal halo model for one purpose and an explicit core-excited basis for another.

The phenomenological need for this extension is clearest in one-neutron halo systems. In ^{11}Be , the ^{10}Be core has a low-lying 2^+ state, and the observed $3/2^+$ continuum strength cannot be represented by a pure $^{10}\text{Be}(0^+) + n$ model. Calculations that include core excitation enhance breakup for light targets and repair part of the missing continuum strength, whereas prompt core excitation in transfer can be only a few percent for ^{11}Be because the same mechanism is filtered by different kinematics [125, 128]. In near-barrier $^{11}\text{Be} + ^{64}\text{Zn}$, the suppressed elastic rainbow and large reaction cross section first identified experimentally require a continuum treatment that is sensitive to both halo breakup and core structure [21, 123]. The later XCDCC plus inclusive-breakup analyses are a direct continuation of this logic [22, 23].

Target excitation is the mirror problem. If the target is deformed or collective, its intrinsic coordinate ξ_A enters the fragment-target interactions, and the CDCC channel functions carry target spin labels as well as projectile continuum labels. The coupled equations retain the form of Eq. (45), but the coupling potential is now an integral over projectile and target intrinsic coordinates. The formal extension reduces exactly to ordinary CDCC when the target excitation space is closed, and it has been benchmarked against AGS calculations with core excitation in deuteron reactions [83, 120, 129]. The open difficulty is mutual excitation. A reaction such as ^9Be on ^6Li can excite both the projectile and the target while also breaking the projectile, and a clean few-body description then rapidly outgrows a three-body CDCC model.

F. Four-body CDCC for three-cluster projectiles

Borromean projectiles and three-cluster weakly bound nuclei require one more Jacobi coordinate. For a projectile $a = b + x + c$, the internal Hamiltonian is

$$h_{bxc} = T_{\mathbf{r}} + T_{\mathbf{y}} + V_{bx}(\mathbf{r}) + V_{bc}(\mathbf{r}_{bc}) + V_{xc}(\mathbf{r}_{xc}), \quad (67)$$

where $\mathbf{r}$ and $\mathbf{y}$ are Jacobi coordinates inside the projectile. The four-body CDCC Hamiltonian is

$$H_{4b} = T_{\mathbf{R}} + h_{bxc} + U_{bA}(\mathbf{R}_b) + U_{xA}(\mathbf{R}_x) + U_{cA}(\mathbf{R}_c). \quad (68)$$

The projected wave function is still a CDCC expansion,

$$\Psi_{JM}^{(+)}(\mathbf{R}, \mathbf{r}, \mathbf{y}) = \sum_{\alpha \in \mathcal{P}_4} \frac{u_{\alpha J}(R)}{R} \left[\Phi_{\alpha}(\mathbf{r}, \mathbf{y}) Y_{L_{\alpha}}(\hat{R}) \right]_{JM}, \quad (69)$$

but the projectile states Φ_{α} now diagonalize a genuine three-body projectile Hamiltonian. In a hyperspherical representation one may write

$$\Phi_{\alpha}(\mathbf{r}, \mathbf{y}) = \rho^{-5/2} \sum_{\beta} R_{\alpha\beta}(\rho) \mathcal{Y}_{\beta K}(\Omega_5), \quad (70)$$

$$\rho = \sqrt{r^2 + y^2},$$

where ρ is the hyperradius, Ω_5 denotes the five hyperangles, K is the hypermomentum, and β collects orbital, spin, and coupling labels. In the Gaussian-expansion representation the same states are constructed from correlated Gaussian functions on the two Jacobi coordinates [130, 131]. After projectile diagonalization, the radial equations in R again have the form of Eq. (45). Defining

$$\mathcal{Y}_{\alpha}^{4,JM}(\mathbf{r}, \mathbf{y}, \hat{R}) = \left[\Phi_{\alpha}(\mathbf{r}, \mathbf{y}) Y_{L_{\alpha}}(\hat{R}) \right]_{JM}, \quad (71)$$

the four-body coupling potentials are

$$V_{\alpha\alpha'}^J(R) = \left\langle \mathcal{Y}_{\alpha}^{4,J} | U_{bA} + U_{xA} + U_{cA} | \mathcal{Y}_{\alpha'}^{4,J} \right\rangle_{\mathbf{r}, \mathbf{y}, \hat{R}}. \quad (72)$$

The four-body extension is not an aesthetic improvement. It changes the physics for Borromean projectiles such as ^6He and ^{11}Li , where no two-body subsystem is bound, and for stable three-cluster nuclei such as ^9Be . It also tests when a simpler cluster reduction is adequate. A parameter-free four-body calculation of ^6Li elastic scattering on heavy targets showed that only a small subset of d - α dominated pseudostates controls the elastic observable, while the much larger set of explicit $n + p + \alpha$ dominated pseudostates has little effect [132]. This retroactively explains why a three-body $d + \alpha + A$ CDCC description can work for many ^6Li elastic and IAV applications, provided the d -target interaction is generated consistently from nucleon-target inputs. The same argument cannot be assumed for ^6He , ^{11}Li , or ^9Be , where the cluster thresholds and continuum multipoles are different.

The main numerical difficulty in four-body CDCC is not the radial projectile-target equation, which retains the CDCC form. It is the construction and interpretation of the projectile pseudostates. The three-body projectile continuum contains different Jacobi partitions, resonant and nonresonant regions, and multipoles that can be localized in different cluster coordinates. A pseudostate expansion represents this continuum by a finite set of square-integrable eigenvectors. To compare with a measured breakup spectrum one must smooth the discrete strength. Formally, if an observable is obtained as

$$\frac{d\sigma}{dE} \simeq \sum_n \sigma_n \delta(E - \varepsilon_n), \quad (73)$$

the physical spectrum requires a smoothing kernel or a projection onto exact continuum states,

$$\frac{d\sigma}{dE} = \sum_n \sigma_n \left| \langle \Psi_E^{(-)} | \Phi_n \rangle \right|^2. \quad (74)$$

The second expression is the conceptually clean one, but it is costly for a three-body projectile. As a result, four-body CDCC is strongest when the observable depends on the net polarization of the projectile continuum, such as elastic scattering or integrated breakup, and most demanding when the experiment resolves fragment-fragment energy correlations.

For ${}^6\text{He}$, the decisive continuum coordinate is often the collective α -to-dineutron or cigar-like dipole motion, because this generates the long-range Coulomb polarization on heavy targets [133]. For ${}^9\text{Be}$, the charged $\alpha + \alpha + n$ breakup creates both Coulomb and nuclear couplings, and the stable beam makes inclusive measurements much more complete than for radioactive halos [134]. These two projectiles therefore stress different parts of the same four-body formalism. ${}^6\text{He}$ asks whether the neutron-halo $E1$ strength and transfer channels have been represented accurately. ${}^9\text{Be}$ asks whether unresolved two-fragment continua and incomplete-fusion channels can be summed without losing fragment identity.

G. Microscopic and relativistic extensions

The CDCC equations above treat the fragment-target potentials as optical inputs. Microscopic CDCC moves part of that input into structure theory by using cluster wave functions with Pauli projection and folding the fragment-target interaction with transition densities. Schematically, a coupling potential can be written as

$$V_{\alpha\alpha'}(R) = \int d\mathbf{s} \rho_{\alpha\alpha'}(\mathbf{s}) v_{\text{eff}}(|\mathbf{R} - \mathbf{s}|), \quad (75)$$

where $\rho_{\alpha\alpha'}$ is a diagonal or transition density between projectile cluster states and v_{eff} is an effective nucleon-nucleon interaction. The Kyushu microscopic calculations showed that explicit virtual breakup removes the need to renormalize double-folding potentials by about a factor of two in ${}^6\text{Li}$ and ${}^7\text{Li}$ scattering: the missing interaction is the repulsive real dynamic polarization generated by continuum coupling [47]. This result is one of the cleanest examples in the field where a phenomenological optical-model anomaly is resolved by restoring the projectile continuum.

Microscopic input also changes what is meant by a projectile state in CDCC. In a purely phenomenological cluster model, the projectile Hamiltonian h_{bx} is adjusted to reproduce separation energies and perhaps a few resonances. In a microscopic cluster model, the Pauli principle between nucleons in different clusters must be enforced. This can be done by orthogonality conditions, by forbidden-state projection, or by using antisymmetrized

cluster wave functions. The reaction consequence is direct: Pauli-forbidden components can appear as deeply bound spurious states in effective fragment interactions, and the way they are removed can affect continuum phase shifts and therefore CDCC couplings [97, 111, 132]. For weakly bound projectiles this is not a small structure-theory concern, because the reaction is driven precisely by the near-threshold continuum.

The microscopic and phenomenological routes should therefore be treated as complementary. A microscopic projectile model gives cleaner transition density and Pauli content, but it still needs fragment-target optical operators. A phenomenological cluster model can be tuned tightly to the separation energy and low-lying resonances, but it may hide Pauli and short-range structure effects in the fitted potential. CDCC does not remove this choice. It makes the choice visible by showing which projectile states and couplings control the observable. This is why four-body ${}^6\text{Li}$ CDCC is so informative: it demonstrates when the effective $d + \alpha$ reduction captures the active continuum and when explicit $n + p + \alpha$ degrees of freedom are only spectators to the elastic observable [132].

At intermediate and relativistic energies the same three-body expansion can be combined with relativistic kinematics and Lorentz-contracted interactions. The coupled equations are still organized by projectile states, but the channel wave numbers and coupling potentials are modified so that the projectile-target relative motion carries the correct relativistic energy and the target field is contracted along the beam direction [12]. These corrections are typically at the 5 to 15 percent level in radioactive beam Coulomb-dissociation analyses. They are smaller than the effect of including the continuum at all, but they are not negligible when extracting dipole polarizabilities or astrophysical S factors from high-energy breakup.

The modern pressure on CDCC comes from uncertainty quantification. Model-space convergence, fragment optical potentials, continuum-continuum couplings, and absorptive off-diagonal matrix elements all carry uncertainties that are larger than the statistical errors in many data sets. Recent work reframes the coupled equations as an object to emulate rather than to rerun from scratch. A reduced-basis Galerkin projection constructs a low-dimensional trial space from full CDCC solutions,

$$u_{\alpha J}(R; \boldsymbol{\theta}) \simeq \sum_{n=1}^{N_{\text{rb}}} c_{nJ}(\boldsymbol{\theta}) \varphi_{\alpha J}^{(n)}(R), \quad (76)$$

and determines the coefficients by projecting Eq. (45) onto the same basis [135]. Here $\boldsymbol{\theta}$ denotes the optical or structure parameters being varied. In a deuteron CDCC problem with 18 optical-potential parameters, this strategy gives roughly a 220-fold speedup at sub-0.1 percent accuracy, making Bayesian calibration of the coupled reaction feasible [65]. The emulator does not remove the need for the convergence checks described above. It

makes those checks cheap enough to include in inference rather than treating them as a pre-processing ritual.

H. Long-range couplings and multipole content

The most common failure mode in CDCC applications is not the absence of the coupled equations. It is an insufficient statement of which couplings have actually been retained. In the two-cluster notation of Sec. II, the projectile-target interaction is

$$U_{bA}(\mathbf{R}_b) + U_{xA}(\mathbf{R}_x),$$

$$\mathbf{R}_b = \mathbf{R} + \frac{m_x}{m_b + m_x} \mathbf{r}, \quad \mathbf{R}_x = \mathbf{R} - \frac{m_b}{m_b + m_x} \mathbf{r}. \quad (77)$$

For a central fragment-target interaction this operator is expanded in coupled spherical harmonics as

$$U_{bA}(\mathbf{R}_b) + U_{xA}(\mathbf{R}_x) = \sum_{\lambda\mu} \mathcal{V}_\lambda(R, r) Y_{\lambda\mu}(\hat{R}) Y_{\lambda\mu}^*(\hat{r}), \quad (78)$$

and the channel coupling in Eq. (48) is obtained by folding $\mathcal{V}_\lambda$ between projectile internal states and the angular-momentum recoupling coefficients [4, 130]. The index λ is not a harmless numerical cutoff. It decides whether dipole Coulomb breakup, quadrupole reorientation, nuclear surface coupling, and continuum-continuum transitions are present in the calculation.

The distinction between Coulomb and nuclear couplings is visible already in the radial range of $\mathcal{V}_\lambda$. Nuclear couplings are controlled by the surface of the target and by the tail of the projectile wave function; they decay once the fragments no longer overlap the target nuclear field. Coulomb couplings are long-ranged and require larger matching radii, finer continuum discretization near threshold, and more careful partial-wave convergence. In a neutron halo such as ^{11}Be or ^6He , the effective charge of the dipole operator is large because the charged core and neutral valence neutrons move against each other. In an isoscalar cluster such as ^6Li approximated as $\alpha + d$, the cluster $E1$ coupling is strongly suppressed, so quadrupole and nuclear couplings become relatively more important. The same small separation energy therefore does not imply the same CDCC model space.

Continuum-continuum couplings are especially easy to underestimate. If only ground-to-continuum couplings are retained, the calculation is a distorted first-order breakup model rather than a full CDCC calculation. All-order effects enter because a breakup bin can be accelerated by the target field, coupled into another bin, and then feed back into the elastic channel. In Coulomb dissociation of ^8B this is precisely the mechanism by which the first-order virtual-photon expression is corrected [136]. In near-barrier ^6Li and ^7Li scattering, continuum-continuum nuclear couplings control part of the repulsive real dynamic polarization potential [10, 47]. A convergence test that varies only the maximum excitation energy while freezing the continuum-continuum coupling scheme is therefore incomplete.

The multipole language also clarifies the role of projectile resonances. A narrow projectile resonance is not a separate reaction mechanism in CDCC; it is a concentration of continuum strength in one or more discretized states with strong couplings to the entrance channel. If the resonance is included as a bin with the correct phase-shift behavior, it can dominate an exclusive breakup spectrum. If the observable is elastic scattering, the same resonance contributes through the DPP in Eq. (60). If the observable is inclusive fragment production, the resonance may be only one path into the source term of Sec. V. Thus a resonance should be reported with the observable it controls, not treated as a universal explanation for all missing flux.

There is a practical consequence for published calculations. A CDCC model space should specify the projectile partial waves, binning prescription, maximum continuum energy, maximum multipolarity, maximum projectile-target orbital angular momentum, matching radius, and whether continuum-continuum couplings are complete or truncated. These are not implementation details. They define the approximation to the three-body Hamiltonian. For weakly bound projectiles the observable may be more sensitive to one of these choices than to the optical-potential parameter set. This is why benchmark calculations that quote only a final elastic angular distribution are less useful than calculations that also publish convergence of breakup bins, DPP components, and integrated absorption.

I. Absorption and flux accounting in CDCC

Because CDCC is usually solved with complex fragment-target optical potentials, its coupled equations are not unitary within the explicitly retained projectile space. The loss is physical: it represents target excitation, transfer, fusion, and other channels that are not explicitly resolved. The partial-wave reaction cross section can be written in the same form as in the optical model,

$$\sigma_R = \frac{\pi}{k_a^2} \sum_J (2J+1) \left[1 - \sum_{\alpha \in P} |S_{\alpha 0}^J|^2 \right], \quad (79)$$

where the sum is over the retained CDCC channels and the entrance channel is 0 [4]. This equation is exact only for the chosen CDCC model space and the chosen optical interactions. It does not say which omitted physical channel took the lost flux.

The volume-integral form exposes the same issue more directly. If $\Psi_{\text{CDCC}}^{(+)}$ is the coupled-channel wave function, the absorptive contribution associated with a fragment-target imaginary potential W_{iA} is schematically

$$\sigma_{W_i} = -\frac{2}{\hbar v_a} \left\langle \Psi_{\text{CDCC}}^{(+)} | W_{iA} | \Psi_{\text{CDCC}}^{(+)} \right\rangle, \quad (80)$$

with $i = b, x$ for a two-cluster projectile [68, 70]. This is the bridge from CDCC to incomplete fusion and inclusive

breakup. If W_{xA} absorbs the participant while b remains outside the strong absorption region, the same flux can be interpreted as nonelastic breakup or incomplete fusion depending on which final residue is measured. If both fragments are absorbed, the event contributes to complete fusion in a near-barrier fusion analysis. The CDCC wave function supplies the amplitude; the observable definition supplies the projection.

Recent flux-decomposition work has made this point sharper by showing that off-diagonal absorptive couplings can matter. If one artificially keeps only diagonal imaginary matrix elements, the decomposition into direct and breakup-related absorption channels can shift by tens of percent in weakly-bound systems [68]. This matters for the interpretation of complete-fusion suppression. A suppressed complete-fusion cross section does not automatically mean that a large exclusive breakup cross section has been measured. The missing flux may be carried by nonelastic breakup, transfer-triggered cluster capture, or absorptive continuum couplings that do not lead to two fragments in the detector.

The CDCC flux ledger also explains why IAV is needed. CDCC can compute elastic scattering, exclusive breakup between discretized bins, and absorptive volume integrals. It does not by itself provide the inclusive energy and angular distribution of an observed spectator after summing over all nonelastic states of the participant-target subsystem. That object requires the Green-function sum rule of Sec. V. Thus the relation between CDCC and IAV is not competition. CDCC constructs the entrance three-body wave function with continuum coupling; IAV applies the inclusive projection appropriate to a detected fragment. The IAV-CDCC formulation is therefore the natural language for reactions in which continuum dynamics prepares the source and absorption of one fragment produces the measured inclusive yield [20, 73, 127].

The conservative reporting standard is to keep three numbers separate. The first is elastic loss from the entrance channel, which is what an optical model sees. The second is explicit breakup probability within the CDCC continuum, which depends on the bin basis and detector cuts. The third is absorption by the fragment-target imaginary potentials, resolved if possible by fragment and by channel. Only the third number is directly connected to fusion or nonelastic breakup, and even that connection depends on the experimental residue definition. Many disagreements in the weakly bound fusion literature can be traced to mixing these three quantities under the single word “breakup”.

V. INCLUSIVE BREAKUP: THE ICHIMURA-AUSTERN-VINCENT THEORY

A. Elastic versus nonelastic breakup

CDCC is naturally exclusive. Once a bin state is populated, one can ask for the asymptotic momenta of both fragments. Many experiments do not measure that observable. They detect one fragment, say b , and sum over every final state of the participant fragment x and the target A . For

$$a(=b+x)+A\rightarrow b+B^*, \quad (81)$$

the observed double-differential spectrum separates into elastic breakup (EBU), in which $x+A$ remains elastically scattered, and nonelastic breakup (NEB), in which x is absorbed, excites, transfers to, or fuses with the target:

$$\frac{d^2\sigma}{dE_b d\Omega_b} = \left. \frac{d^2\sigma}{dE_b d\Omega_b} \right|_{\text{EBU}} + \left. \frac{d^2\sigma}{dE_b d\Omega_b} \right|_{\text{NEB}}. \quad (82)$$

This decomposition is not just semantics. The early breakup measurements and post-form DWBA analyses showed that elastic breakup can be the minority component of the inclusive yield, while nonelastic breakup, often called breakup fusion in the older literature, carries most of the observed spectator-fragment spectrum [137–139].

The exclusive post-form DWBA amplitude that underlies the elastic breakup piece is

$$T_{\text{EBU}}^{\text{post}} = \left\langle \chi_b^{(-)}(\mathbf{R}_b) \chi_x^{(-)}(\mathbf{R}_x) | V_{\text{post}} | \chi_a^{(+)}(\mathbf{R}) \phi_a(\mathbf{r}) \right\rangle, \quad (83)$$

where ϕ_a is the projectile ground-state wave function, $\chi_a^{(+)}$ is the entrance distorted wave, and $\chi_b^{(-)}$, $\chi_x^{(-)}$ are outgoing distorted waves. In the exact post form

$$V_{\text{post}} = V_{bx} + U_{bA} - U_{bB}, \quad (84)$$

with U_{bB} the optical potential used to distort fragment b relative to the final $x+A$ system. In the spectator approximation $U_{bB} \simeq U_{bA}$, so $V_{\text{post}} \simeq V_{bx}$. The EBU spectrum is then

$$\left. \frac{d^2\sigma}{dE_b d\Omega_b} \right|_{\text{EBU}} = \frac{2\pi}{\hbar v_a} \rho_b(E_b) \int d\Omega_x \rho_x(E_x) |T_{\text{EBU}}^{\text{post}}|^2, \quad (85)$$

where v_a is the entrance relative velocity,

$$\rho_i(E_i) = \frac{\mu_i k_i}{(2\pi)^3 \hbar^2} \quad (86)$$

is the density of final states for fragment i , and $E_x = E - E_b$ is the energy available to the unobserved $x+A$ subsystem. Equation (85) is useful, but it is not an inclusive theory: it still asks the calculation to resolve an unobserved outgoing x wave.

The IAV step is to replace that impossible explicit sum over all nonelastic $x+A$ final states by the absorptive

part of the optical potential U_{xA} . The post-form NEB formula is [5, 6, 140]

$$\left. \frac{d^2\sigma}{dE_b d\Omega_b} \right|_{\text{NEB}} = -\frac{2}{\hbar v_a} \rho_b(E_b) \langle \psi_x^{\text{post}} | W_x | \psi_x^{\text{post}} \rangle, \quad (87)$$

where $W_x = \text{Im}U_{xA}$ is negative for an absorptive potential in the convention of Eq. (1). The x -channel wave function is not an asymptotic scattering wave; it is the solution of the inhomogeneous equation

$$(E_x - K_x - U_{xA}) \psi_x^{\text{post}}(\mathbf{R}_x) = \rho_x^{\text{post}}(\mathbf{R}_x), \quad (88)$$

with source

$$\rho_x^{\text{post}}(\mathbf{R}_x) = \left\langle \chi_b^{(-)}(\mathbf{R}_b) | V_{\text{post}} | \Psi^{(+)}(\mathbf{R}, \mathbf{r}) \right\rangle_{\mathbf{R}_b}. \quad (89)$$

In DWBA,

$$\Psi^{(+)}(\mathbf{R}, \mathbf{r}) \simeq \chi_a^{(+)}(\mathbf{R}) \phi_a(\mathbf{r}), \quad (90)$$

whereas in IAV-CDCC the same source is evaluated with the full projected CDCC wave function of Eq. (43) [20, 127]. In operator form,

$$\begin{aligned} \psi_x^{\text{post}} &= G_x(E_x) \rho_x^{\text{post}}, \\ G_x(E_x) &= \frac{1}{E_x - K_x - U_{xA} + i\epsilon}. \end{aligned} \quad (91)$$

Equations (87) to (91) are the core of the IAV theory: the detected fragment b appears explicitly, while every unobserved nonelastic fate of x is summed by the optical theorem.

B. Closure over the unobserved subsystem

The compactness of Eq. (87) can obscure what has been summed. The unobserved subsystem is not a single outgoing particle. It includes elastic $x + A$ scattering, target excitation, transfer, compound formation, and any channel that can be represented by the optical potential U_{xA} . The formal derivation begins with an exclusive transition amplitude to a specific state c of the $x + A$ system,

$$T_{cb,a} = \left\langle \chi_b^{(-)} \Phi_c^{(-)} | V_{\text{post}} | \Psi^{(+)} \right\rangle, \quad (92)$$

where $\Phi_c^{(-)}$ is an exact incoming-wave solution in the $x + A$ channel. The inclusive spectrum would require

$$\sum_{c \neq 0} \rho_c(E_x) |T_{cb,a}|^2, \quad (93)$$

where $c = 0$ denotes the elastic $x + A$ channel and the sum includes discrete and continuum states. Directly evaluating Eq. (93) is precisely what is impossible for a heavy target. IAV replaces it by the optical-model identity

$$\sum_{c \neq 0} \rho_c(E_x) \left| \Phi_c^{(-)} \right\rangle \left\langle \Phi_c^{(-)} \right| = -\frac{1}{\pi} G_x^{(-)\dagger}(E_x) W_x G_x^{(+)}(E_x), \quad (94)$$

where the elastic channel has been removed into the optical potential and $W_x = \text{Im}U_x$ is negative for absorption. Substituting the source ρ_x^{post} on both sides of Eq. (94) gives Eq. (87) after the conventional phase-space factors are included. Thus the NEB formula is not a model for one mechanism. It is the closed-channel sum implied by the optical theorem.

This interpretation also fixes the meaning of the imaginary potential. In a two-body optical-model reaction cross section, one writes

$$\sigma_R^{xA} = -\frac{2}{\hbar v_x} \left\langle \chi_x^{(+)} | W_x | \chi_x^{(+)} \right\rangle. \quad (95)$$

The IAV expression is the same theorem with the incident $x + A$ wave replaced by the source-generated wave ψ_x^{post} . This replacement is the whole three-body content of the theory. The participant fragment x is not sent toward the target as an asymptotic beam; it is created inside the reaction by the breakup or transfer operator and then propagates in the optical field of A . The spectator fragment fixes the source and therefore selects which part of the $x + A$ reaction cross section is sampled.

The distinction between EBU and NEB follows from the same identity. The elastic breakup term keeps the $x + A$ elastic wave explicitly and is therefore quadratic in a two-fragment outgoing amplitude. The NEB term removes all nonelastic $x + A$ states into W_x and is therefore quadratic in the source-generated wave. Their sum is inclusive in the measured fragment b , but not inclusive in b itself: the energy and angle of b are fixed by the detector. This is why IAV is especially suited to inclusive alpha spectra, deuteron-induced proton spectra, and surrogate (d, p) formation probabilities. It matches the observable, rather than forcing an exclusive breakup calculation to approximate an inclusive measurement [5, 6, 26, 141].

C. Closed-form theories and the post-prior controversy

The elegance of Eq. (87) hid a long controversy. The post-form derivations of Austern and Vincent, Kasano and Ichimura, and Ichimura, Austern, and Vincent were challenged by the prior-form Kerman-McVoy and Udagawa-Tamura line [6, 142–144]. In prior form the transition operator is

$$V_{\text{prior}} = U_{xA} + U_{bA} - U_{aA}, \quad (96)$$

where U_{aA} is the optical potential that generates the entrance distorted wave. The prior source is

$$\rho_x^{\text{prior}} = \left\langle \chi_b^{(-)} | V_{\text{prior}} | \chi_a^{(+)} \phi_a \right\rangle, \quad (97)$$

and

$$\psi_x^{\text{prior}} = G_x \rho_x^{\text{prior}}. \quad (98)$$

The Udagawa-Tamura cross section keeps only

$$\left. \frac{d^2\sigma}{dE_b d\Omega_b} \right|_{\text{UT}} = -\frac{2}{\hbar v_a} \rho_b(E_b) \langle \psi_x^{\text{prior}} | W_x | \psi_x^{\text{prior}} \rangle. \quad (99)$$

The missing term is the nonorthogonality wave

$$\psi_x^{\text{NO}} = \langle \chi_b^{(-)} | \chi_a^{(+)} \phi_a \rangle, \quad (100)$$

which appears because the entrance and exit channel bases are not orthogonal after the unobserved degrees of freedom have been eliminated. The IAV identity can be written, in the convention used in the modern numerical work, as [6, 141, 145]

$$\psi_x^{\text{post}} = \psi_x^{\text{prior}} + \psi_x^{\text{NO}}. \quad (101)$$

Substitution into Eq. (87) gives

$$\left. \frac{d^2\sigma}{dE_b d\Omega_b} \right|_{\text{IAV}} = \left. \frac{d^2\sigma}{dE_b d\Omega_b} \right|_{\text{UT}} + \left. \frac{d^2\sigma}{dE_b d\Omega_b} \right|_{\text{NO}} + \left. \frac{d^2\sigma}{dE_b d\Omega_b} \right|_{\text{IN}}, \quad (102)$$

where

$$\left. \frac{d^2\sigma}{dE_b d\Omega_b} \right|_{\text{NO}} = -\frac{2}{\hbar v_a} \rho_b \langle \psi_x^{\text{NO}} | W_x | \psi_x^{\text{NO}} \rangle \quad (103)$$

and

$$\left. \frac{d^2\sigma}{dE_b d\Omega_b} \right|_{\text{IN}} = -\frac{4}{\hbar v_a} \rho_b \text{Re} \langle \psi_x^{\text{prior}} | W_x | \psi_x^{\text{NO}} \rangle. \quad (104)$$

The controversy was therefore not post versus prior as exact theories. It was whether the prior-form implementation was allowed to drop the nonorthogonality and interference terms. Analytically, IAV showed that it was not [6]. Numerically, the equality was checked decades later: for $^{62}\text{Ni}(d, pX)$ at 25.5 MeV and $^{209}\text{Bi}(^6\text{Li}, \alpha X)$ at 36 MeV, post and corrected prior agree to within a few percent, while the bare UT term is typically 30 to 40 percent low at the peak [145]. The same resolution was reached independently in the surrogate-reaction formulation [141]. A further extension to $E_x < 0$ showed that complex binding potentials again break the naive post-prior equality unless the same nonorthogonality and interference terms are retained [146].

D. Solving the source equation

The practical IAV calculation is governed by the inhomogeneous equation (88), not by the compact cross-section formula alone. After coupling the angular momenta of $x + A$ to a channel λ , the source wave can be expanded as

$$\psi_x^{\text{post}}(\mathbf{R}_x) = \sum_{\lambda m} \frac{u_\lambda(R_x)}{R_x} Y_{\lambda m}(\hat{R}_x), \quad (105)$$

and each radial component satisfies

$$\left[E_x + \frac{\hbar^2}{2\mu_x} \frac{d^2}{dR_x^2} - \frac{\hbar^2 \lambda(\lambda+1)}{2\mu_x R_x^2} - U_{xA}^\lambda(R_x) \right] u_\lambda(R_x) = R_x \rho_\lambda(R_x). \quad (106)$$

For $E_x > 0$ the solution has outgoing-wave behavior in the $x + A$ coordinate; for $E_x < 0$ it is exponentially damped if the binding potential is real, and it carries absorption if the bound-state optical potential is complex. This single radial equation is the reason the same IAV framework can describe continuum nonelastic breakup, transfer to bound states, and the transition between them [146, 147].

A Green-function construction makes the boundary condition explicit. Let $f_\lambda(R)$ be the regular solution at the origin and $h_\lambda^{(+)}(R)$ the outgoing solution at large radius for the same optical potential. The radial Green function is

$$G_\lambda(R, R'; E_x) = -\frac{2\mu_x}{\hbar^2} \frac{f_\lambda(R_<) h_\lambda^{(+)}(R_>)}{\mathcal{W}_\lambda}, \quad (107)$$

where $R_< = \min(R, R')$, $R_> = \max(R, R')$, and $\mathcal{W}_\lambda$ is their Wronskian. The radial wave is then

$$u_\lambda(R) = \int_0^\infty dR' G_\lambda(R, R'; E_x) R' \rho_\lambda(R'). \quad (108)$$

Equations (107) and (108) show why the source localization matters. If $\rho_\lambda(R)$ is localized in the surface region, the result is controlled by the optical wave in that region. If the source samples the interior, the calculation becomes sensitive to the part of the optical potential least constrained by elastic scattering.

This source perspective is also the clean way to compare finite-range, zero-range, and semiclassical implementations. Finite range keeps the full Jacobi transformation between $(\mathbf{R}, \mathbf{r})$ and $(\mathbf{R}_b, \mathbf{R}_x)$. Zero range collapses the b - x coordinate and therefore forces the source onto a line in the three-body configuration space. Semiclassical transfer-to-continuum methods replace part of the quantum propagation by a classical trajectory for the spectator, which is often accurate at intermediate energies but must be checked against the fully quantum source when Coulomb barriers, low energies, or narrow resonances are involved [139, 147, 148]. The common object is always the source term, not the language used for the detected fragment.

E. Modern implementations

The modern revival of IAV was numerical as much as formal. The source equation (88) must be solved for every detected fragment energy and angle, with finite range, remnant terms, spin couplings, and unbound as well as bound $x + A$ energies. The finite-range post-form source

in DWBA is

$$\rho_x^{\text{FR}}(\mathbf{R}_x) = \int d\mathbf{R}_b \chi_b^{(-)*}(\mathbf{R}_b) V_{\text{post}}(\mathbf{R}_b, \mathbf{R}_x) \times \chi_a^{(+)}(\mathbf{R}) \phi_a(\mathbf{r}), \quad (109)$$

with $\mathbf{R}$ and $\mathbf{r}$ transformed from $(\mathbf{R}_b, \mathbf{R}_x)$ by the Jacobi relations. In the zero-range limit

$$V_{bx}(\mathbf{r}) \phi_a(\mathbf{r}) \rightarrow D_0 \delta(\mathbf{r}), \quad (110)$$

the source collapses to a one-coordinate integral,

$$\rho_x^{\text{ZR}}(\mathbf{R}_x) = D_0 \chi_b^{(-)*}(\mathbf{R}_x) \chi_a^{(+)}(\mathbf{R}_x), \quad (111)$$

up to the usual recoil and finite-range correction factors. For deuterons the zero-range approximation can be accurate at the few-percent level in many inclusive spectra; for ${}^6\text{Li}$, finite range and the remnant term can change the NEB yield by 10 to 20 percent because the α - d Coulomb interaction extends the projectile structure [148].

IAV-CDCC replaces the DWBA entrance wave in Eq. (109) by the full CDCC wave function:

$$\rho_x^{\text{CDCC}}(\mathbf{R}_x) = \left\langle \chi_b^{(-)} | V_{\text{post}} | \Psi_{\text{CDCC}}^{(+)} \right\rangle_{\mathbf{R}_b}, \quad (112)$$

and then uses the same Green's-function formula

$$\psi_x^{\text{CDCC}} = G_x \rho_x^{\text{CDCC}}, \quad \left. \frac{d^2\sigma}{dE_b d\Omega_b} \right|_{\text{NEB}}^{\text{CDCC}} = -\frac{2}{\hbar v_a} \rho_b \langle \psi_x^{\text{CDCC}} | W_x | \psi_x^{\text{CDCC}} \rangle. \quad (113)$$

This is the form proposed in the 1987 CDCC synthesis and implemented only much later [4, 20, 127]. It makes the old "breakup then capture" picture testable. In ${}^6\text{Li}$ -induced incomplete fusion, the full CDCC entrance wave shows that the dominant mechanism is one-step direct capture of the participant cluster, with a two-step breakup-fusion component below about 12 percent over representative binding energies and incident energies [20]. That result also explains why DWBA-IAV often works better than one might expect: for the inclusive observable, the post-form source filters the entrance wave in a localized region.

Convergence remains delicate. The post-form radial integrals are only conditionally convergent when charged particles appear in both entrance and exit channels, and the Vincent-Fortune contour is the standard way to rotate the radial integration into the complex plane [147, 149]. The recent post-form convergence analysis shows that bin averaging and the complex-contour method agree at the few-percent level when both are pushed to convergence, and that a hybrid finite-range plus zero-range contour treatment is a practical route for heavy systems [147]. Surface dominance should also be tested, not assumed. A radial-cutoff diagnostic shows that ${}^6\text{Li}$ -induced NEB is largely peripheral, while deuteron-induced NEB can retain roughly 25 percent interior contribution and becomes less surface-like with increasing energy [150]. This is an important constraint on

semiclassical, transfer-to-continuum, and Trojan-Horse reductions.

The comparison with data requires one more layer, because experiments rarely measure the fully inclusive quantity over all spectator energies and angles. For a detector acceptance $\mathcal{A}_b$, the measured yield is

$$\sigma_b(\mathcal{A}_b) = \int_{\mathcal{A}_b} dE_b d\Omega_b \frac{d^2\sigma}{dE_b d\Omega_b}. \quad (114)$$

The IAV formula gives the integrand, but the physics interpretation can change across the acceptance. At high spectator energies the source often emphasizes quasi-free breakup and transfer to continuum states. At low spectator energies the same inclusive spectrum can be dominated by participant absorption and compound-like processes. This is why inclusive alpha spectra in ${}^6\text{Li}$ and ${}^7\text{Li}$ reactions are not interchangeable with exclusive breakup measurements. They are different projections of the three-body wave function.

Energy-angle correlations are especially diagnostic. In a pure EBU picture, the kinematics of the two outgoing fragments are tightly correlated by three-body energy conservation and by the breakup continuum energy. In NEB, the detected spectator keeps only part of that memory; the participant fragment has been summed over all nonelastic fates through W_x . The resulting spectrum can be broad, asymmetric, and much larger than the EBU component. The ${}^6\text{Li}+{}^{93}\text{Nb}$ inclusive-alpha analysis is a recent example: CDCC accounts for the elastic-breakup component, but IAV-DWBA is needed for the dominant nonelastic yield [73]. The same separation underlies the earlier ${}^{209}\text{Bi}$ and ${}^{62}\text{Ni}$ tests of the post-prior identity [145, 148].

This acceptance dependence is also a warning for using IAV as a label. Stating that a calculation is "IAV" is not enough. One must specify the participant optical potential, the spectator distorted wave, the treatment of finite range and remnant terms, the entrance wave function, and the experimental integration. The formal sum rule fixes the operator structure, but the numerical spectrum is a convolution of that operator with detector cuts and with the chosen optical-potential family. In a review of weakly bound reactions this distinction is central: many disagreements about breakup, transfer, and incomplete fusion are disagreements about which projection of the same inclusive strength is being compared.

F. Incomplete fusion and complete-fusion suppression

Inclusive breakup is not just a way to reproduce spectator spectra. It gives a microscopic bookkeeping of incomplete fusion. In the IAV language, detecting fragment b and absorbing fragment x is precisely the cross section for all nonelastic fates of $x + A$ conditional on b being a spectator. For a weakly bound projectile, the total

reaction cross section can be organized as

$$\sigma_R = \sigma_{\text{CF}} + \sigma_{\text{inel}} + \sigma_{\text{EBU}} + \sigma_{\text{NEB}}(b) + \sigma_{\text{NEB}}(x), \quad (115)$$

where σ_{CF} is complete fusion, σ_{inel} collects bound inelastic excitation, σ_{EBU} is elastic breakup, and $\sigma_{\text{NEB}}(b)$ and $\sigma_{\text{NEB}}(x)$ are the two spectator-tagged nonelastic breakup components [19]. This is the microscopic counterpart of the older survival-probability formulations of complete and incomplete fusion [151, 152].

The macroscopic fusion phenomenology is well known. Above the Coulomb barrier, complete fusion of weakly bound projectiles such as ${}^6\text{Li}$, ${}^7\text{Li}$, and ${}^9\text{Be}$ is suppressed by roughly 25 to 35 percent relative to expectations from tightly bound systems, and the suppression correlates with the cluster separation energy [11, 153–155]. The IAV interpretation changes the mechanism. Elastic breakup is too small to be the main flux drain. Instead, the projectile cluster structure allows one fragment to be delivered inside its own lower Coulomb barrier while the other fragment escapes as the observed spectator. In ${}^6\text{Li}$ and ${}^7\text{Li}$ on ${}^{209}\text{Bi}$, the large inclusive alpha yield is therefore a Trojan-Horse-like nonelastic breakup channel, and the missing complete-fusion flux is carried primarily by participant capture rather than by free breakup followed by fusion [19, 156]. Coincidence measurements on ${}^7\text{Li}$ -induced reactions later confirmed the same physical picture: residues assigned to incomplete fusion are produced mainly by direct cluster transfer, not by capture of a fragment after asymptotic breakup [72, 157].

The coupled-channels absorption problem has a further layer. If every continuum channel carries a complex optical potential, the total absorption is not simply the sum of diagonal channel losses. Off-diagonal imaginary couplings generate an interference term, giving

$$\sigma_A = \sigma_D + \sigma_B + \sigma_{\text{int}}, \quad (116)$$

where σ_D and σ_B denote direct-channel and breakup-channel absorption components, and σ_{int} is basis independent when the full coupled space is retained [68]. Dropping off-diagonal imaginary couplings can bias the inferred components by 5 to 57 percent in representative systems. A related ingoing-wave-boundary formulation gives

$$\sigma_{\text{abs}} = \sigma_{\text{fusion}} + \sigma_W, \quad (117)$$

separating flux that reaches the inner fusion boundary from flux absorbed peripherally by complex fragment-target potentials [70]. These identities sit naturally beside IAV: both insist that “absorption” is not a single number once the projectile has internal degrees of freedom.

The difference between these ledgers is operational. IAV conditions on the detected spectator and asks how the participant is removed from the elastic channel. The coupled-channel absorption decomposition conditions on the channel basis and asks where the total CDCC flux

disappears. A complete description of weakly bound fusion must eventually use both. The IAV spectator-tagged spectrum identifies which fragment is likely to have been captured or transferred. The CDCC absorption ledger decides whether the flux was lost at small radius, where a complete-fusion boundary is appropriate, or in the surface, where peripheral nonelastic breakup and transfer dominate. This is why the phrase “breakup fusion” is too coarse. It does not specify whether breakup occurred as an asymptotic EBU event before capture, as a virtual continuum admixture inside the entrance wave, or as a direct post-form participant-capture amplitude.

One can state the distinction in probability language. Let $P_{\text{surv}}(R)$ be the probability that the projectile reaches radius R without losing flux into nonelastic channels. A survival model writes schematically

$$\sigma_{\text{CF}} = \int db \, 2\pi b \, P_{\text{fus}}(b) P_{\text{surv}}(b), \quad (118)$$

where b is the impact parameter and P_{fus} is the probability for reaching the fusion region. IAV does not start from this factorization. It computes the participant-removal probability from the quantum source. The survival formula can be a useful summary when breakup is peripheral and incoherent, but it cannot decide whether the lost flux is EBU, transfer, or direct capture. That decision requires the operator-level separation in Eqs. (87), (116), and (117).

G. The Trojan Horse method as a limit of inclusive breakup

The Trojan Horse method (THM) is usually presented as an indirect astrophysical technique, but structurally it is a transfer-to-continuum problem. A projectile $a = b + x$ is brought into the field of A ; fragment b acts as a spectator; the binary subreaction $x + A \rightarrow c + C$ is inferred from the three-body reaction

$$a + A \rightarrow b + c + C. \quad (119)$$

The exact post-form amplitude has the same skeleton as Eq. (83),

$$T_{\text{TH}} = \left\langle \chi_b^{(-)} \Psi_{cC}^{(-)} | V_{\text{post}} | \chi_a^{(+)} \phi_a \right\rangle, \quad (120)$$

where $\Psi_{cC}^{(-)}$ is the final interacting wave in the binary subsystem. The plane-wave impulse approximation replaces distorted waves by plane waves in the quasi-free region and factorizes the triple-differential cross section as [158–160]

$$\frac{d^3\sigma}{dE_b d\Omega_b d\Omega_c} = K_F |W(\mathbf{q})|^2 \left. \frac{d\sigma}{d\Omega_c} \right|_{xA \rightarrow cC}^{\text{HOES}}, \quad (121)$$

where K_F is a kinematic factor, $W(\mathbf{q})$ is the spectator momentum distribution folded with the off-shell vertex,

and HOES denotes the half-off-energy-shell binary cross section. This equation is the practical engine of most THM extractions, but its assumptions are stronger than the inclusive IAV assumptions: plane waves, quasi-free kinematics, a localized b - x vertex, and an on-shell reduction of the binary amplitude.

The astrophysical quantity usually sought is not the laboratory binary cross section itself but the S factor,

$$S(E) = E \sigma_{xA \rightarrow cC}(E) \exp(2\pi\eta_{xA}), \quad (122)$$

where η_{xA} is the Sommerfeld parameter in the charged binary entrance channel. THM is attractive because the three-body reaction can populate the $x + A$ subsystem at low relative energy without forcing the charged particles to tunnel through the entrance Coulomb barrier as an isolated two-body beam. The price is that the extracted binary amplitude is half off shell and is embedded in a distorted three-body reaction. Equation (121) therefore has to do two jobs at once: remove the spectator momentum distribution and replace the off-shell binary amplitude by the on-shell astrophysical one [26, 158].

This is where the IAV viewpoint is useful. It does not deny the quasi-free logic of THM. It asks which operator reductions are needed to reach that logic. The post-form source fixes how the participant x is delivered to the target. The Green function of the $x + A$ subsystem fixes resonant propagation and absorption. Plane-wave impulse approximations then replace the distorted source and final waves by a factorized momentum distribution. Each step can be tested numerically. If a reaction is peripheral and the spectator momentum distribution is sharply selected, the factorization may be accurate. If the reaction is sub-Coulomb, resonant, or dominated by distorted waves, the same factorization can shift absolute normalizations or even the extracted energy dependence.

The IAV spectral representation gives a way to see what is being reduced. If the $x + A$ optical Hamiltonian has an isolated resonant state ϕ_n with energy E_n and width

$$\Gamma_n = -2 \langle \phi_n | W_x | \phi_n \rangle, \quad (123)$$

then the Green's function near that pole gives

$$\left. \frac{d^2\sigma}{dE_b d\Omega_b} \right|_n = \frac{1}{2\pi} \frac{\Gamma_n}{(E_x - E_n)^2 + \Gamma_n^2/4} \frac{d\sigma_n^{\text{DWBA}}}{d\Omega_b}, \quad (124)$$

where the direct population factor is

$$\frac{d\sigma_n^{\text{DWBA}}}{d\Omega_b} = \frac{\mu_a \mu_b}{(2\pi\hbar^2)^2} \frac{k_b}{k_a} \left| \left\langle \chi_b^{(-)} \phi_n | V_{\text{post}} | \chi_a^{(+)} \phi_a \right\rangle \right|^2. \quad (125)$$

This is the bridge to THM: the astrophysical method is a further reduction of the same post-form pole amplitude under quasi-free and plane-wave approximations. It is therefore unsafe to correct a THM extraction by a single multiplicative Coulomb factor unless the reduction from the distorted three-body amplitude to Eq. (121) has

been tested for the reaction at hand. The $^{12}\text{C}+^{12}\text{C}$ dispute illustrates the point: the identical-boson selection rule is a clean spectroscopy constraint, but the Coulomb-renormalization issue requires a numerical distorted-wave or IAV-chain calculation, not only a closed-form scaling [26, 161, 162]. The recent $^{19}\text{F}(p, \alpha\gamma)^{16}\text{O}$ THM measurement pushes the same question to even deeper sub-Coulomb kinematics [163].

For a review of three-body methods, the important conclusion is not that THM is unreliable. It is that THM has a clear approximation hierarchy. At the top is the exact three-body amplitude. Below it is the distorted-wave post-form amplitude with an interacting binary final state. Below that is the pole or spectral reduction of the $x + A$ Green function. Only after those steps does one reach the plane-wave quasi-free expression. Different published THM analyses sit at different points in this hierarchy. Comparing them only by their final extracted $S(E)$ hides the reaction-model content. The same operator discipline used for IAV NEB should be applied to THM normalizations, off-shell corrections, and Coulomb suppression factors.

H. Surrogate reactions

Surrogate reactions use a direct reaction to form a compound nucleus that would otherwise be reached by neutron capture, fission, or another difficult channel. The statistical-decay statement is

$$\sigma_{\alpha\chi}(E) = \sum_{J\pi} \sigma_{\alpha}^{\text{CN}}(E, J\pi) G_{\chi}^{\text{CN}}(E, J\pi), \quad (126)$$

where $\sigma_{\alpha}^{\text{CN}}$ is the formation cross section of a compound nucleus with spin-parity $J\pi$ in entrance channel α , and G_{χ}^{CN} is its decay probability into channel χ [164]. In a deuteron-induced surrogate (d, p) reaction, the formation probability is the nonelastic breakup probability for the neutron participant. In partial waves one may write schematically

$$P_{J\pi}(E_x) \propto - \langle \psi_n^{J\pi} | W_{nA}^{J\pi} | \psi_n^{J\pi} \rangle, \quad (127)$$

which is the same IAV object as Eq. (87) resolved by the spin-parity of the compound subsystem [26, 141]. The surrogate problem then adds a statistical-decay layer on top of the reaction formation layer.

The spin-parity distribution is the reaction-theory bottleneck. A neutron capture reaction and a deuteron-induced surrogate can form the same compound nucleus at the same excitation energy but with different $J\pi$ weights. The surrogate ratio method and the Weisskopf-Ewing approximation reduce this dependence by assuming that the decay probability is insensitive to $J\pi$,

$$G_{\chi}^{\text{CN}}(E, J\pi) \simeq G_{\chi}^{\text{CN}}(E), \quad (128)$$

so that the measured decay probability can be used directly. This approximation can be reasonable for some

fission observables at sufficient excitation energy, but it is not a general theorem. Capture gamma cascades, near-threshold particle emission, and direct-reaction selectivity can retain strong spin-parity dependence [164].

The IAV formulation identifies what the direct reaction must supply. The partial formation probability can be written in terms of the source wave in each $J\pi$ channel, and the decay calculation then acts on that distribution. If the IAV source populates a narrow angular-momentum window, no statistical model can recover a neutron-induced distribution by assumption. Conversely, if the surrogate source populates a broad distribution and the decay probability is smooth in $J\pi$, the surrogate method can work even if the direct reaction mechanism is complicated. The issue is therefore not whether the reaction is direct or compound. It is whether the direct reaction forms the right compound spin-parity ensemble.

This connection is useful because it separates two uncertainties that are often conflated. IAV controls the probability that the direct reaction forms the relevant compound system with a given spin-parity distribution. Hauser-Feshbach decay controls the branching from that distribution into fission, capture, or particle emission. The Weisskopf-Ewing approximation suppresses the spin-parity dependence and can work for some fission ratios above about a MeV of excitation, but it is unreliable for neutron capture and for reactions where the direct mechanism populates a narrow subset of angular momenta [141, 164]. Thus the same moral returns: the unobserved degrees of freedom can be summed, but they cannot be ignored.

I. What the absorptive operator represents

The compact form of Eq. (87) can make IAV look more phenomenological than it is. The imaginary potential W_{xA} is not a detector efficiency and not a fitted branching ratio. It is the absorptive part of the optical operator for the participant-target subsystem, obtained after eliminating all channels of $x + A$ other than the elastic channel. In Feshbach notation, if P_x projects onto elastic $x + A$ motion and Q_x onto all nonelastic channels, the eliminated operator has the same structure as Eq. (3),

$$U_{xA}^{\text{opt}}(E_x) = P_x V P_x + P_x V Q_x \frac{1}{E_x^+ - Q_x H Q_x} Q_x V P_x, \quad (129)$$

and $W_{xA} = \text{Im } U_{xA}^{\text{opt}}$ appears in the IAV expectation value [4, 6, 35]. The nonelastic breakup cross section is therefore an inclusive optical theorem for the participant. The spectator is observed, while the eliminated subsystem accounts for target excitation, transfer, compound formation, particle emission, and any other process represented by the optical absorption.

This interpretation is essential for weakly bound fusion. In a naive breakup picture, the projectile first breaks into $b + x$, then one fragment fuses with the target.

IAV does not require such a classical time ordering. The amplitude in Eq. (89) creates a source for the participant in the field of the target while the spectator is distorted by the residual interaction. Absorption by W_{xA} then sums all nonelastic destinies of that participant. If the measured residue corresponds to capture of x while b is observed or escapes, the same term is naturally identified with incomplete fusion. If the measurement is an inclusive spectator spectrum, the same term is called NEB. The quantum object is the same; the experimental projection changes the language.

The spatial support of the integrand provides a useful diagnostic. Define the radial absorption density

$$\mathcal{A}_x(R_x) = -\frac{2}{\hbar v_a} R_x^2 \int d\hat{R}_x \psi_x^{\text{post}*}(\mathbf{R}_x) \times W_{xA}(\mathbf{R}_x) \psi_x^{\text{post}}(\mathbf{R}_x), \quad (130)$$

so that its integral gives the NEB contribution for the chosen spectator energy and angle. This is a derived diagnostic rather than a new postulate. It shows whether the inclusive yield is generated at the nuclear surface, in the Coulomb tail, or in the strongly absorptive interior. For halo projectiles and THM kinematics, this support test is often more informative than the total cross section. A quasi-free interpretation is plausible only if the source has the expected peripheral geometry and the spectator distortion does not force the participant into a region where the assumed two-body subreaction is no longer isolated.

The same point warns against overinterpreting optical-potential choices. A more absorptive W_{xA} can increase the NEB yield, but it can also move the radial support inward and change the partial-wave content. Two calculations with similar inclusive cross sections may therefore imply different reaction mechanisms. Conversely, a difference in total yield may be an optical-model uncertainty rather than evidence for a new physical path. This is why the UQ discussion in Sec. VII matters for IAV: the participant optical potential is not a background parameter. It defines which nonelastic channels are being summed and with what radial weight.

There is also a boundary to what IAV can say. The standard two-cluster IAV formula sums over all nonelastic states of $x + A$ while observing b . It does not by itself separate neutron transfer to a particular bound state from compound formation followed by evaporation, unless the absorptive operator or the Green function is resolved into those channels. It does not identify exclusive coincidences between b and a specific decay product of $x + A$. Those require either an explicit coupled-channel enlargement of the participant subsystem or an additional statistical-decay calculation. IAV is inclusive by construction. Its strength is the exact sum over unobserved states; its limitation is the same sum when a more differential final-state assignment is demanded.

J. Acceptance, resolution, and reported observables

The theoretical IAV expression is differential in the spectator energy and solid angle, but experiments rarely measure the mathematical point observable. They measure an acceptance-weighted bin. If $\mathcal{A}_{\text{det}}(E_b, \Omega_b)$ denotes the detector acceptance and efficiency, the reported yield corresponds schematically to

$$\sigma_{\text{bin}} = \int_{\Delta E_b} dE_b \int_{\Delta \Omega_b} d\Omega_b \mathcal{A}_{\text{det}}(E_b, \Omega_b) \frac{d^2 \sigma_b}{dE_b d\Omega_b}. \quad (131)$$

This folding is not a late experimental correction if the cross section varies rapidly across the bin. In THM kinematics, quasi-free peaks, spectator-momentum cuts, and subreaction-energy reconstruction can all make the acceptance integral part of the reaction model. In inclusive alpha production from ${}^6\text{Li}$ or ${}^7\text{Li}$, low-energy thresholds and finite angular coverage can similarly bias the apparent balance between EBU and NEB [72, 73].

The same issue appears in the transformation between the calculated variables and the plotted variables. IAV naturally produces a spectator spectrum in the projectile-target center-of-mass frame. Experiments may report laboratory energies, relative energies in a fragment pair, missing mass, reconstructed subreaction energy, or coincidence yields gated on other particles. Each transformation brings a Jacobian. If $z = z(E_b, \Omega_b)$ is a reconstructed variable, then

$$\frac{d\sigma}{dz} = \int dE_b d\Omega_b \delta[z - z(E_b, \Omega_b)] \frac{d^2 \sigma_b}{dE_b d\Omega_b}, \quad (132)$$

before resolution smearing is applied. This formula is elementary, but it is often where physics interpretations drift. A peak in a reconstructed subreaction energy may be produced by a real resonance, by a Jacobian, by acceptance, or by a continuum-coupling effect in the entrance channel. The operator-level IAV calculation answers only the first part unless the same cuts and transformations are applied.

For comparison with data, the safest practice is therefore to publish both the invariant theory object and the folded observable. The invariant object is the double-differential spectator spectrum or the partial-wave formation probability before detector cuts. The folded observable is what is actually fitted or normalized to the data. If the conclusion changes after folding, the reaction mechanism is not robust. If the conclusion survives folding, the result is much stronger because it is no longer an artifact of a convenient but unmeasured projection.

K. Relations to transfer and knockout

IAV also clarifies the boundary between transfer, breakup, and knockout. In ordinary DWBA transfer, the final $x + A$ state is a specified bound or resonant state.

In IAV NEB, the same final subsystem is summed over through the optical Green function. If the imaginary part of the optical potential is replaced by a narrow pole projector, Eq. (124) is recovered and the inclusive strength collapses toward a transfer-like amplitude. If the final subsystem is broad and absorptive, the same source produces a continuum of nonelastic outcomes. Transfer and NEB are therefore not unrelated mechanisms. They are different resolutions of the same participant-target Green function.

Knockout reactions present the same tension in a different language. A spectator core is detected and a nucleon is removed, so the observable is often treated as a structural spectroscopic factor times a reaction single-particle cross section. But the reaction part contains absorption, elastic breakup, diffraction, and final-state interactions that can be organized in the same participant-spectator language. When the extracted reduction factor varies systematically with separation-energy asymmetry, the question is not only whether the shell-model occupancy is quenched. It is also whether the reaction Hamiltonian omits induced polarization or nonadditive terms that alter the removal probability [24, 36, 165, 166]. IAV does not solve the knockout problem by itself, but it supplies the right caution: a sum over unobserved degrees of freedom is a reaction operator, not a structure number.

This viewpoint prevents a common mistake in indirect methods. One should not compare a transfer spectroscopic factor, a knockout reduction factor, and an IAV formation probability as if they were measurements of the same scalar. They are filtered by different transition operators and different eliminated subspaces. Agreement among them is valuable because it is nontrivial. Disagreement is not automatically a failure of nuclear structure; it may be the first signal that the reaction dynamics has not been controlled. The practical way forward is joint modeling with common overlaps and observable-specific reaction operators, not a forced average of incompatible extractions.

VI. BENCHMARK SYSTEMS: WEAKLY BOUND AND HALO NUCLEI

The formal methods of Secs. III-V are best judged on a small set of nuclei for which the structure scale is low, the reaction signal is large, and more than one observable has been measured. These systems are not interchangeable examples. ${}^6\text{Li}$ is stable and almost isoscalar, so elastic scattering is sensitive to nuclear breakup but not to a large Coulomb dipole field. ${}^{11}\text{Be}$ is a one-neutron halo with a low-lying core-excited sector, so the projectile model itself is part of the reaction question. ${}^6\text{He}$ and ${}^9\text{Be}$ are Borromean projectiles, where the continuum is intrinsically three-body. ${}^8\text{B}$ is the proton-halo case in which breakup is tied directly to an astrophysical capture rate. The comparison among them is more useful than any single system, because it separates threshold,

Coulomb, transfer, and incomplete-fusion effects.

A. Deuteron-induced reactions: the reference weak-binding problem

The deuteron is the smallest weakly bound projectile and the historical source of most three-body reaction ideas. Its binding energy,

$$S_{np}(d) = 2.2246 \text{ MeV}, \quad (133)$$

is large compared with halo separation energies but small compared with typical nuclear beam energies. The deuteron therefore breaks up easily, yet its internal structure is simple enough that the continuum can be tested against exact three-body and adiabatic limits. Early neutron spectra from deuteron breakup already showed that a deuteron beam does not remain a single projectile through the collision [37]. The later Johnson-Soper and Johnson-Tandy reductions then turned that observation into a practical reaction model for (d, p) and (d, n) transfer [41, 42].

The deuteron benchmark is conceptually clean because the projectile has only one bound state and a well-known n - p continuum. In the simplest adiabatic picture the reaction is governed by the sum of neutron-target and proton-target optical interactions,

$$U_{dA}^{\text{AD}}(\mathbf{R}, \mathbf{r}) = U_{pA} \left(\mathbf{R} + \frac{m_n}{m_p + m_n} \mathbf{r} \right) + U_{nA} \left(\mathbf{R} - \frac{m_p}{m_p + m_n} \mathbf{r} \right), \quad (134)$$

which is then folded with the short-range part of the deuteron wave function as in Eq. (34). This expression makes explicit why the deuteron optical potential is not a primitive input in a three-body calculation. It is a derived object, and its validity depends on how much of the n - p continuum the observable samples.

Exact comparisons have made the deuteron the calibration case for CDCC. For $d+^{12}\text{C}$, $d+^{58}\text{Ni}$, and related systems, CDCC reproduces AGS elastic and breakup observables rather well at tens of MeV, while transfer accuracy depends more strongly on energy and on the final partition [76, 77]. This mixed result is useful because it is not a slogan. It says that CDCC is a reliable entrance-channel continuum method for many deuteron elastic applications, that ADWA can be a good short-range transfer filter, and that exact AGS remains needed when rearrangement and breakup are both measured in detail.

The deuteron also fixes the modern UQ problem in the most direct way. Transfer reactions are often used to infer spectroscopic factors, but the same calculation depends on proton and neutron optical potentials at energies and radii not uniquely fixed by elastic data. Frequentist and Bayesian OMP-UQ studies of (d, p) reactions show that uncertainties in the distorted waves can

compete with the formal difference between DWBA and adiabatic treatments [167–169]. Thus even the simplest weakly bound projectile already contains the full review theme: three-body dynamics, optical-potential ambiguity, and an extracted structure quantity that should not be interpreted without both.

B. ^6Li : the alpha-deuteron cluster laboratory

The standard three-body representation of ^6Li treats the projectile as $a = \alpha + d$ in the Hamiltonian of Eq. (5). Its small separation energy,

$$S_{\alpha d} = 1.474 \text{ MeV}, \quad (135)$$

is already enough to make continuum coupling a leading effect, but the projectile remains stable and experimentally clean. This is why the ^6Li system occupies a special position: it is the simplest place where CDCC, four-body CDCC, IAV, and exact few-body ideas can be compared against the same physical intuition. The older microscopic-CDCC work showed that breakup generates a repulsive dynamic-polarization potential and removes the need for the empirical factor $N_R \simeq 0.5$ that had been applied to folding potentials for $^6,7\text{Li}$ scattering [47]. The modern four-body calculation, with ^6Li described as $n + p + \alpha$, then showed that only a small set of $d + \alpha$ -dominant pseudostates carries most of the elastic-scattering coupling, even though many four-body pseudostates are present [132]. This is a useful warning for all continuum methods: the number of basis states is not the same as the number of physically active channels.

The same nucleus is also a structural benchmark. In the $n + p + \alpha$ model, momentum-space Faddeev-AGS calculations with separable and nonseparable two-body inputs agree at the four-significant-figure level for the ^6Li binding energy [101]. Varying the short-range d - α interaction along phase-equivalent trajectories produces a universal correlation between the d - α scattering length and the separation energy, a nuclear analog of a Phillips line [117]. In this limited but quantitative sense ^6Li behaves as a deuteron halo: the energy required to separate d from α is of the same order as the deuteron binding energy itself.

The Coulomb-dipole operator explains why ^6Li should not be read as a generic halo. For a two-cluster projectile $a = b + x$, the long-wavelength $E1$ operator in the projectile intrinsic coordinate is

$$\mathcal{M}_{1\mu} = e Z_{\text{eff}}^{(1)} r Y_{1\mu}(\hat{\mathbf{r}}), \quad Z_{\text{eff}}^{(1)} = \frac{Z_x A_b - Z_b A_x}{A_a}. \quad (136)$$

For $\alpha + d$, $Z_\alpha/A_\alpha = Z_d/A_d = 1/2$, and the cluster $E1$ charge vanishes. Four-body CDCC finds the same cancellation in the $n + p + \alpha$ description [132]. Therefore the strong breakup effect in ^6Li scattering is not a long-range Coulomb-dipole effect of the kind seen in ^6He ; it

is mainly nuclear coupling, resonant-continuum coupling, and absorption through nonelastic breakup.

This distinction is decisive for fusion and inclusive alpha production. The measured complete-fusion cross sections for ${}^6\text{Li}$ and ${}^7\text{Li}$ on heavy targets are suppressed above the barrier by roughly the amount expected from the low breakup threshold, while the total fusion is much less suppressed [11, 18, 154]. In the IAV decomposition of Sec. V, the missing complete-fusion flux is not primarily elastic breakup followed by capture, but alpha-producing nonelastic breakup and direct cluster capture [19, 20]. The recent ${}^6\text{Li}+{}^{93}\text{Nb}$ inclusive-alpha measurement makes the same point in a different mass region: CDCC elastic breakup plus IAV-DWBA nonelastic breakup describes the energy-angle correlations, with NEB much larger than EBU [73]. For ${}^6\text{Li}$, the benchmark question is therefore not whether breakup exists; it is which part of the breakup continuum carries the absorbed flux.

Elastic scattering alone gives only part of this story. In ${}^6\text{Li}$ elastic data, the signature of breakup appears as a suppression or reshaping of the Coulomb rainbow and as a repulsive DPP. In inclusive alpha data, the same projectile continuum appears as a spectator-tagged participant absorption probability. In complete-fusion data, it appears as missing flux from the full projectile capture channel. These observables are often discussed under the single word “breakup”, but they are not the same projection. A CDCC calculation that reproduces elastic scattering is testing the back-coupling of continuum states into the entrance channel. An IAV calculation of alpha spectra is testing the source term for deuteron absorption. A fusion suppression analysis is testing whether both clusters reach the inner absorption region. ${}^6\text{Li}$ is valuable because all three projections have been measured and can be compared within one Hamiltonian.

The same benchmark also warns against identifying the deuteron cluster as an ordinary inert fragment. In a $d + \alpha + A$ description, the deuteron-target interaction is itself an effective object generated by proton-target and neutron-target dynamics. Four-body CDCC shows when this reduction is active and when explicit $n + p + \alpha$ degrees of freedom are spectators to the elastic observable [132]. IAV calculations then show that, for participant absorption, the cluster identity remains physically meaningful because the detected alpha selects the absorbed deuteron-like component [20]. This is why ${}^6\text{Li}$ sits at the junction of the exact few-body, CDCC, and IAV parts of the review.

C. ${}^7\text{Li}$: threshold control without changing the target

${}^7\text{Li}$ is the essential comparison partner of ${}^6\text{Li}$ because it changes the cluster threshold while keeping a stable lithium beam and similar fragment charges. The domi-

nant cluster partition is ${}^7\text{Li} = \alpha + t$, with

$$S_{\alpha t} = 2.47 \text{ MeV}, \quad (137)$$

larger than the ${}^6\text{Li}$ $\alpha + d$ threshold. If weakly bound fusion suppression and inclusive alpha production are controlled mainly by the ease of cluster breakup or transfer, ${}^6\text{Li}$ and ${}^7\text{Li}$ should behave similarly but not identically. This is what the systematics show: complete-fusion suppression above the barrier is smaller for ${}^7\text{Li}$ than for ${}^6\text{Li}$, and both are suppressed relative to tightly bound projectiles [11, 18, 154].

The comparison is more than an empirical trend. In the IAV picture, the observed alpha yield in ${}^7\text{Li}$ reactions corresponds to nonelastic removal of the triton participant, while the alpha acts as the spectator. The fragment Coulomb barriers and transfer Q values differ from the ${}^6\text{Li}$ case, so the spectator spectrum changes even when the qualitative mechanism is the same. The inclusive alpha data on heavy targets and the coincidence studies of residues indicate that direct cluster transfer or participant capture dominates over capture of a fragment after asymptotic breakup [72, 156, 157]. This is why the phrase “breakup suppression” is too vague for lithium projectiles. The threshold sets the available continuum, but the measured residue or alpha spectrum selects a specific participant-spectator projection of that continuum.

${}^7\text{Li}$ also guards against overinterpreting a single-projectile calculation. A model that reproduces ${}^6\text{Li}$ elastic scattering by tuning a composite-projectile optical potential may fail to predict ${}^7\text{Li}$ unless the threshold and fragment absorption are treated explicitly. Conversely, a CDCC or IAV calculation that uses nucleon or cluster fragment potentials can move between ${}^6\text{Li}$ and ${}^7\text{Li}$ with fewer phenomenological changes. This is the practical advantage of a three-body framework: isotope trends are carried by structure and fragment dynamics, not only by refitting an entrance optical potential.

D. ${}^{11}\text{Be}$: the one-neutron halo and core excitation

${}^{11}\text{Be}$ is the canonical one-neutron halo. Its valence neutron separation energy is

$$S_n({}^{11}\text{Be}) = 0.504 \text{ MeV}, \quad (138)$$

and the low-lying $1/2^-$ bound excited state and $5/2^+$ resonance make the continuum spectroscopy unusually visible in reaction observables. In a single-particle projectile model, the breakup strength is governed by the continuum response

$$\begin{aligned} \frac{dB(E\lambda)}{dE} &= \sum_{\nu} B_{\nu}(E\lambda) \delta(E - E_{\nu}), \\ B_{\nu}(E\lambda) &= \frac{1}{2J_0 + 1} \left| \left\langle \Psi_{\nu}^{(-)} \parallel \mathcal{M}(E\lambda) \parallel \Psi_0 \right\rangle \right|^2, \end{aligned} \quad (139)$$

with Ψ_0 the projectile ground state and $\Psi_{\nu}^{(-)}$ continuum states normalized on the energy scale. Equation (139)

is the structure input behind Coulomb-dissociation, CDCC, and time-dependent calculations of halo breakup [125, 170].

The ^{11}Be benchmark exposes two limits of simple CDCC at once. On ^{12}C at intermediate energy, the narrow $5/2^+$ resonance peak is a nuclear-breakup signal rather than a Coulomb one [125, 170]. On a proton target, the exact Faddeev-AGS comparison shows that CDCC can describe elastic scattering while still underestimating transfer and giving basis-dependent breakup in restricted kinematic regions [76]. Agreement for one observable is therefore not a certificate of agreement for the whole reaction.

Near the Coulomb barrier, ^{11}Be makes the continuum visible in a more macroscopic way. The $^{11}\text{Be}+^{64}\text{Zn}$ elastic distribution suppresses the Coulomb-rainbow peak, and the reaction cross section is more than twice that for the stable Be isotopes at comparable energies [21]. CDCC calculations show that convergence is much harder at the physical binding energy than at artificially increased binding, because the long-range dipole couplings remain large at tens of fm [123]. The reaction is also where core excitation becomes unavoidable. The $^{10}\text{Be}(2^+)$ component in the projectile wave function and the associated core-excited resonances are outside a single-particle CDCC basis; XCDCC includes them by coupling the valence motion to intrinsic core states, as described in Sec. IV E [118, 119]. The $^{11}\text{Be}+^{64}\text{Zn}$ inclusive-breakup analyses combine this core-excitation physics with the IAV NEB sum rule, making the system a natural bridge between CDCC and inclusive breakup theory [22, 23].

The target dependence of ^{11}Be is a useful diagnostic. On a light target, nuclear breakup and the detailed resonance content of the continuum are prominent. On a medium or heavy target near the barrier, the long-range Coulomb and nuclear polarization fields alter the elastic rainbow. On a proton target, exact AGS benchmarks expose the competition between elastic, breakup, and transfer partitions [76]. These are not contradictory pictures. They are different ways of projecting the same halo wave function. A single-particle halo model can be adequate for one projection and insufficient for another once core excitation or transfer is resolved.

The inclusive ^{11}Be problem is especially important because it connects halo reactions to heavy-isotope synthesis. In a Trojan-Horse-like picture, the weakly bound neutron can be delivered to the target at sub-barrier energies while the core is observed as a spectator. The IAV-CDCC analysis of $^{11}\text{Be}+^{64}\text{Zn}$ therefore asks a different question from elastic breakup: not how often the projectile dissociates into asymptotic $^{10}\text{Be} + n$, but how often the neutron participant is absorbed or transferred while the core escapes [22, 23]. This is exactly the observable-level distinction that a two-body optical potential cannot make.

E. ^6He : the Borromean two-neutron halo

^6He is structurally different from ^6Li and ^{11}Be because no two-body subsystem is bound. With Jacobi coordinates $\mathbf{r}$ and $\mathbf{y}$ for the internal $\alpha+n+n$ motion, a minimal Borromean Hamiltonian is

$$H_{\text{Bor}} = T_{\mathbf{r}} + T_{\mathbf{y}} + V_{nn}(\mathbf{r}) + V_{\alpha n}(\mathbf{r}_{\alpha n_1}) + V_{\alpha n}(\mathbf{r}_{\alpha n_2}), \quad (140)$$

with the defining spectral pattern

$$E_3 < 0, \quad E_{\alpha n} \geq 0, \quad E_{nn} \geq 0. \quad (141)$$

This is the three-body structure problem reviewed by Zhukov and collaborators [171]. In reaction theory it means that a two-cluster projectile model, though often useful as a discretized effective description, cannot be the final theory of the continuum.

The main observable difference between ^6He and ^6Li is the dipole field. For $\alpha + 2n$, Eq. (136) does not vanish, and low-lying $E1$ strength produces a long-range Coulomb polarization on heavy targets. CDCC calculations of $^6\text{He}+^{208}\text{Pb}$ showed that this dipole polarizability suppresses the forward-angle elastic cross section, while the analogous ^6Li calculation lacks such a dipole effect [133]. This is the cleanest structural contrast in the weakly bound stable and radioactive systems: both projectiles have low thresholds, but only the neutron halo has a strong long-range $E1$ response.

Fusion and reaction cross sections show that the extra halo flux does not automatically become complete fusion. In $^6\text{He}+^{209}\text{Bi}$, the dominant ^4He yield was interpreted as direct transfer or breakup rather than ordinary compound fusion [172]. In $^6\text{He}+^{64}\text{Zn}$, transfer plus breakup accounts for most of the reaction cross section near the barrier, with no simple fusion enhancement relative to ^4He [173]. The old question, enhancement below the barrier or suppression above it, is therefore not a yes-or-no property of a halo. It is an accounting problem over CF, ICF, transfer, EBU, and NEB. A future four-body IAV theory would be the natural way to do this accounting without reducing the $\alpha + n + n$ continuum to an effective two-body channel.

The soft dipole response of ^6He also illustrates why structure and reaction cannot be separated too early. A three-body structure model may reproduce the two-neutron separation energy and matter radius, but the reaction probes the continuum transition density, the neutron-neutron correlation, and the relative weight of dineutron-like and cigar-like configurations. Coulomb excitation on a heavy target emphasizes the dipole strength. Nuclear breakup and transfer emphasize shorter distances and fragment absorption. Fusion data integrate over both. Thus a "good" ^6He projectile wave function is not a scalar property; it is judged by which reaction observable is being calculated.

This is also the system where inclusive theory is most visibly incomplete. For a two-cluster projectile, IAV can sum over all nonelastic states of the participant-target

subsystem. For ${}^6\text{He}$, detecting an alpha particle leaves an unobserved $n+n+A$ subsystem with sequential transfer, dineutron transfer, individual neutron capture, and target excitation all mixed together. Treating this as an effective two-body participant may be useful, but it is not the final operator identity. The need for a four-body IAV extension is therefore not abstract. It is forced by the measured alpha yields and transfer-dominated reaction cross sections in the benchmark systems [172, 173].

F. ${}^9\text{Be}$: a stable Borromean nucleus

${}^9\text{Be}$ offers the experimental convenience of a stable beam together with the theoretical inconvenience of a genuine $\alpha + \alpha + n$ Borromean structure. The lowest breakup thresholds,

$$S_n({}^9\text{Be}) \simeq 1.67 \text{ MeV}, \quad S_\alpha({}^9\text{Be}) \simeq 2.55 \text{ MeV}, \quad (142)$$

place it in the same weak-binding class as ${}^6\text{Li}$ and ${}^7\text{Li}$, but the continuum contains two charged fragments and a neutron. The classic experimental signature is complete-fusion suppression on heavy targets: ${}^9\text{Be}+{}^{208}\text{Pb}$ complete fusion is about 68% of the barrier-distribution expectation, with large incomplete-fusion yields [153]. A later high-precision comparison with ${}^{6,7}\text{Li}$ on ${}^{209}\text{Bi}$ found a common pattern: complete fusion is suppressed above the barrier, while the missing flux appears in incomplete fusion [18].

The target dependence remains important. For ${}^9\text{Be}+{}^{64}\text{Zn}$, breakup does not suppress near-barrier fusion nearly as strongly as for ${}^9\text{Be}+{}^{208}\text{Pb}$ [174]. Thus the low breakup threshold is a necessary but not sufficient predictor; the Coulomb field, available transfer Q values, and fragment absorption all enter. The extension of IAV to a three-body projectile in ${}^9\text{Be}$ -induced reactions is a direct response to this complexity. In that setting the inclusive observable is not a single binary breakup channel but an unresolved sum over states of the unobserved cluster pair and the target, exactly the sum-rule structure described in Sec. V [134].

The stability of ${}^9\text{Be}$ makes it experimentally attractive, but it also creates a subtle theory problem. High-quality stable-beam data can separate elastic scattering, fusion residues, transfer, and breakup fragments more completely than many radioactive-beam experiments. The theory must therefore explain more than a total reaction cross section. It must account for how the $\alpha + \alpha + n$ continuum feeds incomplete fusion, neutron transfer, and alpha emission. A two-body ${}^8\text{Be}+n$ or $\alpha+{}^5\text{He}$ reduction can be useful in restricted kinematics, but the Borromean structure means that different reductions emphasize different unbound subsystems.

The ${}^9\text{Be}$ case also shows why target charge matters. On heavy targets, charged-fragment breakup and long-range Coulomb effects enhance peripheral loss and complete-fusion suppression. On medium targets, nuclear couplings and transfer Q values can dominate, and

the apparent suppression is smaller [18, 174]. A universal suppression factor tied only to the projectile separation energy cannot capture this behavior. The three-body interpretation is that the projectile threshold opens the continuum, but the target and fragment optical potentials decide where the flux goes.

G. ${}^8\text{B}$: proton-halo breakup and astrophysical capture

${}^8\text{B}$ is the proton-halo benchmark. Its valence proton separation energy is only

$$S_p({}^8\text{B}) \simeq 0.137 \text{ MeV}, \quad (143)$$

so Coulomb breakup is strong even though the charged valence particle feels a Coulomb barrier. The reason this system became central is not only its structure. The inverse radiative capture reaction ${}^7\text{Be}(p, \gamma){}^8\text{B}$ determines the solar-neutrino S_{17} factor, and Coulomb dissociation seemed to provide an experimentally accessible route to that capture amplitude [175, 176].

The reaction-theory lesson of ${}^8\text{B}$ is that first-order intuition is not enough for a precision indirect method. In first-order Coulomb dissociation, the breakup cross section is related to the photoabsorption strength,

$$\frac{d\sigma_{\text{Coul}}}{dE} = \sum_{\lambda} n_{E\lambda}(E_{\gamma}) \sigma_{\gamma}^{E\lambda}(E_{\gamma}), \quad (144)$$

where $n_{E\lambda}$ is the virtual-photon number and $\sigma_{\gamma}^{E\lambda}$ is the electromagnetic dissociation cross section for multipolarity $E\lambda$. This formula is the starting point, not the final analysis. Higher-order Coulomb effects, nuclear breakup, $E2$ contamination, and continuum-continuum coupling can all affect the extracted S_{17} [136, 177, 178]. CDCC is therefore not a technical correction to Coulomb dissociation. It is the all-order reaction theory that makes the indirect extraction defensible.

${}^8\text{B}$ also completes the contrast with neutron halos. In ${}^{11}\text{Be}$ and ${}^6\text{He}$, low-lying dipole strength produces long-range Coulomb polarization without a valence Coulomb barrier. In ${}^8\text{B}$, the weak binding is even smaller, but the proton-target and core-target Coulomb fields both shape the continuum. The benchmark is therefore sensitive to the separation of nuclear and Coulomb breakup, to the treatment of charged-particle final-state interactions, and to the experimental angular cuts used to select Coulomb dominance. A reaction theory that succeeds for neutron halos cannot simply be carried over by changing the separation energy.

The ${}^8\text{B}$ history is also a useful check on the relation between perturbation theory and CDCC. At sufficiently high energy and forward angle, first-order Coulomb dissociation captures the dominant physics and gives a transparent connection to the radiative capture matrix element. Away from that clean limit, higher-order Coulomb acceleration, nuclear-Coulomb interfer-

ence, and continuum-continuum couplings alter the inferred strength [136]. This is the same pattern seen elsewhere in this review: the simpler formula is not wrong, but its domain has to be established by an all-order calculation. For astrophysical applications the tolerance is tighter because a modest reaction-model correction can become a significant change in the extracted low-energy S factor.

The proton halo further exposes an asymmetry between structure and reaction language. Structurally, ${}^8\text{B}$ is weakly bound and spatially extended. Reactively, the charged valence particle prevents the breakup from being a simple neutron-halo analog. The continuum wave functions carry Coulomb phases, the E1 and E2 multipoles have different angular signatures, and the core-target and proton-target Coulomb fields both act over long range. This is why ${}^8\text{B}$ belongs in a three-body review even though the measured quantity is often discussed as an electromagnetic response.

H. Energy, target, and residue systematics

The benchmark systems above show why weak binding is an organizing clue but not a prediction. A projectile separation energy sets the lowest continuum scale, yet the observed mechanism is selected by beam energy, target charge, fragment charges, transfer Q values, and the residue or fragment actually measured. This is why ${}^6\text{Li}$, ${}^7\text{Li}$, and ${}^9\text{Be}$ all show above-barrier complete-fusion suppression on heavy targets, but with different magnitudes and with different balances between prompt breakup, direct cluster transfer, and incomplete fusion [18, 72, 153, 154]. The threshold is only the first entry in the ledger.

Near the Coulomb barrier, the target charge controls the time scale and range of the coupling. A heavy target generates long-range Coulomb polarization before the projectile reaches the nuclear surface. This favors dipole breakup in neutron halos and charged-fragment reorientation in cluster projectiles. A medium target reduces the Coulomb lever arm and gives nuclear coupling, transfer, and absorption a larger relative role. The ${}^9\text{Be}$ comparison between ${}^{208}\text{Pb}$ and ${}^{64}\text{Zn}$ is the clean example: the same stable Borromean projectile gives strong complete-fusion suppression on the heavy target and a much weaker effect on the medium target [18, 174]. A theory that assigns a universal suppression factor to ${}^9\text{Be}$ alone has already lost the relevant physics.

The energy dependence is equally diagnostic. Below the barrier, complete fusion is exponentially sensitive to tunneling and to any coupling that changes the barrier distribution. A transfer channel with positive or mildly negative Q value can enhance sub-barrier fusion or produce large spectator fragment yields without requiring strong elastic breakup. Above the barrier, the same coupling can remove flux from complete fusion and feed incomplete fusion or direct reaction residues. This is the

origin of the long-standing language of enhancement below the barrier and suppression above it [11, 66]. Three-body theory refines that language by asking which channel carries the flux. CDCC can show how continuum coupling polarizes the entrance channel. IAV can show whether the detected spectator comes from nonelastic participant absorption. Flux-decomposition methods can separate complete and incomplete absorption when the imaginary potentials are defined consistently [20, 70].

Residue selection is often the hidden variable. Measuring inclusive alpha particles from ${}^6\text{Li}$ or ${}^7\text{Li}$ is not the same as measuring complete fusion residues, and neither is the same as measuring coincident breakup fragments. The inclusive alpha spectrum admits elastic breakup, neutron-transfer-induced alpha emission, deuteron or triton capture, target excitation, and compound decay. A complete-fusion residue excludes events in which only one fragment is captured, but an incomplete-fusion residue selects exactly those events. Thus two experiments on the same projectile-target system can both be correct while emphasizing different projections of the same many-body final space. The theoretical observable must match the experimental residue definition before one asks whether breakup enhances or suppresses fusion.

The transfer Q value is the other hidden variable. For ${}^6\text{Li}$ and ${}^7\text{Li}$, alpha production can be dominated by capture or stripping of the deuteron or triton cluster rather than by preformed elastic breakup [72, 73]. For halo systems, neutron transfer can compete with Coulomb breakup even when the projectile has strong low-lying dipole strength. For ${}^{11}\text{Be}$, the halo neutron can be removed, transferred, or absorbed while the ${}^{10}\text{Be}$ core emerges as a spectator. The same core spectrum may therefore be interpreted as breakup in a channel-resolved measurement or as NEB in an inclusive one. This is exactly where IAV-CDCC is needed: CDCC prepares the continuum-coupled entrance source, and IAV sums the unobserved neutron-target outcomes.

A useful benchmark comparison should therefore report at least four quantities. First, the elastic angular distribution and its DPP, because they show how the entrance channel is polarized. Second, the exclusive breakup spectrum for the measured fragment pairs, because it tests the continuum discretization and resonance content. Third, the inclusive spectator spectrum, because it measures the nonelastic participant sum. Fourth, the residue-resolved fusion or transfer yield, because it determines which part of the absorbed flux is experimentally visible. If a model fits only one of these quantities, it may still be useful, but it has not closed the reaction mechanism.

The stable weakly bound systems are especially valuable for this program because they allow more complete experiments than most radioactive beams. ${}^6\text{Li}$, ${}^7\text{Li}$, and ${}^9\text{Be}$ can be measured with elastic scattering, breakup fragments, transfer channels, and fusion residues in the same laboratory campaign. They are therefore not merely calibration systems for halo reactions. They are

the systems in which the three-body flux ledger can be tested most severely. Halo beams then add the extreme dipole response, core excitation, Borromean structure, or astrophysical capture connection. The mature benchmark strategy is to learn the bookkeeping on stable projectiles and then carry the same observable definitions to radioactive systems, rather than treating every halo data set as a separate special case.

I. Cross-cutting observables

The common observables are elastic scattering, exclusive breakup, inclusive fragment production, complete and incomplete fusion, transfer, and knockout. The first two are channel-resolved and are natural CDCC observables. The inclusive observables require a sum over unobserved target and fragment states, which is why the IAV expression in Eq. (87) is not an optional refinement but the correct observable-level theory. Complete-fusion suppression is likewise not a direct measure of elastic breakup. The ${}^6\text{Li}$ and ${}^7\text{Li}$ data, the Cook direct-transfer result for ${}^7\text{Li}$, and the IAV-CDCC calculation of partial fusion all point to one-step cluster capture and NEB as the dominant mechanisms rather than a simple two-step breakup-fusion picture [20, 72].

The proton-halo case adds a second lesson. Coulomb dissociation of ${}^8\text{B}$ was originally attractive because it seemed to turn the radiative capture ${}^7\text{Be}(p, \gamma){}^8\text{B}$ into a measured breakup observable. In practice the extraction of S_{17} depends on higher-order Coulomb effects, $E2$ contamination, nuclear interference, and continuum-continuum couplings [136, 175–178]. The all-order CDCC result is not a minor correction to first-order perturbation theory; it is the reason Coulomb dissociation can be made quantitatively consistent with direct capture. This is the same methodological message as in the ${}^{11}\text{Be}$ and ${}^6\text{Li}$ benchmarks: a reaction observable can be used as a structure probe only after the three-body dynamics that forms the observable has been controlled.

Across all benchmark systems, the useful organizing variable is not simply the breakup threshold. It is the pair of questions: which fragment is observed, and which fragment or subsystem is summed over? Elastic scattering observes no fragment and sums over none explicitly, but it feels all channels through a DPP. Exclusive breakup observes both fragments and resolves their relative energy. Inclusive breakup observes one fragment and sums the participant over absorption, transfer, and excitation. Fusion observes residues and sums over entrance-channel histories. Transfer observes a final bound or resonant state and filters the entrance continuum through a transition operator. The same projectile can therefore rank differently in different observables. A mature three-body analysis should state the observable projection before naming the method.

The minimum observable set depends on the physics claim. To claim complete fusion suppression as a breakup

effect, one needs more than complete-fusion residues. One needs either exclusive breakup or inclusive spectator data showing where the missing flux goes, and preferably incomplete-fusion residues that close the charge balance. Without that information, the suppression can be caused by transfer-triggered capture, nonelastic breakup, or an entrance-channel polarization effect. This is why the stable systems ${}^6\text{Li}$, ${}^7\text{Li}$, and ${}^9\text{Be}$ remain central: they allow coincidence and residue measurements that can test the ledger directly.

To claim a halo structure effect, one needs an observable that distinguishes the halo degree of freedom from a generic optical renormalization. Elastic scattering alone can show a suppressed Coulomb rainbow or an enlarged reaction cross section, but it does not by itself prove the mechanism. Coulomb dissociation, transfer to bound states, inclusive core spectra, and core-excitation observables probe different components of the halo wave function. For ${}^{11}\text{Be}$, a single-particle halo model may succeed for one projection and fail for another once the ${}^{10}\text{Be}$ core is allowed to excite. For ${}^6\text{He}$, two-neutron correlations and Borromean continuum structure cannot be reduced to a neutron separation energy. A halo benchmark is therefore credible only when the reaction observable actually resolves the halo degree of freedom being discussed.

To claim an astrophysical capture constraint from breakup, one needs control of the inverse-reaction mapping. For ${}^8\text{B}$ Coulomb dissociation, this means separating first-order electromagnetic strength from higher-order Coulomb coupling, nuclear interference, and $E2$ contamination. For THM, it means checking the path from the distorted three-body amplitude to the quasi-free factorized binary amplitude. In both cases the benchmark is not merely that the measured spectrum has the right shape. It is that the reaction theory has been shown to isolate the same matrix element as the desired low-energy capture or fusion process.

To claim a spectroscopic factor or ANC, one needs the reaction filter under control. Transfer angular distributions, breakup momentum distributions, and knockout cross sections can all be sensitive to the same overlap, but they weight radii, partial waves, and final-state interactions differently. Agreement among several probes is powerful because the filters are different. Disagreement is also powerful if it is analyzed as a reaction-model problem rather than averaged away. The benchmark systems reviewed here are therefore not only tests of weak binding. They are tests of whether reaction theory can turn a measured observable into a structure statement without hiding the filter that performed the extraction.

This perspective suggests a useful order of analysis. First, fit or calibrate the elastic channel only to establish the entrance distortion and its uncertainty. Second, compute the DPP or coupled-channel flux ledger to locate which continuum sectors affect the elastic channel. Third, calculate the exclusive breakup distribution if the fragments are measured, and the IAV inclusive distribution if only one fragment is measured. Fourth, com-

TABLE I. Weakly bound benchmark systems and the reaction signal that makes each one diagnostic. Thresholds and quoted suppression/enhancement statements are representative values from the cited benchmark literature.

System	Structure scale	Leading long-range field	Diagnostic reaction signal	Methodological stress test
d	$n + p$, $S_{np} = 2.2246$ MeV	Weak Coulomb through proton fragment	Transfer and breakup test ADWA, CDCC, and AGS on the same Hamiltonian [41, 76, 77]	Exact benchmark for entrance-channel breakup and transfer rearrangement
${}^6\text{Li}$	$\alpha + d$, $S_{\alpha d} = 1.474$ MeV	Cluster $E1$ cancels	CF suppression on heavy targets; inclusive alpha yield dominated by NEB rather than EBU [19, 73, 154]	CDCC continuum selection; IAV separation of EBU, NEB, and ICF
${}^7\text{Li}$	$\alpha + t$, $S_{\alpha t} = 2.47$ MeV	Cluster breakup with higher threshold than ${}^6\text{Li}$	Smaller CF suppression than ${}^6\text{Li}$; alpha yields tied to triton removal and direct cluster transfer [18, 72]	Threshold systematics and spectator-tagged participant capture
${}^{11}\text{Be}$	${}^{10}\text{Be} + n$, $S_n = 0.504$ MeV	Strong dipole plus core-excited sectors	Suppressed Coulomb rainbow and reaction cross section more than twice stable Be on ${}^{64}\text{Zn}$ [21, 123]	XCDCC core excitation; CDCC versus exact AGS transfer and breakup
${}^6\text{He}$	$\alpha + n + n$, $S_{2n} = 0.975$ MeV	Strong neutron-halo $E1$ polarizability	Transfer and breakup carry most of the near-barrier reaction flux on medium and heavy targets [133, 172, 173]	Four-body CDCC and the missing four-body IAV sum rule
${}^9\text{Be}$	$\alpha + \alpha + n$, $S_n \simeq 1.67$ MeV	Coulomb plus charged-fragment breakup	About 30% CF suppression on ${}^{208}\text{Pb}$, weaker on ${}^{64}\text{Zn}$ [153, 174]	Three-body projectile breakup with unresolved two-fragment final states
${}^8\text{B}$	${}^7\text{Be} + p$, $S_p = 0.137$ MeV	Proton-halo Coulomb breakup	Coulomb dissociation constrains S_{17} only after higher-order and continuum-continuum effects are controlled [136, 175, 176]	All-order breakup for indirect astrophysical capture

pare residue-resolved fusion or transfer yields with the absorbed flux. Only after these steps should one assign a mechanism name such as breakup fusion, Trojan-Horse transfer, halo enhancement, or dynamical quenching. The order is slow, but it prevents the mechanism from being chosen before the observable has been defined.

VII. OPTICAL POTENTIALS AND UNCERTAINTY QUANTIFICATION

Every calculation in Secs. III-V inherits an optical-potential problem. In Eq. (5), the three-body dynamics is explicit, but the fragment-target interactions U_{bA} and U_{xA} are still energy-averaged, complex, nonlocal many-body objects represented in practice by fitted two-body potentials. This is the main hidden uncertainty of modern three-body reaction theory. A CDCC calculation may converge beautifully with respect to bins, an IAV calculation may satisfy post-prior equivalence, and an AGS calculation may solve the three-body equations exactly, yet all three can still move if the input optical potentials move within the range allowed by elastic data.

The question is therefore not whether UQ belongs in a three-body review, but where it enters the formalism.

A. Global phenomenological potentials

The practical default for nucleon fragment-target interactions is a local Woods-Saxon optical potential. Suppressing the proton Coulomb term for notational clarity, a representative central plus spin-orbit form is

$$\begin{aligned}
 U_{NA}(r; E) = & -V_v(E)f_v(r) - iW_v(E)f_w(r) \\
 & - i4a_sW_s(E)\frac{df_s(r)}{dr} \\
 & + V_{so}(E)\frac{1}{r}\frac{df_{so}(r)}{dr}\boldsymbol{\ell} \cdot \mathbf{s}, \quad (145)
 \end{aligned}$$

$$f_i(r) = \left[1 + \exp\left(\frac{r - r_i A^{1/3}}{a_i}\right) \right]^{-1}.$$

The global parametrizations of Becchetti and Greenlees, CH89, and Koning-Delaroche differ in details of energy, mass, and isospin dependence, but they share the same reduction: many elastic-scattering data sets are compressed into a smooth function $\theta_{\text{glob}}(E, A, Z)$ [28–30].

This compression made the three-body program possible. The CDCC and IAV calculations reviewed above usually need fragment potentials at energies where no dedicated elastic data exist; without global OMPs, most applications would have no input.

The cost of this convenience is that a global best fit is not a probability distribution. It gives a point in parameter space, not a credible region, and it often has no statistically meaningful covariance. Bayesian reanalyses of CH89 and Koning-Delaroche, and newer dripline-oriented global OMPs, replace the single parameter vector by an ensemble

$$\boldsymbol{\theta}^{(s)} \sim p(\boldsymbol{\theta} \mid D_{\text{el}}, M_{\text{OMP}}), \quad (146)$$

where D_{el} denotes the elastic-scattering database and M_{OMP} the chosen optical-potential model [179, 180]. Equation (146) is a small formal change with a large reaction-theory consequence: any downstream CDCC, IAV, ADWA, or AGS observable can now be evaluated over the posterior ensemble instead of at a single best-fit point.

B. Likelihoods, discrepancy, and posterior ensembles

The phrase "Bayesian optical potential" can mean rather different things in practice. A posterior ensemble is only as meaningful as the likelihood and model-discrepancy structure used to create it. For elastic-scattering data d_i and model predictions $y_i(\boldsymbol{\theta})$, the simplest likelihood is

$$\begin{aligned} \log p(\mathbf{d} \mid \boldsymbol{\theta}) = & -\frac{1}{2} [d_i - y_i(\boldsymbol{\theta})] C_{ij}^{-1} [d_j - y_j(\boldsymbol{\theta})] \\ & - \frac{1}{2} \log |C| + \text{const.} \end{aligned} \quad (147)$$

with C the experimental covariance. Real global OMP calibration usually needs an additional discrepancy term, because the radial form itself is not exact. A common diagonal fractional form is

$$C_{ij} = C_{ij}^{\text{exp}} + \delta_{ij} (\eta y_i(\boldsymbol{\theta}))^2, \quad (148)$$

where η is inferred or assigned as a nuisance parameter [13, 179, 180]. This expression is crude, but it is already better than pretending that all residuals are experimental noise. For three-body reaction predictions, the distinction is decisive. A narrow posterior caused by an overconfident likelihood will propagate into a narrow CDCC or IAV band that looks quantitative but only reflects a statistical assumption.

Different modern global OMPs make different choices at this point. The KDUQ and CHUQ analyses use posterior ensembles to reexpress the older Koning-Delaroche and CH89 forms with calibrated uncertainty, including a likelihood rescaling designed to avoid overweighting very large databases [179]. The East Lansing Model uses a

standard posterior and adds dripline-oriented isovector structure, including information from charge-exchange constraints that ordinary elastic fits do not contain [180]. For a downstream three-body calculation, these choices are not bookkeeping. They determine whether the uncertainty band in a transfer or breakup observable is mostly experimental, mostly discrepancy, or mostly the consequence of extrapolating a global potential beyond the region where data exist.

A useful way to read posterior ensembles is therefore not as "the" error bar on an optical potential, but as conditional distributions:

$$p(\boldsymbol{\theta} \mid D_{\text{el}}, M_{\text{OMP}}, M_{\text{disc}}), \quad (149)$$

where M_{disc} is the discrepancy model. Changing the discrepancy model changes the posterior just as surely as changing the Woods-Saxon form. This is not a weakness of Bayesian analysis. It is an honest statement of what has been assumed. The same clarity is needed when such ensembles are used inside CDCC or IAV: the final reaction band is conditional on the OMP family, the discrepancy model, and the three-body truncation.

C. Microscopic and dispersive optical potentials

Phenomenological OMPs absorb all many-body physics into fitted radial forms. Microscopic models try to compute at least part of the same object from nucleon-nucleon interactions and nuclear densities. In first-order Watson-folding language the nonlocal optical potential has the schematic form

$$U(\mathbf{r}, \mathbf{r}'; E) = \sum_{\alpha=p,n} \int d\mathbf{s} d\mathbf{s}' \tau_{\alpha}(\mathbf{r}, \mathbf{s}; \mathbf{r}', \mathbf{s}'; E) \rho_{\alpha}(\mathbf{s}, \mathbf{s}'), \quad (150)$$

where τ_{α} is an effective projectile-nucleon scattering operator and ρ_{α} is the one-body density matrix of the target [32, 181, 182]. Brueckner g -matrix folding, chiral first-order folding, and global microscopic OMPs differ in how τ_{α} is constructed and how nuclear matter input is localized, but they all preserve two features that local phenomenology hides: nonlocality and off-shell dependence.

Dispersive optical potentials add a second constraint. Causality links the real and imaginary parts through the dispersion relation already introduced in Eq. (4) [33]. This matters for three-body reactions because the fragment optical potentials in Eq. (5) are sampled at several effective energies and in off-shell matrix elements. A local, energy-dependent fit may reproduce elastic scattering at each energy, but it need not represent a single underlying nonlocal operator. The difference is invisible in elastic observables and visible in prior-form transfer and NEB matrix elements, where the interior wave function and off-shell form factors enter directly [103, 183–185]. Thus OMP uncertainty has two layers: parameter un-

certainty within a chosen radial form, and functional or off-shell uncertainty between phase-equivalent operators.

D. Nonlocal and off-shell ambiguity

Elastic scattering constrains the on-shell S matrix. Transfer, prior-form DWBA, and IAV source terms can depend on wave functions and transition operators inside the nuclear surface. These are not fixed uniquely by the elastic S matrix. A simple nonlocal optical kernel has the Perey-Buck form

$$U(\mathbf{r}, \mathbf{r}') = H(\mathbf{r} - \mathbf{r}') U_L \left(\frac{\mathbf{r} + \mathbf{r}'}{2} \right), \quad (151)$$

$$H(\mathbf{s}) = \frac{1}{\pi^{3/2} \beta^3} \exp \left(-\frac{s^2}{\beta^2} \right),$$

where β is the nonlocality range [183]. Under restrictive conditions this kernel can be represented by a local-equivalent potential and a Perey damping factor in the interior [184]. But the equivalence is not a general statement that nonlocality is irrelevant. It is a statement about a particular family of kernels and a particular elastic observable.

Separable optical-potential constructions make the point from the opposite direction. Starting from a local potential, an EST representation can be made phase equivalent at chosen support energies, but the resulting coordinate space operator is intrinsically nonlocal [91, 92, 103]. Near the diagonal it may resemble a Gaussian Perey-Buck kernel; away from the diagonal and in higher partial waves it need not. Thus "local equivalent" and "phase equivalent" are not the same equivalence. The first is a mapping inside a chosen nonlocal family. The second is a statement about on-shell scattering. Three-body matrix elements can distinguish them.

This is why off-shell ambiguity matters for the methods in this review. In post-form breakup dominated by V_{bx} , the transition operator is short ranged in the projectile coordinate, and some off-shell sensitivity is filtered by the source. In prior-form DWBA, the transition operator contains differences of optical potentials, and interior distorted waves enter directly. In IAV, the source-generated participant wave is propagated by the participant optical potential and absorbed by its imaginary part. A family of phase-equivalent optical potentials can therefore leave elastic data unchanged while moving transfer or NEB cross sections [185]. This uncertainty is not captured by sampling parameters of a local Woods-Saxon form. It is a functional uncertainty in the operator class.

The practical implication is that OMP-UQ should be layered. The first layer is a posterior inside a local parametrization. The second is a comparison between local, nonlocal, dispersive, and microscopic operator families. The third is a reaction-level test of which family-dependent differences survive in the observable. For strongly absorbed heavy systems, the off-shell difference

may be screened by Coulomb and centrifugal barriers. For light, low-energy, or prior-form transfer cases, it can be leading. This hierarchy prevents a false sense of precision: a narrow local-parameter posterior does not eliminate an operator ambiguity that the data never constrained.

E. Parameter ambiguities and information content

The old Igo ambiguity says that strong-absorption elastic scattering fixes the potential only in the surface region [61]. Modern information geometry gives the same statement in parameter space. Let $\mathbf{d}$ denote elastic observables and $\mathbf{y}(\boldsymbol{\theta})$ the model prediction computed from an OMP parameter vector $\boldsymbol{\theta}$. For Gaussian experimental errors with covariance C_d , the local sensitivity matrix and Fisher information matrix are

$$J_{i\alpha} = \frac{\partial y_i}{\partial \theta_\alpha}, \quad F_{\alpha\beta} = J_{i\alpha} (C_d^{-1})_{ij} J_{j\beta}. \quad (152)$$

This is the nuclear-reaction version of the sloppy-model geometry developed for nonlinear multiparameter models [186–188]. Large eigenvalues of F are stiff combinations of the OMP parameters; small eigenvalues are directions that elastic data barely see.

A prior-whitened form makes the effective dimensionality explicit. If F is transformed to coordinates in which the prior covariance is the unit matrix, one convenient measure is

$$D_{\text{eff}} = \text{tr} [F(F + I)^{-1}] = \sum_n \frac{\lambda_n}{\lambda_n + 1}, \quad (153)$$

where λ_n are the eigenvalues of the whitened Fisher matrix. For nucleon optical potentials, this number is of order one to two even when the potential has nine to thirteen adjustable combinations [64]. This is not a failure of a particular optimizer. It is the quantitative form of the surface-dominance limit. In the three-body context it explains why changing the OMP in ways that leave elastic scattering almost unchanged can still move transfer, breakup, and NEB observables: those observables project onto directions that elastic scattering did not constrain.

F. Observable classes and experimental design

Information content is not the same for all observables. Elastic angular distributions mostly constrain surface combinations of the real and imaginary central potential. Analyzing powers add spin-orbit sensitivity. Reaction cross sections compress absorption into one number. Transfer and breakup observables probe distorted waves in different radial regions and therefore see parameter combinations that elastic scattering can miss. A

simple local measure of the information added by a proposed data set D_{new} is the expected posterior covariance reduction,

$$\Delta C_\theta = C_\theta - (C_\theta^{-1} + J_{\text{new}}^T C_{\text{new}}^{-1} J_{\text{new}})^{-1}, \quad (154)$$

where C_θ is the current parameter covariance and J_{new} is the sensitivity matrix of the proposed observable. This linear expression is only an approximation to the full Bayesian design problem, but it already shows why adding a new observable class can matter more than reducing error bars on an already redundant angular distribution [13].

The full Bayesian version chooses an experiment by maximizing an expected utility,

$$\mathcal{U}(D_{\text{new}}) = \int d\mathbf{d}_{\text{new}} p(\mathbf{d}_{\text{new}} | D) u[p(\boldsymbol{\theta} | D, \mathbf{d}_{\text{new}})], \quad (155)$$

where u may be the information gain, the reduction of a target observable uncertainty, or the ability to discriminate between potential families. For a three-body program the target utility should not be the OMP parameters themselves. It should be the downstream observable: a spectroscopic factor, an inclusive alpha spectrum, a breakup angular distribution, or a fusion absorption component. The best elastic measurement is the one that reduces the uncertainty in the reaction conclusion, not necessarily the one that shrinks the largest number of Woods-Saxon parameters.

This view changes how benchmark data should be planned. Nearby-energy elastic data can reduce interpolation uncertainty more effectively than simply increasing the angular density at one energy, while spin observables may carry information orthogonal to the unpolarized cross section [13, 169]. For weakly bound reactions, the most valuable data set may combine elastic scattering, inclusive spectator spectra, and transfer or fusion residues on the same target. Such a data set constrains the entrance optical potential, the breakup coupling, and the absorption ledger at once. It also exposes model discrepancy: if the same OMP posterior cannot describe elastic scattering and the IAV spectator spectrum, the problem is not a larger error bar but a missing degree of freedom.

G. Bayesian calibration

Bayesian calibration promotes this information-content statement to a predictive distribution. A minimal posterior for a reaction model M is

$$p(\boldsymbol{\theta}, \boldsymbol{\eta} | \mathbf{d}, M) \propto \exp \left[-\frac{1}{2} \Delta_i(\boldsymbol{\theta}) C_{ij}^{-1}(\boldsymbol{\eta}) \Delta_j(\boldsymbol{\theta}) \right] \pi(\boldsymbol{\theta}) \pi(\boldsymbol{\eta}), \quad (156)$$

with residuals $\Delta_i(\boldsymbol{\theta}) = d_i - y_i(\boldsymbol{\theta})$ and nuisance or discrepancy parameters $\boldsymbol{\eta}$ entering the covariance C

[13, 167, 169, 179]. The frequentist covariance propagation used in early OMP-UQ work is recovered only in the linear-Gaussian limit. For realistic OMPs the posterior is non-Gaussian, and the Bayesian and frequentist uncertainty bands can have different empirical coverage [169].

Equation (156) is also the point where computation becomes physics. If each likelihood evaluation requires a full coupled-channel or Faddeev solve, posterior sampling becomes prohibitive. Gradient-free ensemble samplers are adequate for low-dimensional R-matrix or OMP fits, but they scale poorly and can fail silently in high dimension [189–191]. Hamiltonian Monte Carlo uses gradients of the log posterior to move through parameter space. With an auxiliary momentum $\mathbf{p}$ and mass matrix M_p , the HMC dynamics is

$$\begin{aligned} \mathcal{H}(\boldsymbol{\theta}, \mathbf{p}) &= -\log p(\boldsymbol{\theta} | \mathbf{d}, M) + \frac{1}{2} \mathbf{p}^T M_p^{-1} \mathbf{p}, \\ \dot{\boldsymbol{\theta}} &= \frac{\partial \mathcal{H}}{\partial \mathbf{p}}, \quad \dot{\mathbf{p}} = -\frac{\partial \mathcal{H}}{\partial \boldsymbol{\theta}}, \end{aligned} \quad (157)$$

as reviewed by Neal and used adaptively in the no-U-turn sampler [192, 193]. The bottleneck is therefore not only a fast forward model but a differentiable one. This is the practical reason that modern reaction emulators must be designed with gradients in mind, rather than merely with cross-section speedup factors in mind.

The scaling argument is simple. For many random-walk or ensemble schemes, the number of likelihood evaluations needed to explore a posterior grows rapidly with dimension and with posterior anisotropy. HMC is not magic, but in smooth high-dimensional targets its stable step size decreases only as a mild power of the dimension, while gradients point proposals along the posterior geometry rather than across it [192, 193]. If a reaction model has ten to twenty optical-potential parameters, plus normalizations, discrepancy terms, and perhaps structure parameters, this scaling is no longer a numerical detail. It decides whether the posterior is actually sampled or only decorated by a trace plot.

For three-body reactions this leads to a strict requirement. The emulator must reproduce the observable and its derivatives with respect to the parameters being inferred. A fast table interpolation that gives accurate cross sections but noisy gradients may be adequate for plotting uncertainty bands; it is not adequate for HMC. Conversely, a Galerkin emulator whose residual is slightly more expensive but differentiates through the same linear solve can be more valuable for inference. This is the distinction between a surrogate calculator and a statistical engine. The former speeds up forward predictions. The latter changes which posteriors are computationally reachable [65, 194].

H. Emulators

Most emulator constructions used in nuclear reactions are projection-based model reductions. After discretization, the coupled equations of Sec. IV or the two-body scattering equation underlying an OMP fit can be written schematically as

$$A(\boldsymbol{\theta}) \mathbf{x}(\boldsymbol{\theta}) = \mathbf{b}(\boldsymbol{\theta}), \quad (158)$$

where $\mathbf{x}$ contains wave-function coefficients or channel amplitudes. Snapshots at selected parameters define a reduced basis V , and the Galerkin approximation

$$\mathbf{x}(\boldsymbol{\theta}) \simeq V \mathbf{c}(\boldsymbol{\theta}), \quad V^\dagger A(\boldsymbol{\theta}) V \mathbf{c}(\boldsymbol{\theta}) = V^\dagger \mathbf{b}(\boldsymbol{\theta}) \quad (159)$$

is the common algebra behind eigenvector continuation, reduced-basis scattering emulators, and POD-Galerkin coupled-channel emulators [14, 15, 135, 195–197]. In affine problems, the reduced matrices can be assembled exactly and cheaply. Realistic optical potentials are usually non-affine in geometry and energy, so empirical interpolation, operator interpolation, or direct differentiable assembly is needed [65, 198, 199].

The natural error indicator is the residual of the full equation evaluated on the reduced solution,

$$\mathbf{r}(\boldsymbol{\theta}) = \mathbf{b}(\boldsymbol{\theta}) - A(\boldsymbol{\theta}) V \mathbf{c}(\boldsymbol{\theta}). \quad (160)$$

For a coercive Hermitian problem this residual can be converted into rigorous state-error bounds. Scattering with complex optical potentials is less friendly: the operator is non-Hermitian, boundary conditions matter, and Kohn anomalies or near-singular reduced matrices can produce localized failures [15, 194, 196]. Still, the residual is the right diagnostic. It tests the equation, not only a selected observable. Projection error in an elastic cross section may look small while the interior wave function, the transfer source, or the IAV absorption matrix element is less accurate.

The singular-value spectrum of the snapshot matrix gives a complementary diagnostic. If full solutions collected at training parameters are arranged as columns of a matrix X , the POD basis is obtained from

$$X = U \Sigma V^\dagger, \quad V_{\text{rb}} = U_{1:N_{\text{rb}}}, \quad (161)$$

and the discarded singular values estimate how compressible the solution manifold is [14, 197]. This test is useful before Bayesian inference begins. A slow singular-value decay means that the parameter variation is not low-dimensional in the chosen coordinates, and a small reduced basis will extrapolate rather than emulate. In CDCC this can happen when the continuum thresholds, absorption radii, or Coulomb tails move with the parameters. The emulator then needs either a larger basis, a better parameterization, or an exact block treatment of the stiff part of the operator.

Affine structure is the dividing line between exact algebra and numerical compression. If

$$A(\boldsymbol{\theta}) = \sum_{q=1}^Q \Theta_q(\boldsymbol{\theta}) A_q, \quad (162)$$

then all reduced matrices $V^\dagger A_q V$ can be built offline, and the online calculation only combines Q small matrices. Woods-Saxon geometries, energy-dependent optical terms, and channel couplings generated by calibrated fragment potentials are typically non-affine. Empirical interpolation can recover an approximate affine form, but it adds its own error. This is why a CDCC emulator used for inference must report not only speedup but also direct-solve validation over the posterior support [65, 135, 198, 199].

For Bayesian calibration, the important derivative is obtained by differentiating Eq. (159),

$$\frac{\partial \mathbf{c}}{\partial \theta_\alpha} = A_r^{-1} \left(\frac{\partial \mathbf{b}_r}{\partial \theta_\alpha} - \frac{\partial A_r}{\partial \theta_\alpha} \mathbf{c} \right), \quad (163)$$

$$A_r = V^\dagger A V, \quad \mathbf{b}_r = V^\dagger \mathbf{b}.$$

This formula is simple, but it changes the inference regime. A coupled-channel model that was formerly a forward calculator becomes a gradient source for HMC. The differentiable CDCC calibration of $d+^{58}\text{Ni}$ is one example: the posterior over eighteen optical-potential parameters is sampled directly, and the data pull the deuteron surface absorption above the global-potential value [65]. The physics message is that an emulator should not be judged only by how many forward cross sections per second it produces. The decisive question is whether it preserves the derivatives, channel couplings, and error controls needed for posterior inference.

I. Uncertainty propagation into three-body observables

Once the posterior over optical-potential and model-discrepancy parameters is available, propagation into a three-body observable O is conceptually straightforward:

$$p(O | \mathbf{d}) = \int d\boldsymbol{\theta} p(O | \boldsymbol{\theta}, M_{3b}) p(\boldsymbol{\theta} | \mathbf{d}, M_{\text{OMP}}), \quad (164)$$

where M_{3b} may be CDCC, IAV, ADWA, or an exact three-body solver. In practice the integral is evaluated by posterior samples,

$$\bar{O} = \frac{1}{N_s} \sum_{s=1}^{N_s} O(\boldsymbol{\theta}^{(s)}),$$

$$\text{Cov}(O_a, O_b) = \frac{1}{N_s - 1} \sum_s \Delta O_a^{(s)} \Delta O_b^{(s)}. \quad (165)$$

This elementary formula has already changed how several reaction conclusions should be read. In deuteron transfer, honest OMP uncertainties can be large enough that

ADWA and DWBA are not statistically distinguishable for some extracted spectroscopic factors [168]. In near-barrier weakly bound scattering, uncertainty in fitted optical potentials can turn an apparent breakup threshold anomaly into a precision artifact [17]. In coupled-channel deuteron scattering, Bayesian calibration of the three-body model itself identifies which fragment optical potential components the data actually want to move [65].

There remains a deeper uncertainty that Eq. (164) does not remove. The posterior samples a chosen model family. If two nonlocal operators are phase equivalent in elastic scattering but differ off shell, the elastic posterior cannot choose between them. This is precisely the uncertainty that enters prior-form DWBA and NEB matrix elements, because those observables involve interior distorted waves and transition operators rather than on-shell elastic amplitudes [103, 185]. Thus the UQ program for three-body reactions has two distinct tasks. The first is now operational: propagate parameter uncertainty from calibrated OMP ensembles through CDCC, IAV, and transfer observables. The second is still open: quantify functional, nonlocal, and omitted-channel uncertainty in the optical operators themselves.

A complete error budget should therefore be written as a hierarchy rather than as a single band,

$$\text{Var}(O) \simeq \text{Var}_\theta(O) + \text{Var}_{\text{disc}}(O) + \text{Var}_{3b}(O) + \text{Var}_{\text{str}}(O), \quad (166)$$

where the terms denote optical-parameter uncertainty, model discrepancy of the optical operator, approximation error of the three-body reaction method, and uncertainty in structure inputs such as overlaps or transition densities. The decomposition is schematic because the terms can be correlated, but it is useful precisely because it prevents a common mistake: treating an OMP posterior band as the whole theory uncertainty. For a CDCC elastic calculation, Var_θ may dominate once the continuum model space is converged. For a halo transfer calculation, Var_{3b} and Var_{str} can be just as important. For a prior-form DWBA or NEB calculation, nonlocal off-shell uncertainty may not appear in any local OMP posterior at all.

The practical closure tests follow the formalism already developed in this review. CDCC calculations should report convergence in the elastic channel, exclusive breakup bins, and the DPP or absorption observable being used. IAV calculations should check post-prior equivalence where both forms can be computed, finite-range versus zero-range sensitivity, and the radial support of the source. Bayesian calibrations should validate emulator predictions against direct solves over posterior samples, not only over a training grid. Exact AGS or Faddeev benchmarks should be used when they are available to assign the reaction-model component of Eq. (166). This is the route from attractive error bands to defensible uncertainty quantification.

J. Model comparison and posterior predictive checks

Parameter estimation is only the first layer of UQ. A reaction analysis also chooses a model class: a local Woods-Saxon OMP or a nonlocal one, a two-body projectile or an XCDCC projectile, a CDCC entrance wave or a DWBA entrance wave, a diagonal absorptive coupling or a full coupled absorptive matrix. These choices cannot be represented by a covariance matrix over parameters inside one model. In Bayesian notation the evidence for a model M is

$$p(\mathbf{d} | M) = \int d\boldsymbol{\theta} p(\mathbf{d} | \boldsymbol{\theta}, M) \pi(\boldsymbol{\theta} | M), \quad (167)$$

and the ratio of evidences compares model classes rather than parameter points [194, 196]. This expression is demanding for expensive reaction models, but the conceptual distinction matters even when one does not compute the evidence explicitly. A narrow posterior inside the wrong model family is not a precise physical conclusion.

For three-body reactions, model comparison should be tied to observables rather than to formal elegance. If a two-body ^{11}Be model and an XCDCC model both fit elastic scattering but predict different core-excitation transfer yields, the evidence from elastic data alone cannot choose the structure model. If CDCC and ADWA agree for a deuteron transfer angular distribution within OMP uncertainty, the data do not justify claiming a continuum-coupling effect in the extracted spectroscopic factor [168]. If a local and a nonlocal phase-equivalent optical potential both reproduce elastic data but differ in a prior-form DWBA matrix element, the elastic likelihood cannot remove the off-shell ambiguity [103, 185]. The observable that breaks the ambiguity must be part of the comparison.

Posterior predictive checks provide a practical version of this logic. After calibrating a model to data $\mathbf{d}$, one computes the distribution of a new observable $\tilde{\mathbf{d}}$,

$$p(\tilde{\mathbf{d}} | \mathbf{d}, M) = \int d\boldsymbol{\theta} p(\tilde{\mathbf{d}} | \boldsymbol{\theta}, M) p(\boldsymbol{\theta} | \mathbf{d}, M). \quad (168)$$

The check is not whether the posterior mean follows the data by eye. It is whether replicated data drawn from Eq. (168) look like the withheld observable. For a weakly bound projectile, a serious check would calibrate to elastic scattering and then predict breakup, or calibrate to elastic plus breakup and then predict inclusive alpha spectra or fusion residues. Failure is informative. It indicates that the calibrated optical potential or projectile model does not carry the physics needed by the withheld observable.

This is where model discrepancy should be treated carefully. A large discrepancy term can make posterior predictive checks pass by inflating the uncertainty band, but that does not explain the mechanism. Conversely, a small discrepancy term can force the posterior to distort

physical parameters in order to fit data that the model class cannot represent. The correct response depends on the pattern of residuals. Smooth angular residuals may indicate an optical-potential shape error. Residuals localized near diffraction minima may indicate angular normalization or spin-orbit effects. Residuals that appear only in inclusive spectra may indicate missing NEB or transfer channels. UQ should classify these failures instead of turning them into a single nuisance variance.

K. Information gain and experimental design

The Fisher and effective-dimensionality diagnostics discussed earlier become more useful when they are connected to experiment design. A measurement should be judged not only by smaller pointwise error bars but by the information it adds about the combinations of parameters and operators that control the physics question. One way to state this is the expected information gain,

$$\mathcal{I} = \int d\tilde{\mathbf{d}} p(\tilde{\mathbf{d}} | M) D_{\text{KL}} \left[p(\boldsymbol{\theta} | \tilde{\mathbf{d}}, M) \parallel \pi(\boldsymbol{\theta} | M) \right], \quad (169)$$

where D_{KL} is the Kullback-Leibler divergence between the posterior and prior [14, 196]. This formula is rarely evaluated literally for full CDCC or IAV calculations, but it gives a clean criterion. The best new data are not necessarily the most precise data at an angle already over-constraining one stiff direction. They are the data that rotate or enlarge the constrained subspace relevant to the observable.

Optical-model ambiguities make this point concrete. Elastic angular distributions often constrain only a small number of effective parameter directions, even when the potential has many Woods-Saxon parameters [61, 64]. Measuring more angles at the same energy can reduce statistical noise without increasing the effective dimension. Adding reaction or total cross sections, analyzing powers, breakup spectra, or transfer data can constrain different directions because the observables sample different radii and different combinations of real and imaginary terms. The experimental design problem is therefore coupled to the reaction model: one must know which observable sees which part of the operator.

For weakly bound systems, the most valuable design is often a correlated data set rather than a single high-precision observable. Elastic scattering fixes the entrance distortion and DPP. Exclusive breakup fixes continuum bins and resonances. Inclusive spectator spectra fix the participant absorption sum. Fusion residues fix which absorbed flux becomes complete or incomplete fusion. Transfer fixes overlap and final-state optical potentials. A joint experiment that measures several of these observables with common normalization can reduce model degeneracy far more effectively than a separate precision measurement of only one channel. This is the statistical version of the flux-ledger argument in Secs. IV I and VII.

Information diagnostics also help decide when an emulator is accurate enough. If an emulator error lies in a sloppy direction that the likelihood does not constrain and that the final observable does not use, it may be harmless. If the same absolute error lies along a stiff posterior direction, it can bias the inference even when the cross section error looks visually small. The appropriate emulator validation metric is therefore not only a maximum relative error in the training box. It is the projection of the emulator error onto the likelihood gradients and onto the posterior-predictive observable. This is why differentiable emulators are so useful: they allow one to check not only values but sensitivities [65].

L. Correlated systematics and normalization

Many reaction data sets are dominated by correlated systematics rather than independent pointwise errors. Absolute normalization, target thickness, detector efficiency, beam integration, and angular calibration can move whole sections of an angular distribution coherently. Treating such uncertainties as independent diagonal errors overweights the shape information and understates normalization ambiguity. A simple hierarchical likelihood introduces a normalization nuisance parameter N ,

$$d_i = N y_i(\boldsymbol{\theta}) + \epsilon_i, \quad N \sim \mathcal{N}(1, \sigma_N^2), \quad (170)$$

with ϵ_i carrying the point-to-point covariance. This is standard statistics, but it has specific reaction consequences. A normalization shift can mimic absorption strength in elastic scattering, while an angular-shape change is more likely to indicate a radius, diffuseness, spin-orbit, or coupling effect.

When several observables are measured in the same experiment, their normalization errors may be correlated. An elastic distribution and an inclusive alpha spectrum may share beam current and target thickness but have different detector efficiencies. A fusion-residue yield may share the beam normalization but not the charged-particle efficiency. Ignoring these correlations can make a joint posterior artificially tight or can create spurious tension between observables. In the multi-observable closure program of Sec. VIII G, the covariance model is therefore part of the physics analysis. It determines whether a discrepancy is a real failure of the reaction Hamiltonian or a bookkeeping error in the data.

The same caution applies to old benchmark data. Many classic weakly bound fusion and breakup measurements remain essential because they map systems that are hard to revisit. Their quoted errors, however, may not include the full correlated covariance needed for modern inference. A responsible UQ analysis should preserve the historical value of those data while avoiding a false precision claim. One practical route is to introduce conservative normalization and discrepancy hyperparameters, then test whether the physics conclusion survives

their marginalization. If complete-fusion suppression or NEB dominance disappears only when an unrealistically large normalization error is allowed, the conclusion is robust. If it disappears under a modest normalization prior, the old qualitative statement should be softened.

M. Reporting standards for reaction UQ

A mature three-body UQ calculation should report four layers. The first is the input posterior: which optical, structure, and discrepancy parameters were calibrated, to which data, with which priors and covariance. The second is the solver error: continuum convergence, radial matching, partial-wave truncation, emulator residuals, and benchmark deviations. The third is the observable propagation: posterior predictions for the measured and withheld observables, with correlations when possible. The fourth is the model-class uncertainty: alternative optical forms, nonlocality, projectile structure, and reaction approximation. Without this separation, a single band can hide whether the dominant uncertainty is statistical, numerical, structural, or conceptual.

This reporting standard is demanding but realistic. Exact AGS benchmarks can anchor the reaction approximation in selected systems. CDCC convergence studies can quantify continuum truncation. IAV post-prior checks can test source consistency. Emulator validation can be automated over posterior samples. Bayesian calibration can carry optical-potential correlations rather than hand-tuned parameter sets. What remains hardest is model-class uncertainty, especially nonlocal off-shell behavior and omitted many-body channels. That difficulty should be stated explicitly rather than hidden by precise-looking error bars.

The payoff is clarity. A three-body calculation with UQ should be able to say which part of the conclusion is driven by data, which part by the chosen reaction operator, which part by the optical model, and which part by structure input. That is the standard needed for indirect astrophysical rates, spectroscopic factors, and weakly bound fusion mechanisms. It is also the standard that lets theory guide new measurements rather than merely fitting old angular distributions.

VIII. OPEN PROBLEMS AND OUTLOOK

The field now has three mature answers to the same three-body Hamiltonian. Faddeev-AGS theory is exact within the chosen pair interactions, CDCC is the workhorse approximation for exclusive continuum dynamics, and IAV gives the inclusive sum rule for unobserved nonelastic breakup. The open problems are therefore not caused by a lack of formalism. They arise where these formalisms must be combined with Coulomb fields, four-body projectiles, microscopic structure, optical-potential uncertainty, and indirect reaction

observables. This section identifies the points where a review of the existing theory naturally turns into a research program.

A. Exact benchmarks beyond the easy Coulomb cases

The most basic need is a new generation of exact benchmarks. The existing CDCC-versus-Faddeev comparisons are invaluable precisely because they are uncomfortable: deuteron scattering can be close to exact in elastic observables, while ^{11}Be transfer and breakup expose basis and channel-selection dependence [76, 77]. But the heaviest and most Coulomb-dominated systems, the systems most relevant for radioactive beams and near-barrier fusion, remain outside routine exact AGS calculations. Screening and renormalization work beautifully in controlled three-nucleon and light-cluster cases, but become numerically severe for high- Z targets [76, 80]. Coulomb-distorted AGS formulations were designed to remove this bottleneck, but they have not yet become the production benchmark for heavy-target transfer [83, 84].

The benchmark problem should not be stated as “make CDCC exact”. CDCC is a projection of the three-body wave function onto projectile continuum states, and it answers a particular set of questions well. The sharper task is to map its domain of validity by observable: elastic scattering, transfer, exclusive breakup, inclusive fragment production, fusion absorption, and polarization potentials need not converge at the same rate. A useful modern benchmark would therefore compare the full vector of observables, not only an elastic angular distribution. It should include halo core excitation, heavy Coulomb fields, and the same fragment optical potentials used in practical analyses. Without such benchmarks, error bars in Sec. VII quantify input uncertainty but not the approximation error of the three-body method itself.

B. Four-body projectiles and inclusive breakup

The clearest missing theory is the merger of four-body CDCC with IAV inclusive breakup. Four-body CDCC already treats ^6He , ^6Li , and other Borromean or three-cluster projectiles by discretizing their internal three-body continuum [130–132]. IAV, in contrast, supplies the inclusive sum over unobserved final states for a two-cluster projectile. The natural generalization for an observed fragment b and an unobserved subsystem $X = x_1 + x_2 + A$ has the same operator form,

$$\frac{d^2\sigma_b^{\text{NEB}}}{dE_b d\Omega_b} = -\frac{2}{\hbar v_a} \rho_b(E_b) \langle \rho_X | W_X | \rho_X \rangle, \quad (171)$$

with ρ_X generated by a source that contains the full three-body projectile wave function rather than a two-body bound state. The obstacle is not the prefactor. It is the definition and numerical construction of W_X ,

which must represent absorption in a three-fragment subspace without double counting sequential decay, transfer, and fragment capture. This is exactly why the problem matters: ${}^6\text{He}$ and ${}^9\text{Be}$ benchmarks show that inclusive alpha yields and incomplete fusion are dominated by unresolved channels, not by the exclusive breakup channel alone [18, 134, 172, 173].

A working four-body IAV theory would also close a conceptual loop in the weakly bound fusion problem. Older language often separated prompt breakup, transfer, incomplete fusion, and complete-fusion suppression as if they were distinct mechanisms that could be assigned by fragment counting. The operator view says they are different projections of the same absorbed three-body or four-body wave function. The Line-F absorption decompositions, the IAV-CDCC partial-fusion result, and the ${}^6\text{Li}+{}^{93}\text{Nb}$ alpha data already point in this direction [20, 68, 70, 73]. The next step is a single quantum calculation in which CF, ICF, EBU, NEB, and transfer are not competing stories but separate terms in a conserved flux ledger.

C. Exact polarization potentials from continuum elimination

Dynamic polarization potentials are often discussed as local radial corrections extracted after a coupled-channel calculation. That practice is useful, but it hides the exact object. If the elastic channel is separated from all CDCC breakup channels by projectors P and Q , then eliminating the breakup channels gives the nonlocal, energy-dependent operator

$$\Delta U_{\text{CDCC}}(E) = H_{PQ} \frac{1}{E^+ - H_{QQ}} H_{QP}. \quad (172)$$

This is the Feshbach optical potential of Eq. (3) written inside a discretized continuum model [4, 35]. It is also a Schur complement of the coupled-channel matrix. The local TELP used in phenomenology is therefore a representation of Eq. (172), not the object itself.

Computing Eq. (172) channel by channel would answer several questions that are currently phrased qualitatively. Which continuum bins generate the long-range imaginary tail? How much of the repulsive real DPP is resonant and how much is nonresonant? Which part of the DPP is lost when it is made local? How does the answer change when the eliminated channels are transfer channels rather than breakup channels? The same operator would connect the channel-importance metrics used in CDCC, the generalized optical-theorem decompositions of absorption, and the old threshold-anomaly phenomenology [16, 17, 47, 67, 71]. This is a concrete calculation, not only a conceptual program.

D. Optical potentials from structure to reactions

The long-term structure-reaction problem is still unsolved. A reaction analysis often combines a microscopic overlap or ANC with phenomenological fragment optical potentials whose uncertainties and off-shell content are not derived from the same Hamiltonian. Bayesian global OMPs and microscopic folding potentials improve this situation, but neither yet provides the full nonlocal, absorptive, energy-dependent operator required by Eq. (5) for unstable targets and weakly bound projectiles [179, 180, 182]. The problem is most acute in transfer and knockout, where the extracted spectroscopic information is precisely the quantity that should test structure theory.

Off-shell ambiguity is not a philosophical concern. Phase-equivalent nonlocal potentials can have the same elastic S matrix and different interior distorted waves, and prior-form DWBA or NEB matrix elements are sensitive to those interior functions [103, 183–185]. The only satisfactory arbiter is a microscopic optical potential whose nonlocality and absorption are tied to the same many-body theory used for the overlap. Until that exists in a form usable by CDCC, IAV, and transfer calculations, UQ must report a functional uncertainty in addition to parameter error bars.

E. Bayesian three-body reaction theory

The UQ program has moved from the question “can one attach an error band to an optical potential?” to the harder question “can one infer the inputs of a three-body reaction model from the data that the model predicts?” The answer now depends on emulators and differentiability. Projection-based emulators make repeated coupled-channel solves cheap; differentiable implementations make HMC and NUTS feasible in parameter spaces large enough to describe realistic OMPs [65, 135, 194, 196, 197]. The remaining frontier is not elastic scattering alone. The posterior should be constrained simultaneously by elastic, breakup, transfer, and inclusive data, because each observable sees a different combination of fragment potentials and continuum couplings.

The exact-emulator idea based on Woodbury or Schur identities is promising but has a hard scope boundary. For affine, low-rank updates at fixed energy, few-body scattering can be reduced exactly to a small system [109]. Realistic CDCC with Woods-Saxon geometries and calibrated fragment potentials is non-affine, and the breakup coupling blocks themselves move with the parameters. The honest target is therefore a hierarchy: operator-level exactness where the algebra allows it, controlled projection or interpolation where it does not, and posterior validation against direct solves in the observables that carry the physics. Anything less risks turning a computational approximation into an unquantified theory error.

F. Indirect methods and spectroscopic quenching

Indirect reaction methods are where three-body theory has the highest leverage and the highest risk. Coulomb dissociation, THM, surrogate reactions, and knockout analyses all infer structure or astrophysical information from a reaction that is not the desired reaction itself [12, 26, 160, 164]. The review literature contains successful cases, but also unresolved disputes. The $^{12}\text{C}+^{12}\text{C}$ THM controversy shows that an approximate plane-wave surrogate formula can move an astrophysical rate by orders of magnitude if used outside its domain [26, 161, 162]. The IAV reduction of THM provides a way to state the approximation hierarchy at the operator level, but the decisive calculation is a full numerical surrogate chain benchmarked against the commonly used factorized formula.

The spectroscopic-quenching problem is similar in structure. Knockout, transfer, and (p,pN) reactions do not always return the same reduction factors, and the observed R_s trend with separation-energy asymmetry may contain reaction dynamics as well as structure information [24, 25, 165, 166]. In Feshbach language, additive fragment optical potentials omit induced nonadditive and polarization terms generated by eliminating channels. Whether those induced terms account for the observed quenching systematics is an open reaction question, not a matter to settle by adjusting spectroscopic factors after the fact [36]. This is exactly the kind of problem for which a three-body review should insist on the Hamiltonian first and the extracted structure number second.

G. Multi-observable closure

The next decisive data sets will not be single angular distributions. They will be closure experiments in which elastic scattering, breakup, transfer, inclusive fragment spectra, and fusion or residue yields are measured for the same projectile-target system over overlapping energies. Such data are hard, but they match the formalism. CDCC constrains the elastic and exclusive breakup channels. IAV constrains the inclusive spectator spectra. Flux decomposition constrains where absorption occurs. Transfer or knockout constrains the structure overlap. If these observables cannot be described with one set of fragment optical potentials and one projectile model, the failure is diagnostic rather than inconvenient.

The statistical form of this closure is a joint posterior,

$$p(\boldsymbol{\theta} \mid D_{\text{el}}, D_{\text{bu}}, D_{\text{inc}}, D_{\text{fus}}) \propto \prod_c p(D_c \mid \boldsymbol{\theta}, M_c) \pi(\boldsymbol{\theta}), \quad (173)$$

where the data classes c are computed by different reaction observables but share the same physical inputs. The point is not to force all data into one overconstrained fit. The point is to expose incompatibilities. If elastic data prefer one surface absorption, inclusive alpha spectra re-

quire another, and fusion residues require a third, then the model is missing a channel or using the wrong optical operator. This is a stronger statement than saying that each observable can be fitted separately.

For weakly bound fusion, a closure experiment would measure elastic scattering and inclusive fragment spectra together with complete and incomplete fusion residues. For halo transfer, it would combine elastic scattering, breakup, and transfer to bound and continuum final states. For THM, it would benchmark the quasi-free three-body reaction against direct capture or a well-constrained surrogate in an energy region where both are possible. These programs are experimentally demanding, but they are the only way to turn the operator identities in this review into falsifiable reaction-model tests.

H. A minimal benchmark suite

The field would benefit from a small, stable benchmark suite rather than a large collection of unrelated case studies. The suite should be selected by physics role. A deuteron case should test the connection between exact AGS, ADWA, CDCC, and transfer. A stable weakly bound cluster case, such as ^6Li or ^7Li on a medium and a heavy target, should test the split between elastic breakup, NEB, transfer, and incomplete fusion. A one-neutron halo case, such as ^{11}Be , should test core excitation, Coulomb polarization, and transfer partitions. A Borromean case, such as ^6He or ^9Be , should test four-body CDCC and the missing inclusive four-body sum rule. A proton-halo case, such as ^8B , should test all-order Coulomb dissociation for an astrophysical observable.

Each benchmark should define the Hamiltonian, fragment optical potentials, projectile structure model, continuum truncation, and measured observables. The same input should then be run through multiple approximations. For example, the deuteron case should compare AGS, ADWA, CDCC, and DWBA transfer on one set of $n+A$ and $p+A$ interactions. The ^6Li case should compare CDCC elastic and breakup observables with IAV spectator spectra and residue-resolved absorption. The ^{11}Be case should compare inert-core CDCC, XCDCC, and exact calculations where available. The purpose is not to declare one method universally best. It is to map where each approximation breaks and which observable notices first.

The benchmark suite should include negative tests. A method that succeeds for elastic scattering may fail for transfer. A two-body projectile model may fit the elastic rainbow while missing core-excited breakup. A first-order Coulomb-dissociation formula may work at very forward angles and fail once continuum-continuum coupling is included. Publishing only the successful projection teaches too little. A useful benchmark records the failure mode, because that is what tells future users when they must pay the cost of a larger calculation.

The suite should also separate numerical and physi-

cal convergence. Numerical convergence asks whether increasing radial boxes, partial waves, bin numbers, and quadrature points changes the answer. Physical convergence asks whether adding a channel, a multipole, core excitation, nonlocality, or an absorptive off-diagonal coupling changes the interpretation. Both are needed. A calculation can be numerically converged inside a physically inadequate model space, and a physically rich calculation can still be numerically underresolved. The reporting standard should make that distinction explicit.

I. From local potentials to reaction-ready operators

A second practical frontier is the construction of optical operators that are ready for three-body calculations, not only for two-body elastic fits. Most global potentials were designed to reproduce elastic scattering. They are local or nearly local, energy dependent, and parameterized in forms that are convenient for Schrödinger solvers. A three-body calculation asks more of the operator. It uses the potential at off-shell momenta, inside transition matrix elements, between continuum bins, and sometimes at energies far from those where elastic data constrain it. The same elastic fit can therefore be insufficient for transfer, NEB, or knockout.

Reaction-ready optical operators should carry at least four pieces of information. They should specify the non-local kernel or justify the local equivalent approximation. They should provide uncertainty and correlations for both real and imaginary parts. They should state the energy and mass domain where the parameterization is constrained by data or by microscopic input. They should define how absorption is partitioned among target excitation, transfer, compound formation, and other omitted channels when that partition is used physically. A local elastic OMP without these metadata may still be useful, but it should not be treated as a complete operator for three-body reaction theory.

Microscopic and dispersive approaches are the natural long-term path because they tie nonlocality, absorption, and energy dependence to many-body structure [33, 179, 180]. But the near-term path is hybrid. Global phenomenological potentials remain indispensable because they cover wide mass and energy ranges. The immediate improvement is to attach posterior ensembles, covariance information, and benchmarked non-local extensions to those potentials. A CDCC or IAV user could then propagate not only a best-fit parameter set but an operator ensemble through the reaction calculation. This is the minimum needed before extracted spectroscopic factors or astrophysical rates can be called uncertainty quantified.

The same principle applies to projectile structure. A single overlap function, ANC, or separation energy is not enough for all observables. CDCC needs continuum phase shifts and transition densities. XCDCC needs core transition densities and deformation lengths. IAV needs

source functions with the correct radial support. THM needs a spectator momentum distribution and an off-shell subreaction amplitude. A structure calculation that provides only the bound-state overlap may be adequate for one transfer observable and inadequate for breakup or inclusive spectra. Reaction-ready structure input should therefore be distributed with the continuum information needed by the reaction method.

J. Theory triage for new measurements

Not every new data set needs the most expensive theory. The useful triage starts with the observable. If the measurement is an elastic angular distribution for a weakly bound projectile, CDCC with documented continuum convergence is usually the correct first theory. If the measurement is an exclusive two-fragment breakup spectrum, CDCC remains natural, but detector cuts and continuum-bin resolution become central. If one fragment is observed inclusively and the other is absorbed, transferred, or unobserved, IAV or an IAV-CDCC formulation is the right starting point. If the projectile has three clusters and the measured observable depends on the internal three-body continuum, four-body CDCC is the starting point and four-body IAV is the missing endpoint. If the system is light enough and Coulomb is manageable, exact AGS should be used as a benchmark rather than as an aspiration.

This triage prevents two common mistakes. The first is using a simple two-body optical model as the reaction theory for every observable, then interpreting the missing strength as structure. The second is using a large CDCC calculation for an inclusive observable and then reading the exclusive breakup probability as the measured yield. Both mistakes confuse method with observable. The correct question is: what degrees of freedom are observed, and which are summed over? Once that is answered, the method choice follows from the projection.

The triage should also include a stop condition. If a low-cost calculation and a high-cost calculation differ by less than the experimental and optical model uncertainty for the observable of interest, the low-cost calculation may be sufficient. If they differ in a way that changes the inferred spectroscopic factor, fusion suppression, or astrophysical S factor, the larger calculation is required. This is not a philosophical preference for complexity. It is a cost-aware rule tied to the conclusion being drawn.

K. Publication standards for formulae and notation

Three-body reaction papers are unusually vulnerable to notation drift. The same symbol may mean a projectile state in one section, a participant-target state in another, and a compound-nucleus channel in a third. For a review paper this is more than a stylistic problem. AGS, CDCC, IAV, and DWBA use different natural partitions

of the same Hamiltonian, and an incorrect translation between notations can change a sign, a boundary condition, or a phase-space factor. A useful standard is therefore to define the Hamiltonian and partitions once, then force every method chapter to use that notation. That is the reason Sec. II introduced a common $a = b + x$ notation before the formal sections.

Formula reporting should be equally explicit. The AGS equations should be printed, not merely cited. The CDCC ansatz and coupled equations should be printed with the channel coupling definitions. The DWBA breakup matrix element and the IAV post-form NEB expression should be printed with the source equation and Green-function interpretation. When a formula is adapted from a known source but the phase, normalization, or spin averaging has not been checked line by line, the uncertainty should be marked rather than hidden. This is a practical standard for preventing review articles from propagating wrong prefactors for decades.

The same standard should be applied to approximations. If ADWA is used, the adiabatic replacement of the projectile continuum should be stated. If THM is used, the chain from distorted three-body amplitude to pole approximation to quasi-free plane wave should be stated. If an equivalent local potential is used for a nonlocal kernel, the conditions of the equivalence should be stated. If a CDCC calculation omits continuum-continuum couplings or absorptive off-diagonal terms, the omission should be stated as a physical approximation, not buried in a numerical section. This is how formulae become auditable physics rather than symbolic decoration.

L. Training data for future theory

The rise of emulators and Bayesian calibration changes the value of old reaction calculations. A well-documented CDCC or IAV calculation is not only a result; it is training data for future reduced models, sensitivity analysis, and posterior prediction. But it is useful for that purpose only if the inputs and convergence metadata are preserved. A published angular distribution without the optical potentials, bin definitions, coupling truncations, and normalization convention is difficult to reuse. A published calculation with those details can become part of a cumulative benchmark base.

This does not require every paper to release a full production code. It does require the physics inputs to be reconstructible. For CDCC, that means the projectile Hamiltonian, continuum bins, multipole truncations, fragment optical potentials, and matching conditions. For IAV, it means the entrance wave, spectator optical potential, participant optical potential, source definition, finite-range or zero-range treatment, and detector folding. For AGS, it means pair interactions, screening radius and renormalization where Coulomb is present, partial-wave truncations, and convergence accelerations.

For UQ, it means priors, covariance, posterior samples or enough summaries to reconstruct them, and emulator validation points. These are the data that make a review field cumulative.

The long-term payoff is a map of method validity. Once enough benchmark calculations are preserved, one can learn where CDCC fails against AGS, where IAV differs from exclusive breakup plus transfer sums, where local OMPs are adequate, and where nonlocal off-shell effects dominate. Such a map would not replace physics judgment. It would sharpen it by showing which approximation has failed before in a nearby regime. That is the practical route from individual reaction models to a predictive reaction theory.

M. Verification standards for emulators

Emulators will become unavoidable as soon as multi-observable posteriors are attempted. That makes verification standards part of reaction theory. A useful emulator should report direct-solve errors on the observables used in the likelihood, residuals of the underlying coupled equations, derivative checks for gradient-based sampling, and failures near thresholds or channel openings. A single global speedup factor is not enough. It depends on the baseline solver, the number of right-hand sides, the channel count, and the observable being extracted. The relevant question is whether the emulator preserves the physics needed by the inference problem.

The verification target can be written as an inequality on posterior support,

$$\max_{\theta \in \mathcal{R}_p} \frac{|O_{\text{emu}}(\theta) - O_{\text{full}}(\theta)|}{\delta O_{\text{exp}}} \ll 1, \quad (174)$$

where $\mathcal{R}_p$ is a high-probability posterior region and δO_{exp} is the experimental uncertainty of the observable entering the likelihood. This criterion is more meaningful than requiring a tiny error over a broad prior box that the posterior never visits, and more honest than validating only at training points. It also connects emulator accuracy to the physics conclusion: an emulator error that is negligible for elastic scattering may not be negligible for an IAV source or a transfer matrix element.

Exact benchmarks should be used in the same spirit. AGS comparisons do not need to cover every target and energy to be valuable. They need to sample the places where CDCC is expected to fail: rearrangement-dominated observables, low-energy breakup with strong component interference, and Coulomb-heavy systems where the basis or boundary condition is stressed [76, 77]. Once those cases are mapped, the emulator and UQ program can carry not only parameter uncertainty but also a calibrated reaction-model error.

N. A practical vision

The practical vision is a closed reaction workflow. The structure model provides overlaps, thresholds, transition densities, and, eventually, microscopic fragment-target optical operators. The reaction model solves the appropriate three-body or four-body problem, choosing AGS when exact benchmarks are feasible, CDCC when exclusive continuum dynamics is the target, and IAV when the measured observable is inclusive. The statistical layer propagates calibrated optical-potential and model-discrepancy posteriors through the same equations rather than through a simplified surrogate. The diagnostic layer checks flux conservation, post-prior equivalence, continuum convergence, emulator residuals, and benchmark deviations against exact solutions where they exist.

This program is ambitious, but it is not speculative. Every piece already exists in partial form. What is missing is integration: exact benchmarks for the regimes where CDCC is used most heavily, inclusive theory for Borromean projectiles, nonlocal optical potentials with uncertainty, and differentiable emulators for the observables that actually determine the physics. The field will make progress when those pieces are treated as parts of one operator problem rather than as separate communities.

The practical workflow should begin with a written observable contract. Before choosing a solver, the analysis should state the detected particles, the unobserved subspace, the energy and angular cuts, the residue definition, and the structure quantity, if any, to be inferred. Only then should the reaction model be selected. Elastic scattering points to CDCC or an optical model with a DPP diagnostic. A spectator spectrum points to IAV. A transfer spectroscopic factor points to DWBA, ADWA, CDCC-BA, or AGS benchmarks, depending on the needed accuracy. A Borromean inclusive spectrum points to the missing four-body IAV problem. This contract is a simple discipline, but it would prevent many cases in which a calculation is technically correct for one observable and then interpreted as if it had computed another.

The second step is an input contract. Fragment optical potentials should be listed with their uncertainty, energy prescription, and role in the calculation. Projectile structure input should include not only bound-state quantities but the continuum information used by the reaction model. Detector cuts and resolution should be part of the theory comparison, not a separate afterthought. For an indirect method, the contract should also state the final low-energy quantity being inferred and the chain of approximations that maps the measured reaction to that quantity. In THM this means stating where the pole approximation and quasi-free factorization enter. In Coulomb dissociation this means stating where first-order electromagnetic theory is replaced by all-order breakup.

The third step is a failure contract. A serious analysis

should say in advance what would falsify the proposed mechanism. If complete-fusion suppression is attributed to elastic breakup, then an inclusive spectrum dominated by NEB should change the interpretation. If a halo radius is inferred from elastic scattering, then core-excitation breakup or transfer data that require a different structure model should change the conclusion. If a Bayesian calibration claims that elastic data determine a fragment potential well enough for transfer, then a posterior predictive failure in transfer should be treated as a model failure, not as a new spectroscopic factor. These falsification criteria are how reaction theory becomes predictive rather than descriptive.

The final step is archival. The field has many important calculations that cannot be reconstructed from the published curves. A modern review should therefore encourage a minimal reproducibility package: input potentials, projectile spectra, continuum bins or pseudostates, coupling truncations, normalization conventions, and enough posterior or emulator validation data to reproduce the uncertainty statement. This is not a demand for a single software ecosystem. It is a demand that the physics calculation be reconstructible. Theoretical disagreements are much easier to resolve when one can change a coupling, an optical potential, or a detector fold and see which part of the conclusion moves.

In this sense, the next decade of three-body reaction theory will be judged less by the invention of new acronyms than by the quality of its interfaces: between structure and reaction, between exact benchmarks and workhorse approximations, between exclusive and inclusive observables, between optical fits and nonlocal operators, and between deterministic solvers and statistical inference. The equations reviewed here already contain those interfaces. The work ahead is to make them explicit enough that weakly bound reaction data can constrain nuclear structure and astrophysics without carrying invisible reaction assumptions.

IX. CONCLUSIONS

Three-body reaction theory is no longer a collection of loosely related approximations. It is a hierarchy built around the same Hamiltonian, Eq. (5), and around the same question: which degrees of freedom must remain explicit for the measured observable? Faddeev-AGS theory answers this question by retaining all three-body rearrangement channels exactly within the chosen pair interactions. CDCC answers it by projecting the projectile continuum onto a finite, controllable set of bins or pseudostates. IAV answers it by summing inclusively over the unobserved nonelastic channels of one fragment. DWBA and adiabatic models remain useful reductions when the operator they approximate and the observable they target are stated clearly.

The central physics lesson is that weak binding changes the observable before it changes the interpretation. Elas-

tic scattering acquires a dynamic polarization potential, transfer and breakup cease to be perturbative side channels, inclusive fragment yields are dominated by NEB rather than EBU in many benchmark systems, and complete-fusion suppression is not simply a measure of prompt elastic breakup. The benchmark nuclei make this visible in different ways: ${}^6\text{Li}$ isolates nuclear breakup and cluster absorption, ${}^{11}\text{Be}$ exposes halo and core-excitation dynamics, ${}^6\text{He}$ and ${}^9\text{Be}$ force a four-body continuum, and ${}^8\text{B}$ tests whether all-order breakup theory can support an astrophysical inference.

The modern change is that convergence is no longer enough. A coupled-channel calculation can be converged in bins while remaining uncertain in the fragment optical potentials. A transfer calculation can be formally post-prior equivalent while remaining sensitive to off-shell non-locality. A Bayesian posterior can quantify parameter uncertainty while missing a functional ambiguity in the optical operator. This is why optical-potential UQ, information geometry, and emulators belong in the same review as AGS, CDCC, and IAV. They do not replace reaction theory; they determine how much of a reaction-theory conclusion is actually constrained by data.

The next useful advances are therefore specific. Exact three-body benchmarks must be extended toward the Coulomb-heavy and halo cases where CDCC is used most often. Four-body CDCC must be joined to an inclusive IAV-type sum rule for Borromean projectiles. Dynamic polarization potentials should be computed as nonlocal Schur complements before being localized. Optical potentials must carry both posterior parameter uncertainty and off-shell functional uncertainty. Differentiable emulators should be judged by whether they enable posterior inference for transfer, breakup, and inclusive observables, not only by raw speed.

The common thread is simple: write the operator first, then decide the approximation. When that discipline is followed, three-body reaction theory can connect elastic scattering, breakup, transfer, fusion, surrogate reactions, and spectroscopic quenching without changing language at every chapter boundary. That is the route from useful models to quantitative reaction theory.

This conclusion is deliberately conservative. It does not say that every weakly bound reaction must be solved by the most exact available method. It says that the approximation has to match the observable. Elastic scattering of a deuteron at tens of MeV may be well served by CDCC, and a transfer matrix element may be well served by an adiabatic or Weinberg reduction. Inclusive alpha spectra from ${}^6\text{Li}$ require IAV because the unobserved participant states are the observable. Borromean projectiles require four-body structure at least until one has shown which effective two-cluster reduction is active. Coulomb dissociation and THM require a hierarchy from the exact three-body amplitude down to the factorized indirect formula. The same Hamiltonian can support all these reductions, but it does not license them interchangeably.

The review also points to a change in standards. A

calculation that gives one good angular distribution is no longer enough for a precision claim. The field now has the tools to ask whether the same inputs describe elastic scattering, breakup, transfer, inclusive spectra, and absorption ledgers; whether the optical potentials are constrained by the data used; whether the continuum truncation has been tested in the observable of interest; and whether an emulator is accurate over the posterior support rather than only at training points. These checks are not bureaucratic. They are the way to separate a structure conclusion from a reaction-model artifact.

The best near-term progress will likely come from systems that are not maximally exotic. Deuterons, ${}^6,{}^7\text{Li}$, ${}^9\text{Be}$, and ${}^{11}\text{Be}$ on well measured targets are demanding enough to expose the three-body physics but simple enough to support redundant observables and benchmark calculations. They can anchor the methods before the same machinery is pushed to the dripline, to heavy radioactive beams, or to astrophysical indirect reactions. In that sense the familiar benchmark nuclei remain the field's most useful laboratory. They are where exact theory, CDCC, IAV, and UQ can still meet on the same page.

The longer-term goal is not a universal code or a universal potential. It is a consistent operator chain: a structure model that supplies the projectile degrees of freedom, fragment-target optical operators with quantified parameter and functional uncertainty, a reaction solver whose approximation is matched to the observable, and a statistical layer that propagates all of this to the measured quantity. When that chain is explicit, a disagreement between theory and data becomes informative. It can point to a missing continuum coupling, an off-shell optical ambiguity, an inadequate structure overlap, or a breakdown of an indirect-method factorization. Without that chain, the same disagreement is only another adjustable normalization.

The review has also tried to separate four meanings of "inclusive", which are often mixed. The optical model is inclusive over eliminated target and projectile excitations but reports only an elastic amplitude. CDCC is exclusive in the retained projectile continuum but may contain absorption through complex fragment-target potentials. IAV is inclusive over participant-target nonelastic states while remaining differential in the observed spectator. Fusion data are inclusive over many entrance histories but exclusive in the residue class if complete and incomplete fusion are separated. These distinctions are not semantic. They decide whether the right object is an S matrix, a coupled-channel wave function, a source function, or an absorptive expectation value.

The same separation is needed for uncertainty. Parameter uncertainty, continuum truncation error, optical functional ambiguity, and structure model error are different objects. Adding them into one band before understanding their origin can hide the relevant physics. A halo reaction may be limited by core excitation rather than by optical-potential statistics. A transfer reaction

may be limited by nonlocal off-shell sensitivity rather than by the fitted elastic covariance. A CDCC Bayesian calibration may constrain surface absorption well while leaving interior matrix elements poorly determined. A useful UQ result should therefore say which uncertainty controls the conclusion, not only display a shaded region.

The most important experimental implication is that single-observable success is becoming less decisive. Elastic scattering remains necessary, because it fixes the entrance distortion and exposes polarization effects. But it is not sufficient for mechanisms that are defined by unobserved fragments or residues. The next generation of weakly bound reaction studies should be designed as closure measurements wherever possible: elastic, exclusive breakup, inclusive spectator spectra, transfer, and residue yields on the same system, with correlated normalizations. Such data are more difficult than another isolated angular distribution, but they are exactly what the operator structure demands.

The most important theoretical implication is that exact and approximate methods should be used in a division of labor. AGS and Faddeev calculations should anchor the approximations where the Hamiltonian is still tractable. CDCC should carry the heavy-target and radioactive-beam continuum dynamics, with model-space convergence reported in the observables that matter. IAV should be used whenever the measured fragment is inclusive over the fate of the other fragment. DWBA and adiabatic reductions should remain in use, but with their approximated operator stated explicitly. Emulators should make these calculations statistically usable, not replace their physics content.

The formal checklist is correspondingly short but strict. For an exact benchmark, write the AGS transition-operator equations and the breakup operator, then state how Coulomb and optical absorption are handled. For a CDCC calculation, write the projectile expansion, the coupled radial equations, the coupling potentials, and the continuum discretization. For an inclusive breakup calculation, write the post-form source equation and the NEB expectation value with the sign convention for the absorptive potential. For a transfer or elastic-breakup calculation, write the DWBA matrix element and the post or prior transition operator. If any of these formulae is only quoted in words, the approximation cannot be audited. If the notation changes between sections, the comparison cannot be trusted.

This checklist is not pedantry. It is how one distinguishes three different levels of approximation. The first level is kinematic: masses, Jacobians, phase-space factors, and boundary conditions. Mistakes here can shift normalizations without changing the apparent physics. The second level is dynamic: which couplings, partitions, and absorptive terms are retained. This level decides whether the calculation describes elastic breakup, NEB, transfer, or fusion absorption. The third level is statistical: which inputs are calibrated, which are fixed, and which uncertainties are propagated. A paper can be cor-

rect at the first level and incomplete at the second, or complete at the second and overconfident at the third. A mature review should keep the three levels visible.

The benchmark nuclei also define a compact memory of the field. The deuteron teaches that the simplest weakly bound projectile already requires rearrangement channels. ${}^6\text{Li}$ teaches that elastic breakup is not the same as incomplete fusion. ${}^7\text{Li}$ teaches that a modest threshold shift can change the transfer and alpha-production balance without changing the target. ${}^{11}\text{Be}$ teaches that halo structure and core excitation must be tested by observable, not assumed from binding energy. ${}^6\text{He}$ teaches that a Borromean projectile cannot be reduced forever to an effective two-body participant. ${}^9\text{Be}$ teaches that stable beams can be the hardest tests of flux accounting. ${}^8\text{B}$ teaches that indirect astrophysical information is only as clean as the all-order reaction mapping that extracts it.

The final standard is therefore reproducibility of interpretation. It should be possible for a reader to identify the Hamiltonian, the projection, the observable, the approximations, the optical inputs, and the uncertainty budget without guessing. It should also be possible to say what calculation would change the conclusion: an AGS benchmark, an XCDCC extension, a four-body IAV calculation, a nonlocal optical potential, a different detector fold, or a joint posterior over several observables. When a result has that structure, disagreement is useful. When it does not, disagreement usually produces only another round of parameter adjustment.

This is also a warning about terminology. A statement such as “breakup suppresses fusion” is incomplete unless it says which breakup, which fusion, and which projection connects them. A statement such as “the optical model fits the data” is incomplete unless it says whether the fitted potential is being used only for elastic scattering or also inside a transfer or NEB matrix element. A statement such as “the spectroscopic factor is quenched” is incomplete unless reaction-induced quenching and structure quenching have been separated. The field has enough formal machinery that these shortcuts are no longer necessary.

The constructive alternative is to treat every reaction claim as a map from an operator to an observable. The map begins with the Hamiltonian and the chosen partitions. It passes through the optical operators, continuum basis, source term, transition amplitude, or absorptive expectation value. It ends only after the detector projection and statistical inference have been specified. This is a longer sentence than the usual mechanism label, but it is the level at which weakly bound reaction physics is actually decided. It also makes clear where future work should enter. A better structure model changes the overlap or continuum density. A better optical potential changes the fragment-target operator. A better reaction solver changes the projection of the same Hamiltonian. Better data change the likelihood and the observable constraints. Keeping these roles separate is what allows the pieces to improve without confusing their meanings.

The review therefore ends with a modest criterion for success. A future analysis of a weakly bound or halo reaction should be able to answer six questions in one notation. What is the projectile partition? Which channels are explicit? Which channels are eliminated into an optical operator? Which fragment or residue is observed? Which final states are summed? Which uncertainties are propagated to the final physics number? If those questions can be answered, the method choice becomes transparent. If they cannot, the calculation may still fit a data set, but it has not yet explained the reaction.

There is no contradiction between this standard and pragmatic modeling. Many useful calculations will still use local potentials, truncated continua, zero-range approximations, or first-order reductions. The point is to make their role explicit. A local potential can be the right elastic surrogate while still being insufficient for an off-shell transfer matrix element. A truncated CDCC basis can be adequate for an elastic DPP while still missing a breakup spectrum. A zero-range IAV source can be a useful estimate while still requiring finite-range validation for a precision spectator spectrum. A first-order Coulomb-dissociation formula can be the correct transparent limit while still needing all-order checks for an astrophysical extraction. Approximations become dangerous only when they are unnamed, or when success in one projection is exported to another.

That is why the phrase “complete review” in this subject should mean more than an inventory of methods and

systems. It should give the reader enough operator detail to move between methods, enough benchmark detail to know where each method has been tested, and enough uncertainty language to know which conclusions are data constrained. The review has been written with that standard in mind: exact equations where they define the method, explicit notation shared across sections, benchmark nuclei tied to observables, and formula uncertainties marked where the literature convention still needs a line-by-line check before submission.

Three-body reaction theory has reached the point where its central challenge is integration. The formal equations are known, the benchmark systems are known, the inclusive sum rules are known, and the first generation of Bayesian and reduced-basis tools is in place. What remains is to use them together, with enough discipline that the extracted nuclear-structure and astrophysical information is not carrying hidden reaction assumptions. That is the standard this review has tried to make explicit.

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